



SKG16.1 Control Theory and Applications (Part 2)

LAYER 3 Advanced Regulatory Controls

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Advanced Regulatory Control

General Notes

- Advanced Regulatory Controls or ARC refer to a group of technologies and applications that typically involve :
 - Multiple inputs and multiple outputs (MIMO)
 - Multiple inputs with a single output (MISO)
 - More complex calculations
 - Fast response
- The complexity is ALWAYS derived from placing **MORE PROCESS INFORMATION** into the control system
- Due to the complexity of such systems, Control Narratives are mandatory requirements (to understand such systems)

Advanced Regulatory Control Scope

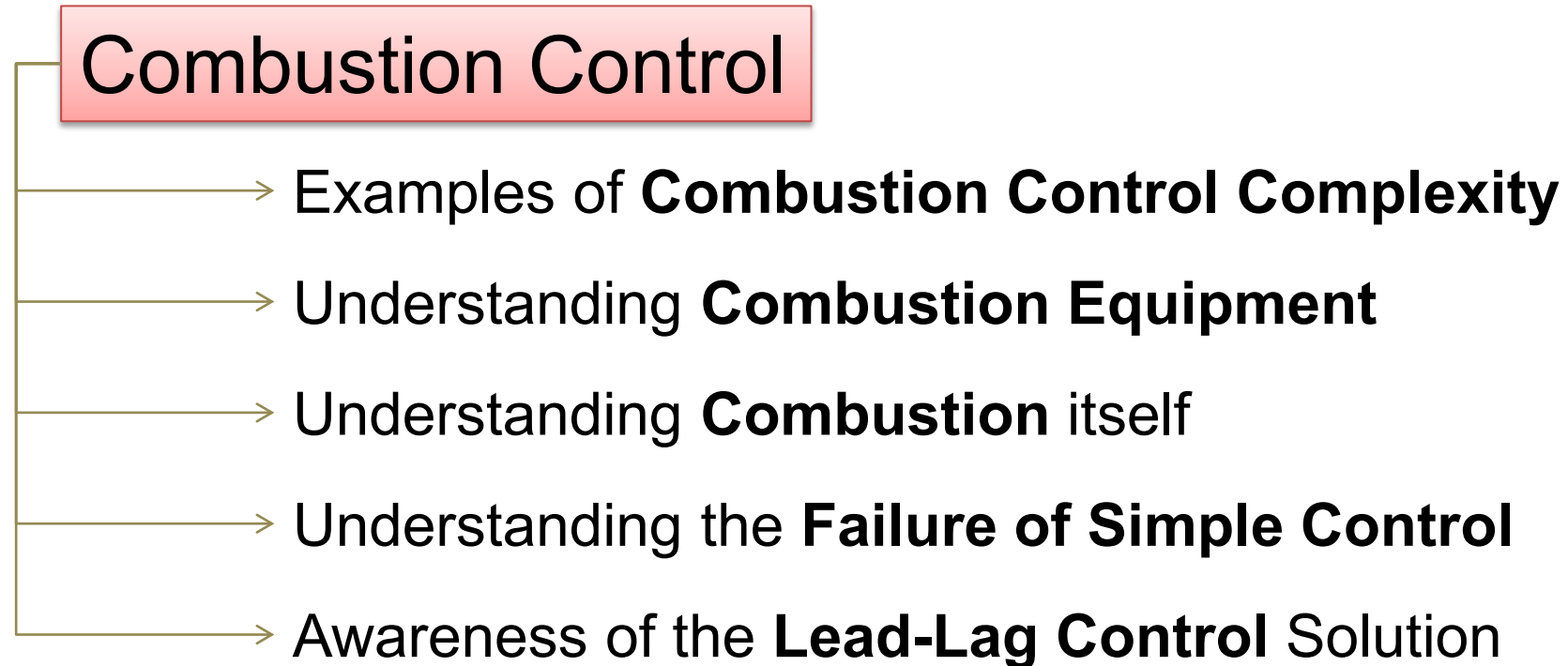
The following topics are usually covered under the topic of ARC:

- Combustion Control and Boiler Waterside Controls
- Compressor Antisurge Control
- Distillation Control

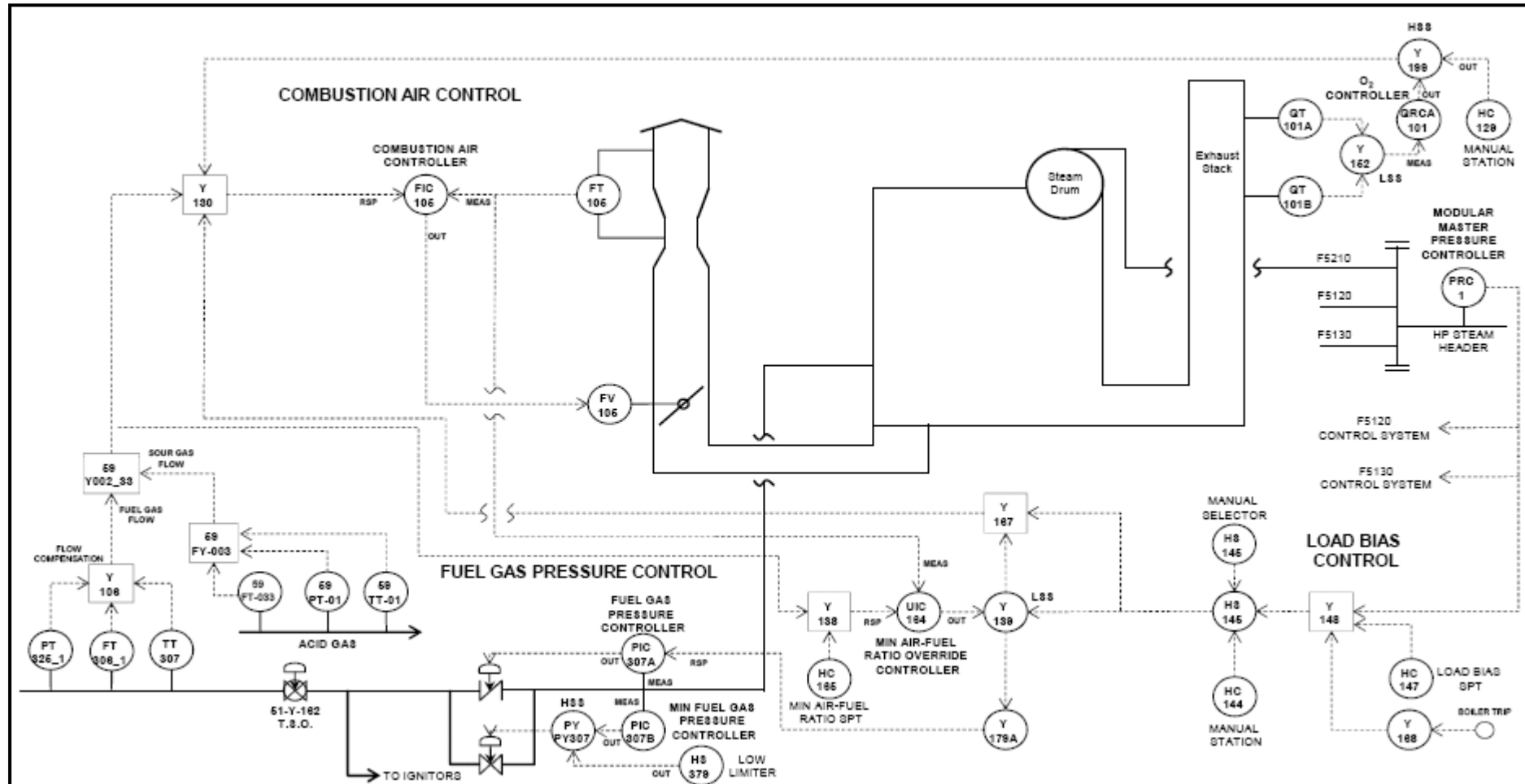
Combustion and Boiler Controls

Layer 3 Controls

Combustion Control

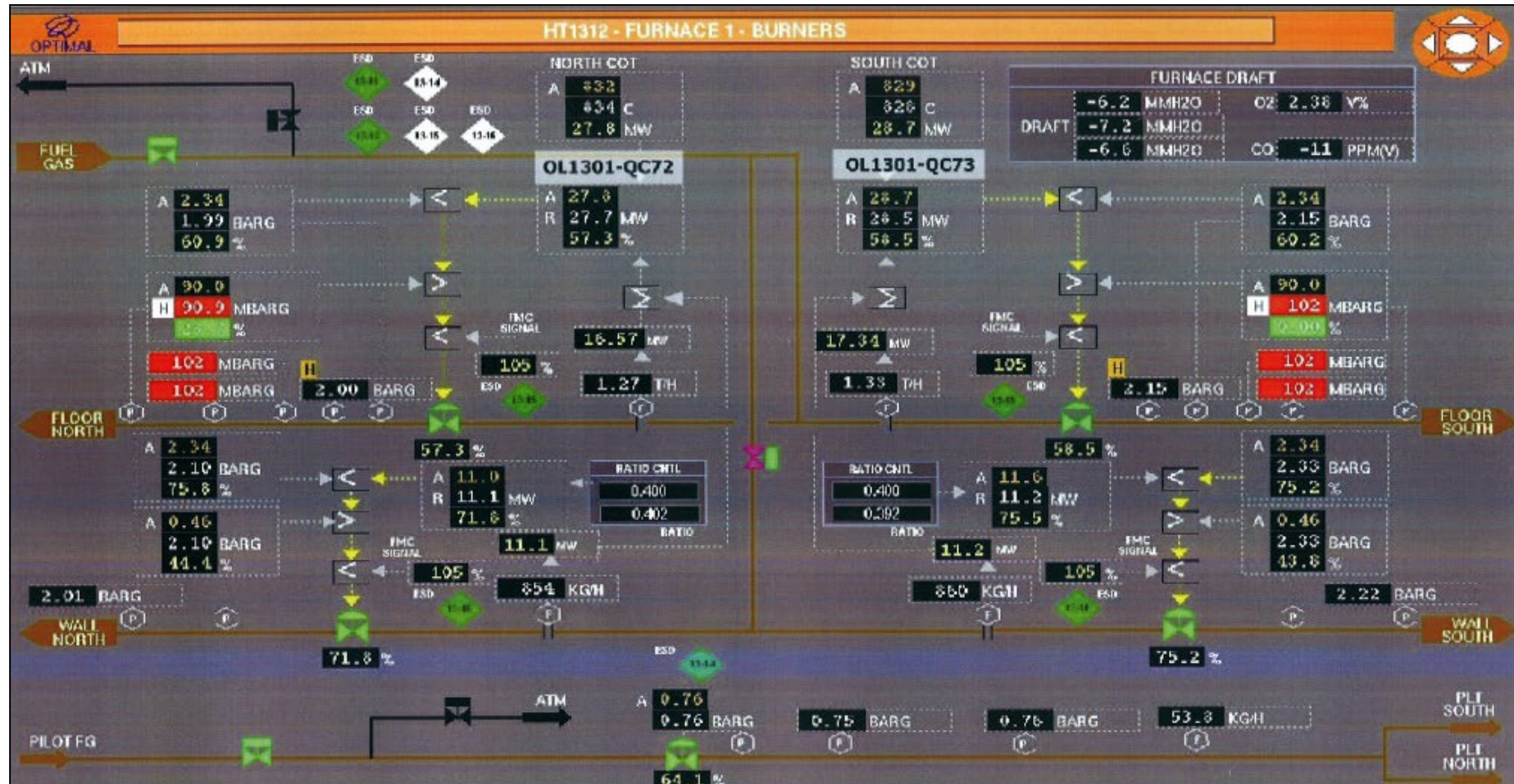


Example of complexity 1) Combustion Controls - MLNG



Combustion control systems typically comprise of many interconnected control blocks that do a variety of actions. The necessity of the complexity derives from the complexity of the combustion process.

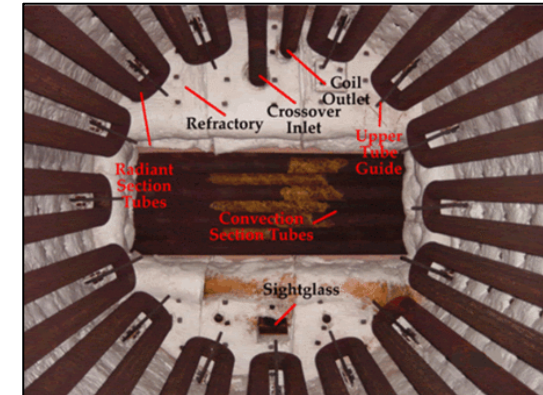
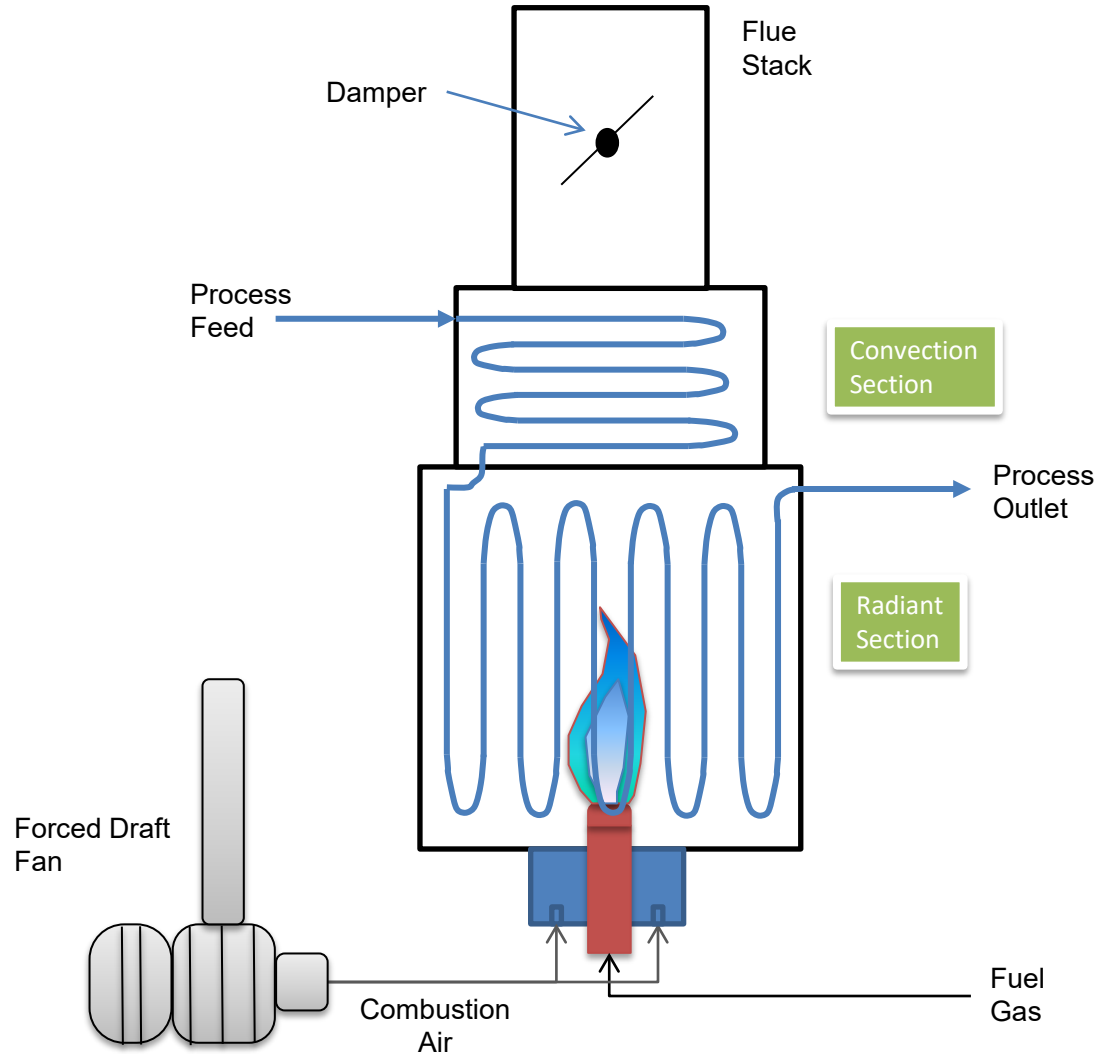
Example of complexity 2) Combustion controls - Optimal



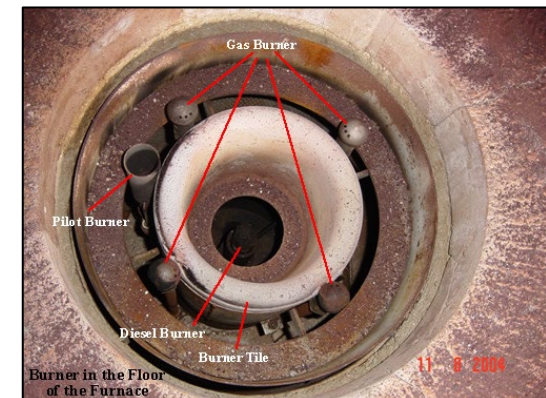
To understand the complexity of control, it is important to understand i) the **combustion equipment** and ii) the **combustion process** and its constraints.

Understanding combustion equipment ..1

- the basic furnace box



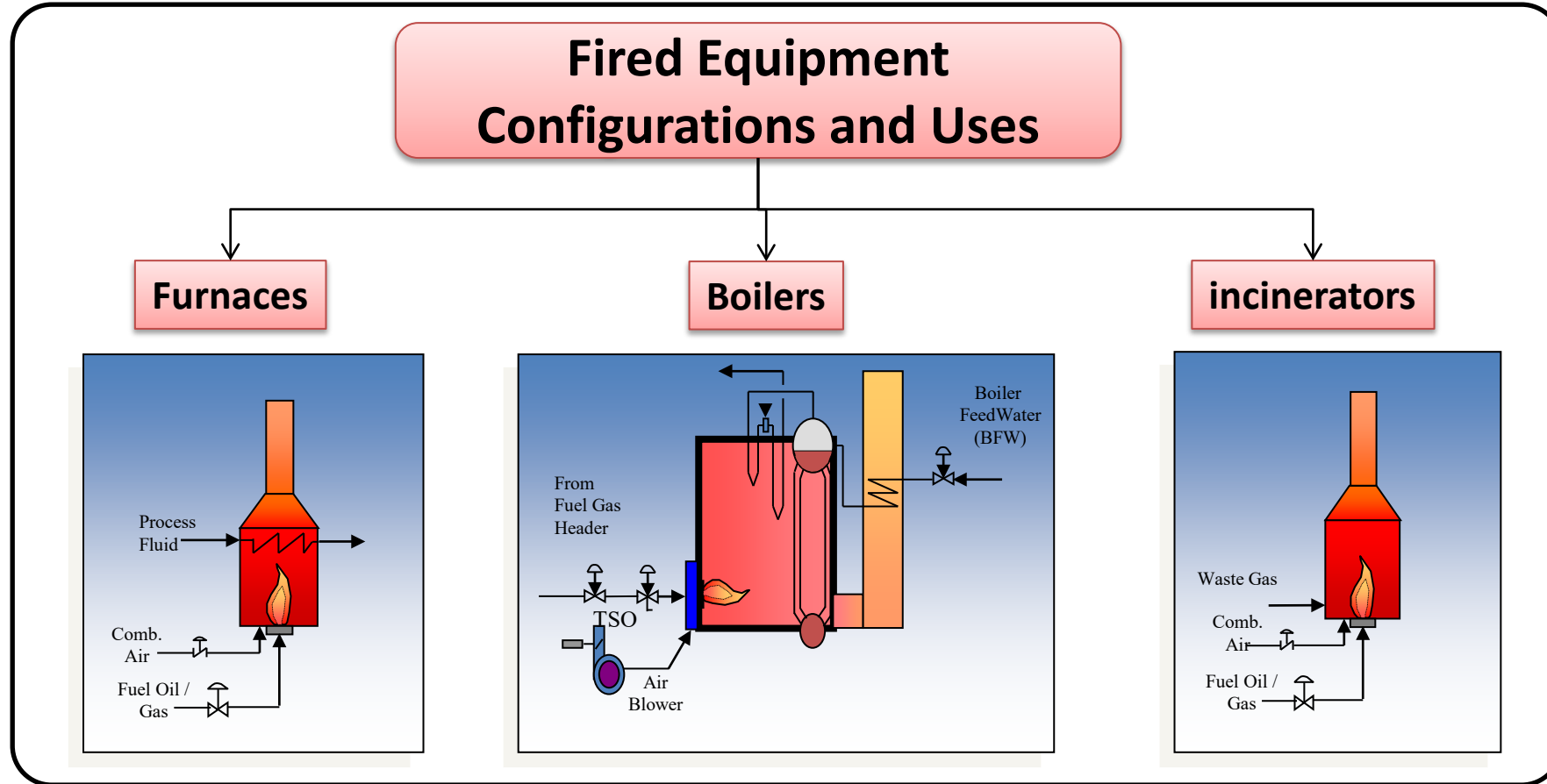
Looking up from burner floor



Looking down to the burner

Understanding Combustion Equipment .. 2

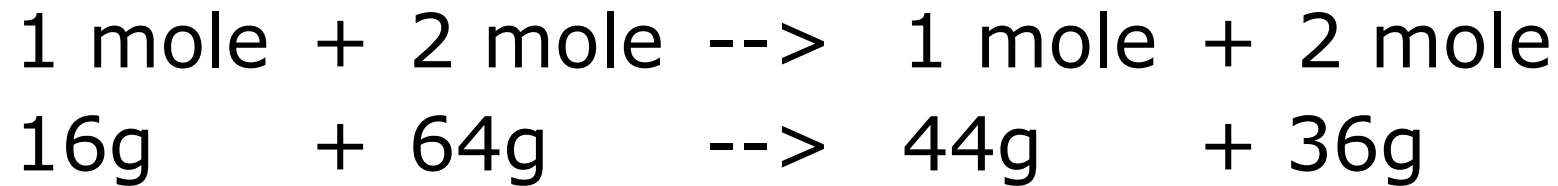
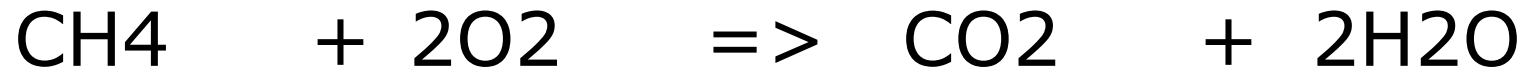
- Configuration and Uses



Heat in a furnace box is utilised either for heating (typical furnace), phase transition (boilers) and thermal decomposition (incineration). However, the basic process of combustion in the furnace box in each of the three different equipment is identical.

Understanding Combustion ...1

Stoichiometry



Ratio : 1 ton CH₄ : 4 tons O₂
Ratio ~1 ton CH₄ : 20 tons Air

Conclusion

1 ton of methane requires approximately 20 tons of air for complete and stoichiometric combustion.

Reference

Compositional Information

% (mass) of oxygen in (dry) air = 23.2 % ^{*1}

^{*1}) Source <http://www.engineeringtoolbox.com/>

Properties

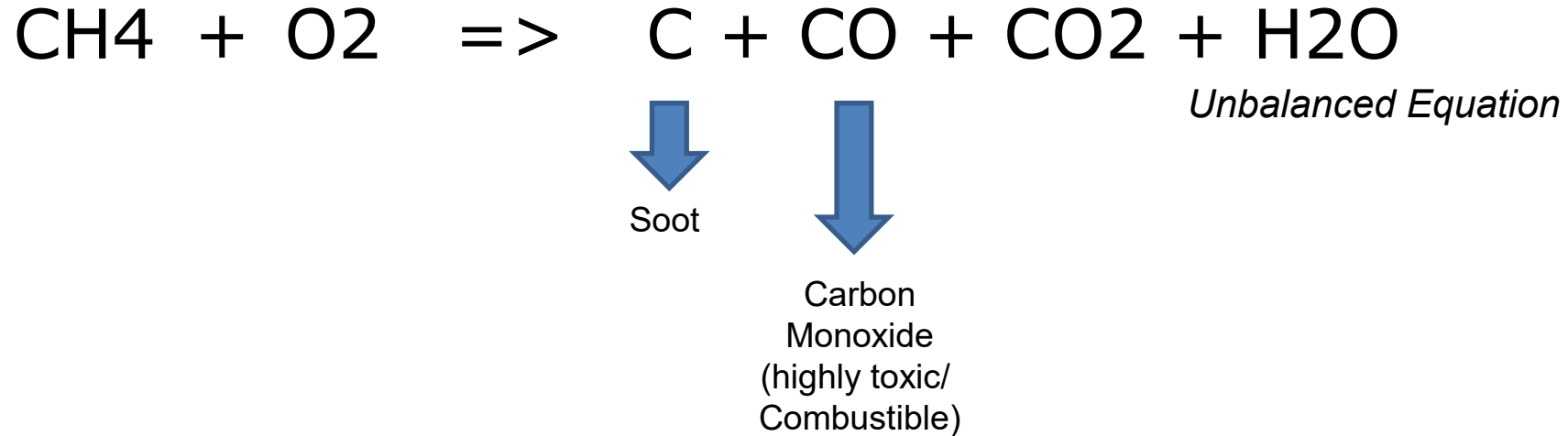
Flame Temperature of Methane (in air) = 1950 C ^{*1}

Autoignition temperature (Methane) = 580 C ^{*1}

Autoignition temperature (CO) = 609 C ^{*1}

Understanding Combustion2

Sub-stoichiometry



“Process heaters and boilers should NEVER be operated in this (substoichiometric) mode. Suddenly adding air to such a hot mixture could result in explosion.”

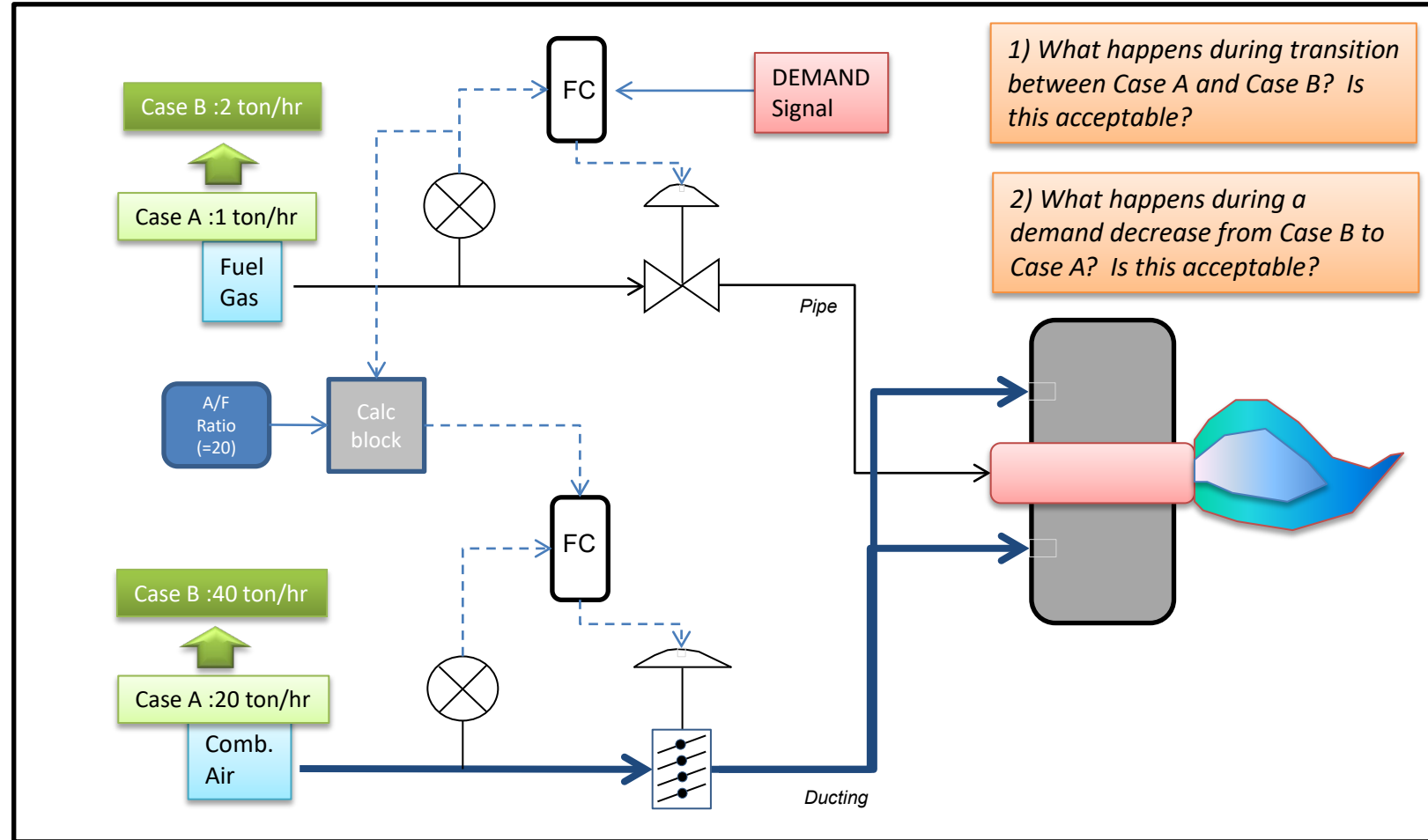
The Coen and Hamworthy Combustion Handbook

Critical Process Conclusion (and impact on control system design)

Combustion systems shall not be allowed to operate in sub-stoichiometric conditions. It is the objective of an effective combustion control system to prevent sub-stoichiometric conditions.

Preventing sub-stoichiometric firing

- Using a Fuel Leading system

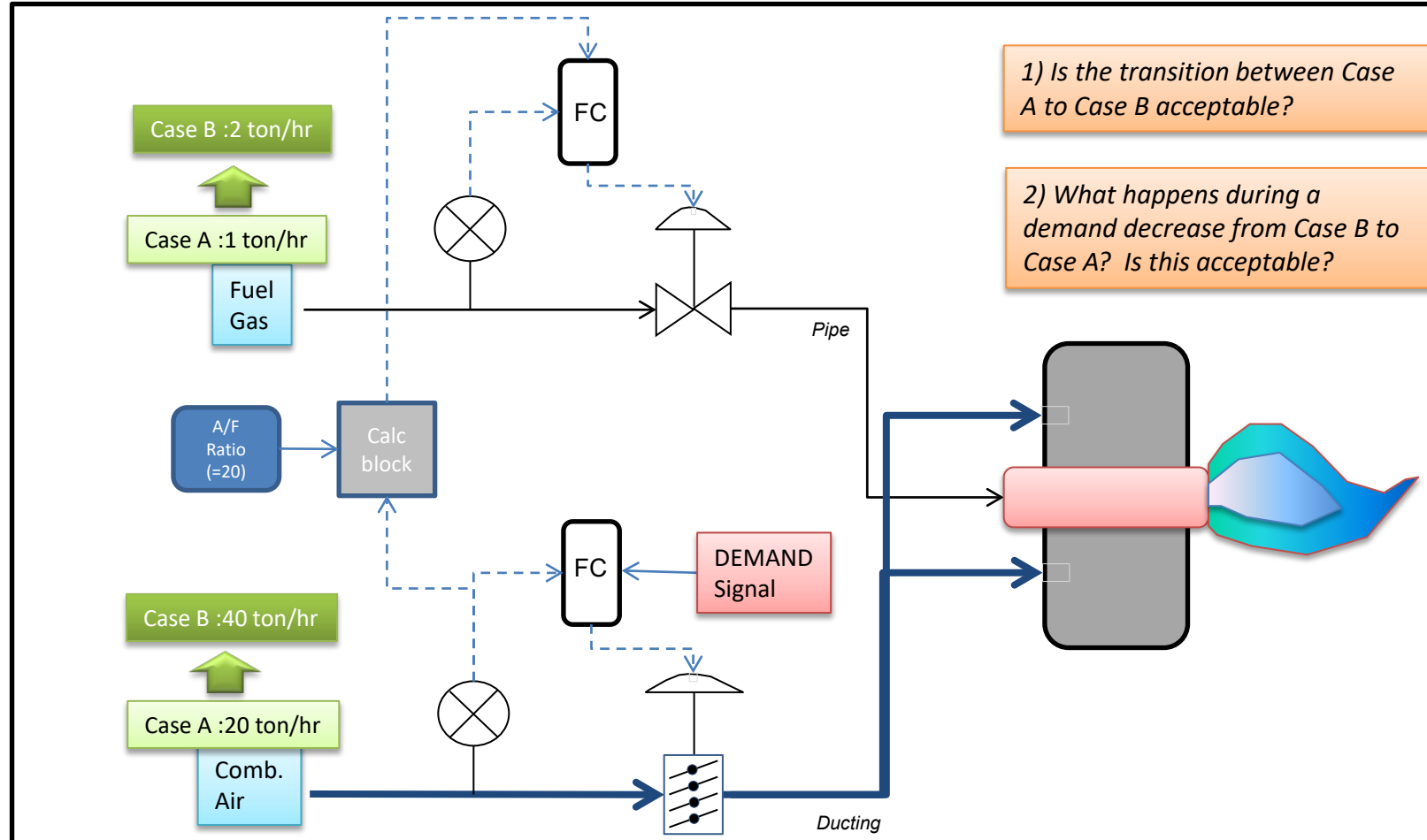


Conclusion

A fuel leading system will result in sub-stoichiometric firing on DEMAND INCREASE. However, it performs correctly on DEMAND DECREASE.

Preventing sub-stoichiometric firing

- Using an Air-Leading system



Conclusion

An air-leading system will perform correctly on DEMAND INCREASE
However, it will result in sub-stoichiometric firing on DEMAND DECREASE.

Conclusions from Air and Fuel Leading control systems

Demand Action	Fuel Leading System	Air Leading System
Demand Increase	Not OK	OK
Demand Decrease	OK	Not OK

IMPLICATION FOR DESIGN

An EFFECTIVE Air-Fuel Ratio Control System shall meet the following criteria:

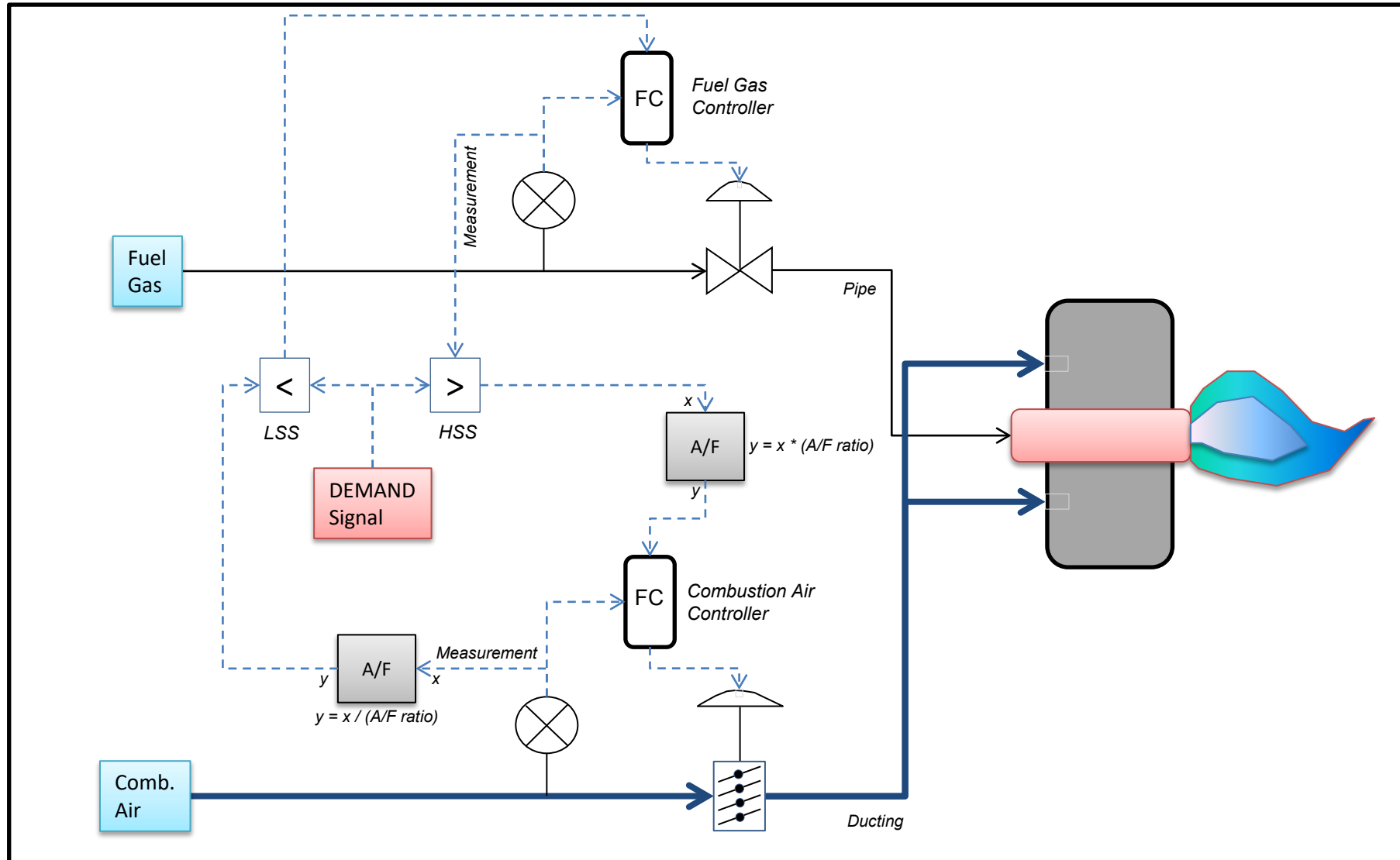
Demand Condition	First Action	Second Action
Demand Increase	Air must 'Lead" upward first	Fuel must "Lag" behind air
Demand Decrease	Fuel must "Lead" downward first	Air must "Lag" behind fuel

Conclusion

There is a need for a "Lead-Lag Control System" for effective combustion control.

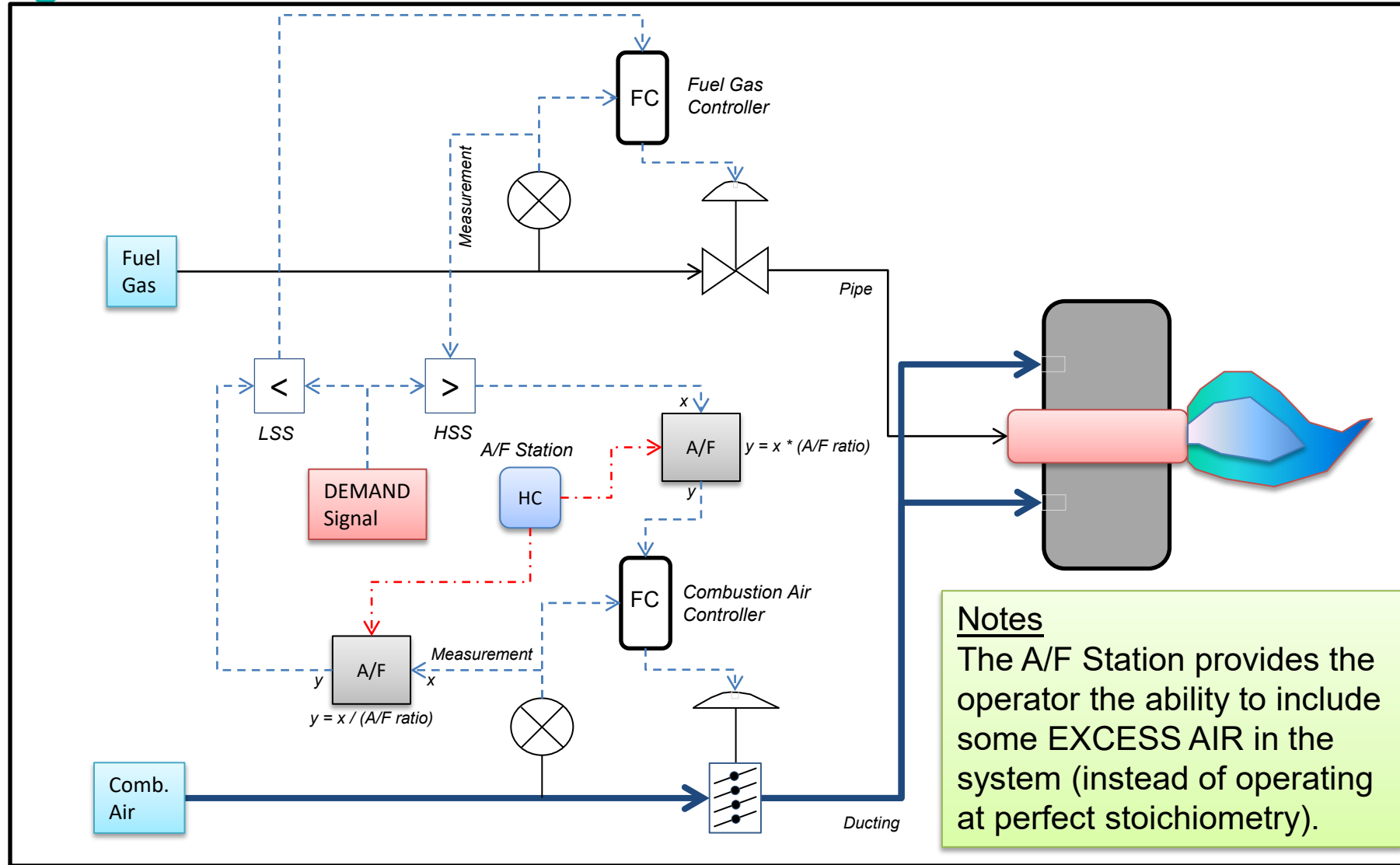
Preventing sub-stoichiometric firing

- Using a Lead-Lag Control Scheme



Using a Lead-Lag Control Scheme

- Adjusting the **Air-Fuel Ratio**

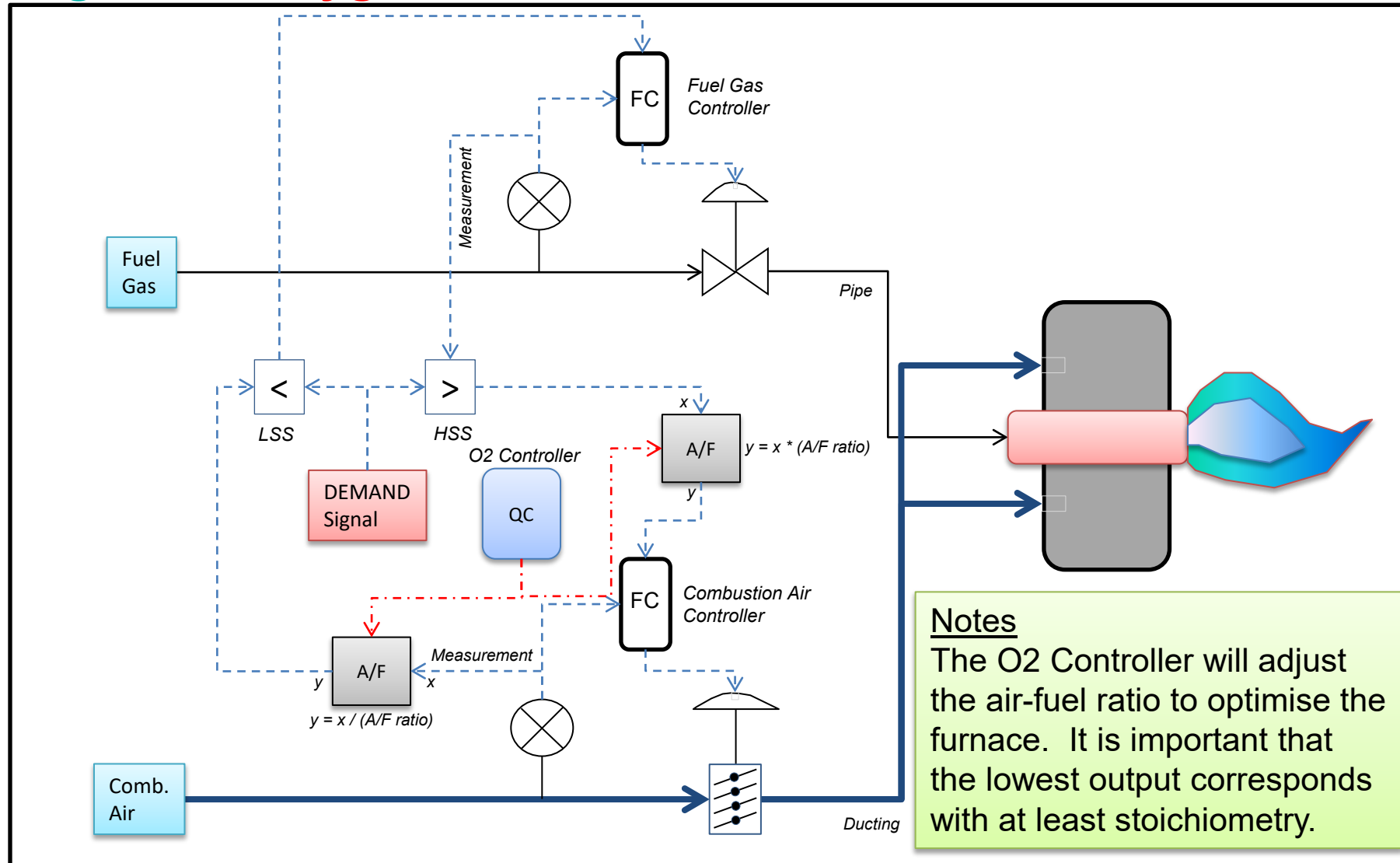


Excess Air

- Combustion systems are typically subject to
 - imperfect mixing at the burner
 - Asymmetrical piping geometry
- The result is that if the combustion system is run at the stoichiometric ratio, it is possible for the burner to be firing sub-stoichiometrically.
- To prevent this, the combustion system is provided 'Excess Air'. This is combustion air that is over and above the stoichiometric requirement
- Excess air has the following constraints
 - Too little Excess air
 - Results in potential sub-stoichiometric firing at the burner – Safety Issue
 - Too much Excess Air
 - Results in cooling of the furnace box and loss of efficiency – Optimisation Issue
- To control the amount of Excess Air, an oxygen control loop is installed. This control loop will then 'trim' the combustion air to meet the optimum targeted stack O₂.

Using a Lead-Lag Control Scheme

- Optimising Stack Oxygen



Notes

The O2 Controller will adjust the air-fuel ratio to optimise the furnace. It is important that the lowest output corresponds with at least stoichiometry.

Retrieval Practice ...1

1. The Combustion Control scheme for a forced draft furnace is fluctuating resulting in a swinging outlet temperature control. The operator decides to place the combustion control in manual. The operator has been told to increase the feed to the furnace. While increasing the feed, he also increases the fuel gas flow manually.
 - a) What is the issue with this?
 - b) What should he do first?

2. If the above scheme is tuned correctly and then placed fully automatic, how would the lead lag scheme behave if the feed flow to the furnace coils suddenly drops and the outlet temperature increases.
 - a) It would first _____
 - b) Then it would _____

3. Is having a stack oxygen controller vital to the safe functioning of a Combustion Control scheme?
 - a) Yes
 - b) No

Retrieval Practice

4. Because of a previous incident of “black smoke”, the panel operator decides to keep air flow in excess. He applies about 30% of excess air. This is safe but over the period of a month, there is a complaint from the Energy and Loss group that the cost of furnace operation has increased. Explain why.

- _____

5. Heat from a combustion process can be used for one of three actions:

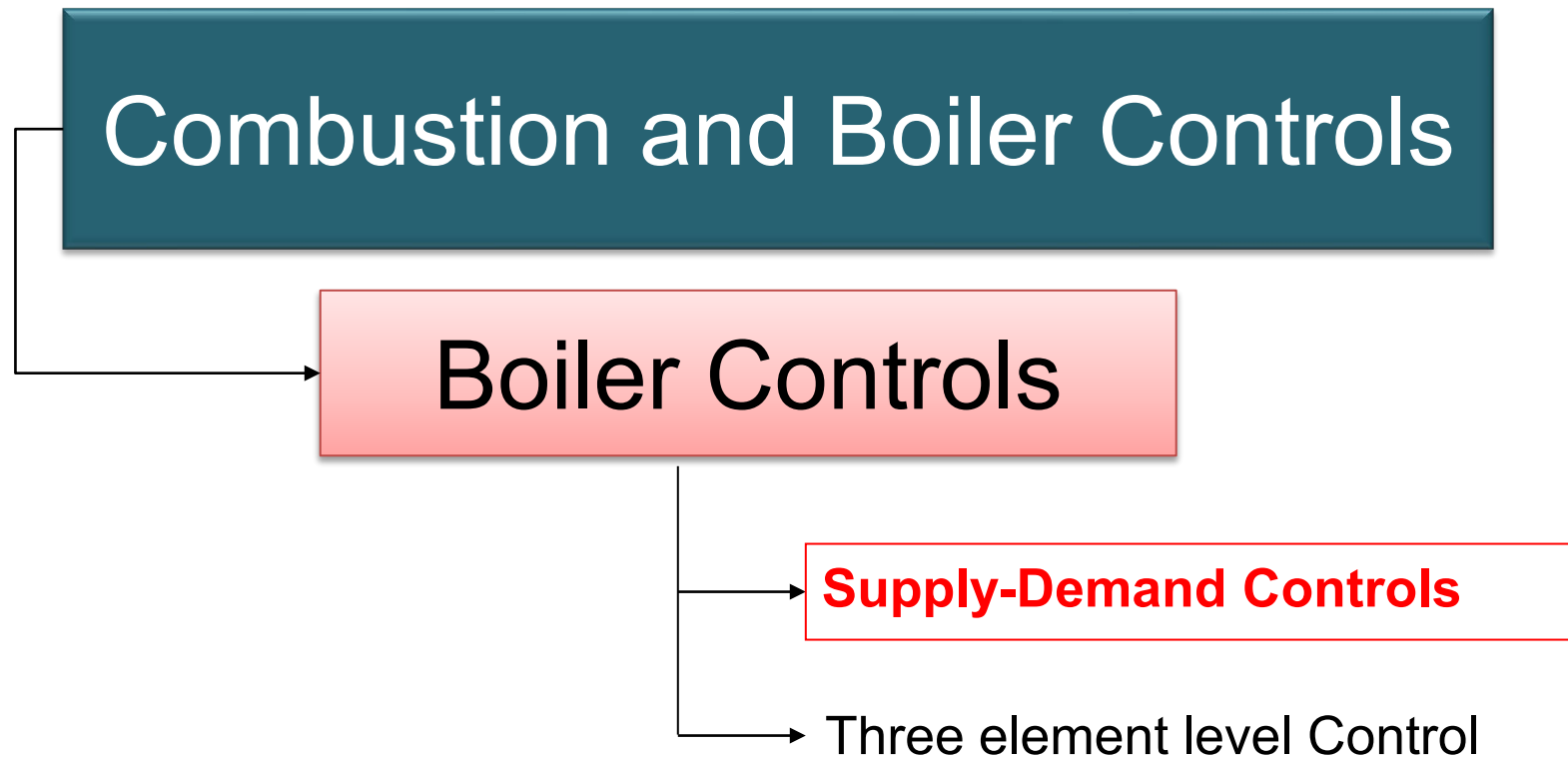
a) _____
b) _____
c) _____

6. The key objective of an effective combustion control system (such as lead-lag controls) is to prevent the presence of _____ inside the _____ box especially during _____ conditions.

Key Takeaways

- The combustion control system is perceived as complex due to the complexity of the combustion process itself.
- The key objective of the combustion control system is to prevent combustibles from entering the furnace box.
- Uncontrolled entry of combustibles could result in an explosion if air is suddenly introduced.
- Basic combustion control systems such as Fuel-leading and Air-leading control systems can only handle either a demand increase or a demand decrease BUT NOT BOTH.
- A lead-lag control system achieves safe operation by leading air and lagging fuel or vice-versa on a demand increase or demand decrease respectively.

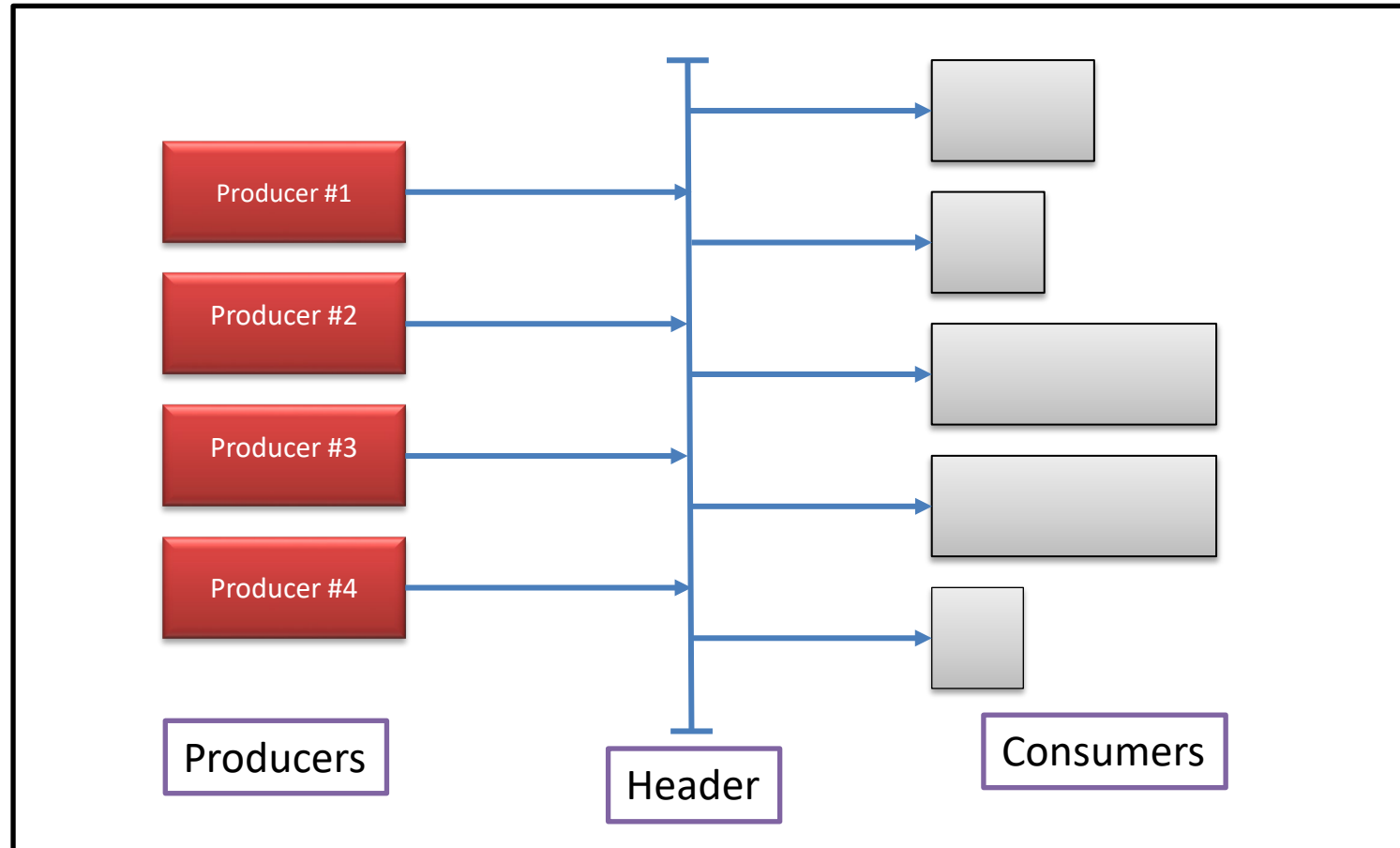
Advanced Regulatory Controls



Supply-Demand Controls

- Supply-Demand controls are utilized for systems where there are multiple consumers of a Utility (such as steam, instrument air, nitrogen, etc.).
- These systems are characterized by a producer/producers of the service supplying into a common Utility header.
- The consumers will then receive the Utility by a tie-up to the header.
- The area of concern is to balance supply and demand.

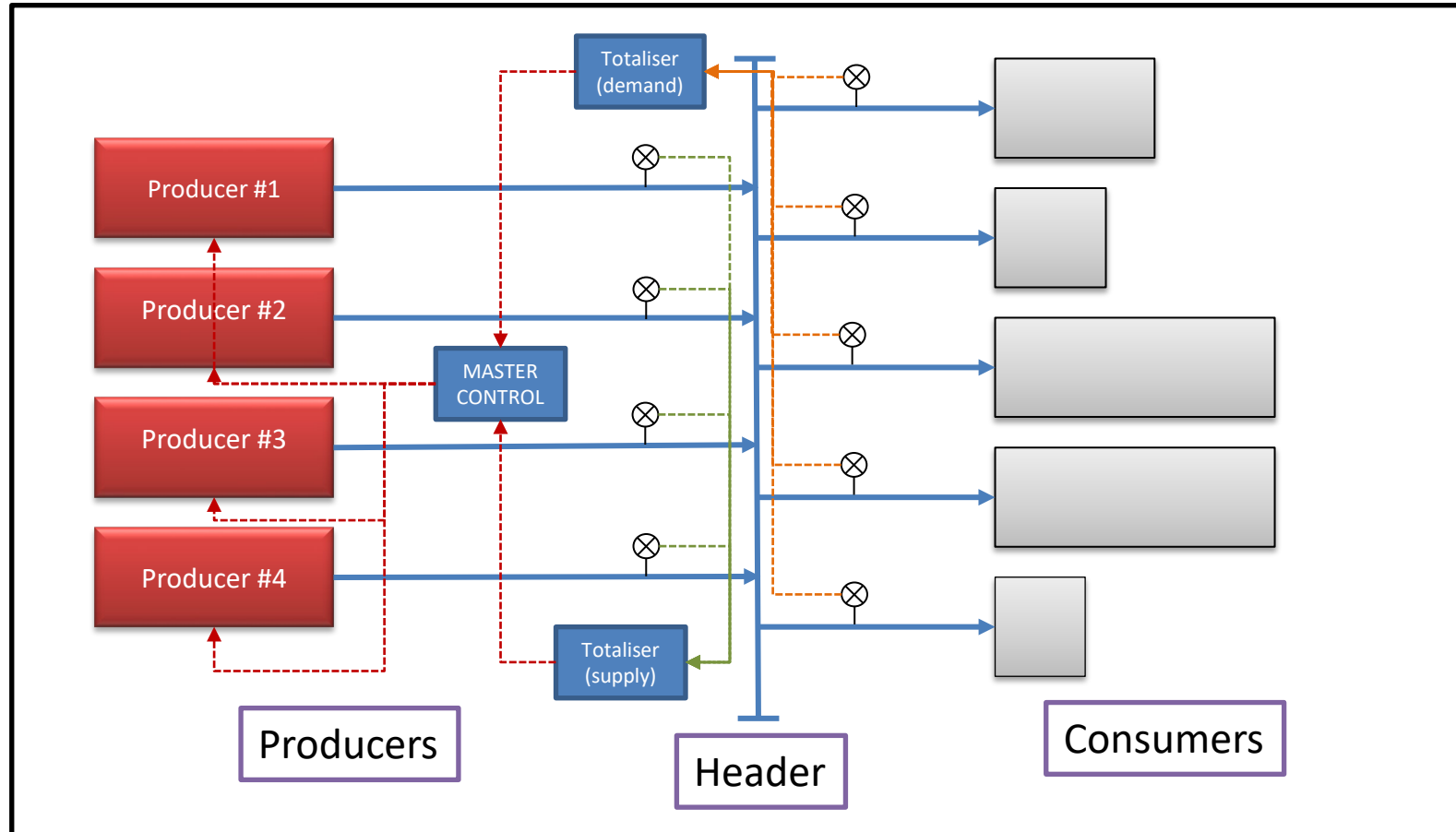
Typical Utility Systems



The key issue in such utility supply systems is to balance the production with the demand from the consumers. Failure to balance supply-demand may lead to outage (trip) of the consumer/s or overpressure of the header.

Typical Utility Systems

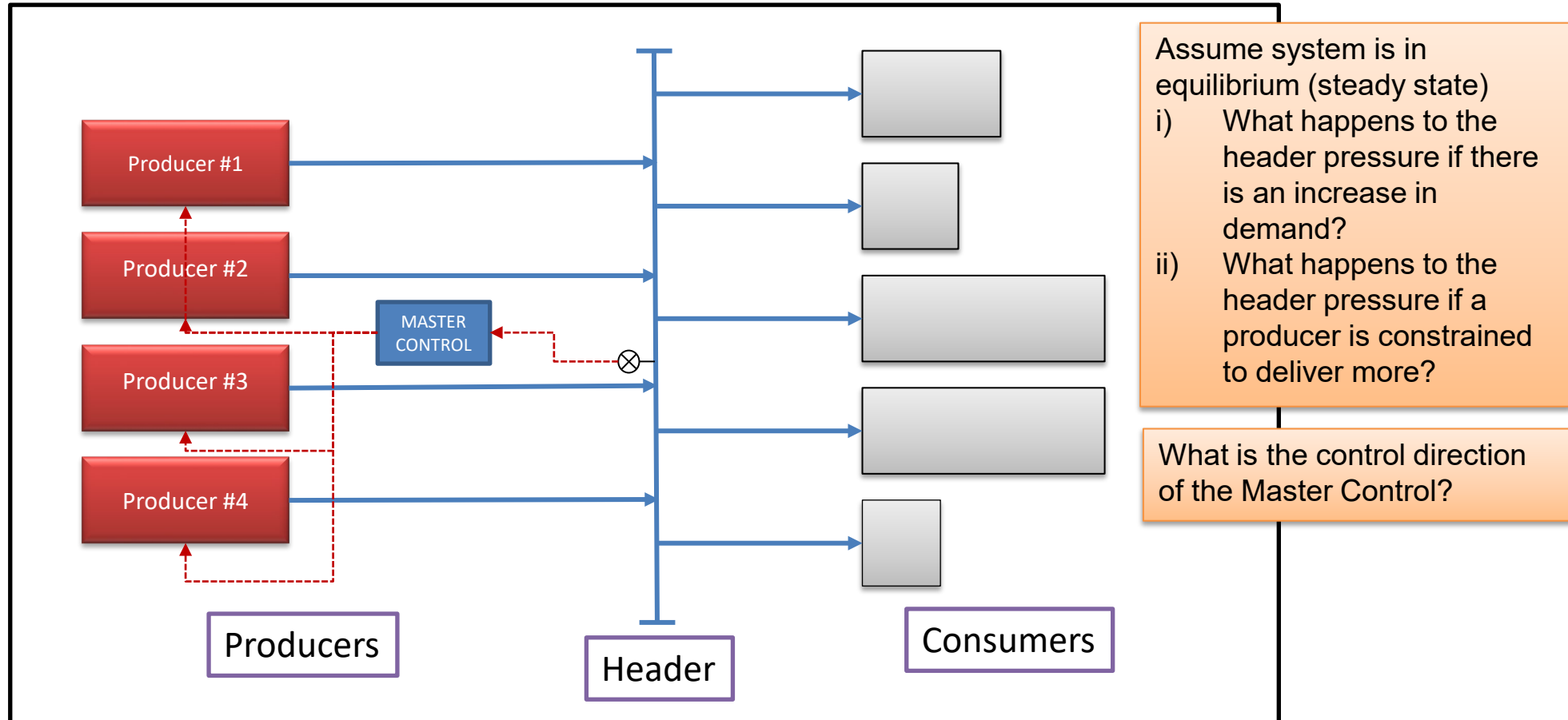
Option 1 for Supply-Demand Control



One option of Supply-Demand Control is to measure the flows to all the consumers and from all the producers and to utilize a Master Demand Controller to balance the flows

Typical Utility Systems

Option 2 for Supply-Demand Control



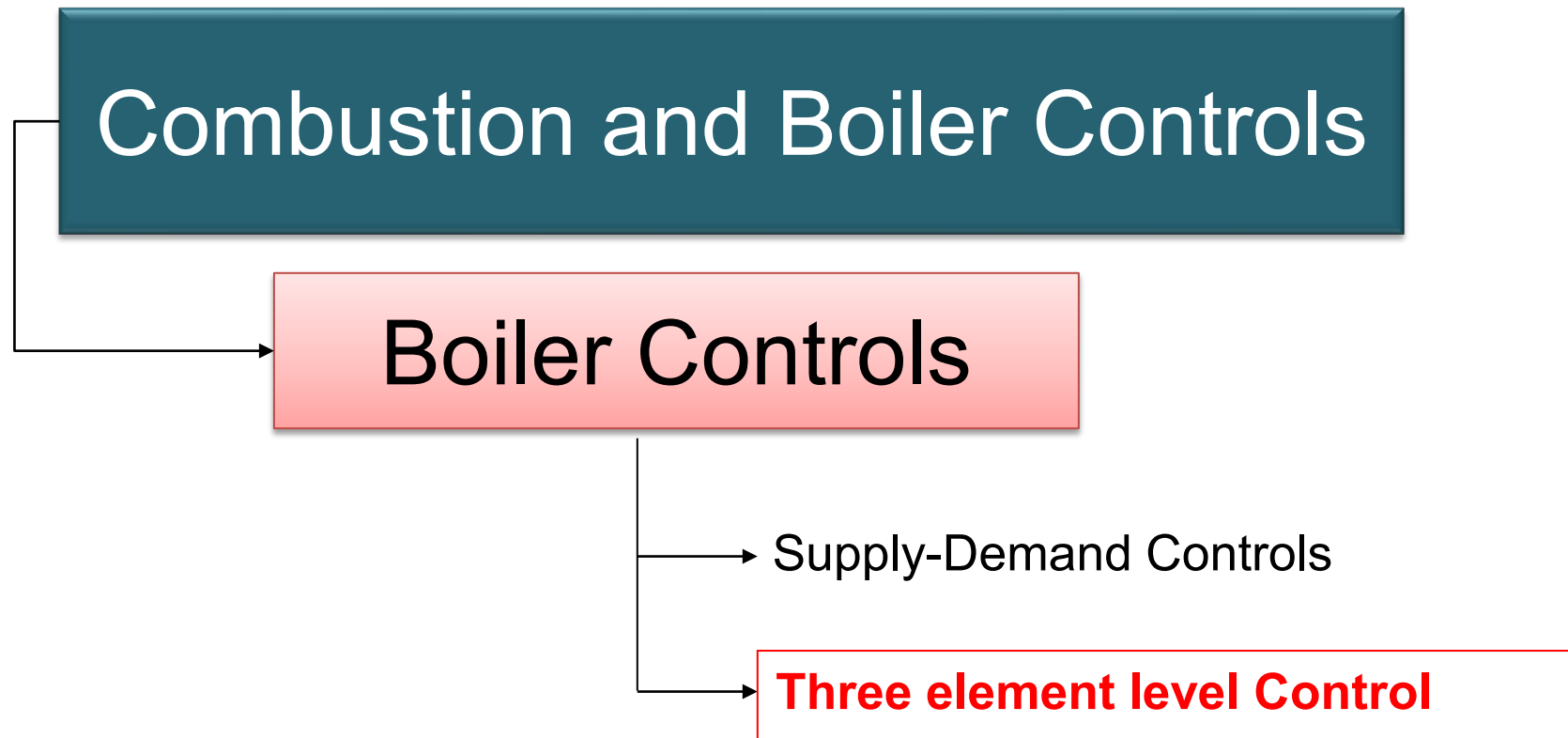
A second and simpler option is to utilize a pressure transmitter to detect changes in supply and demand. What are the concerns?

Issues

Primary issues affecting the demand controller:

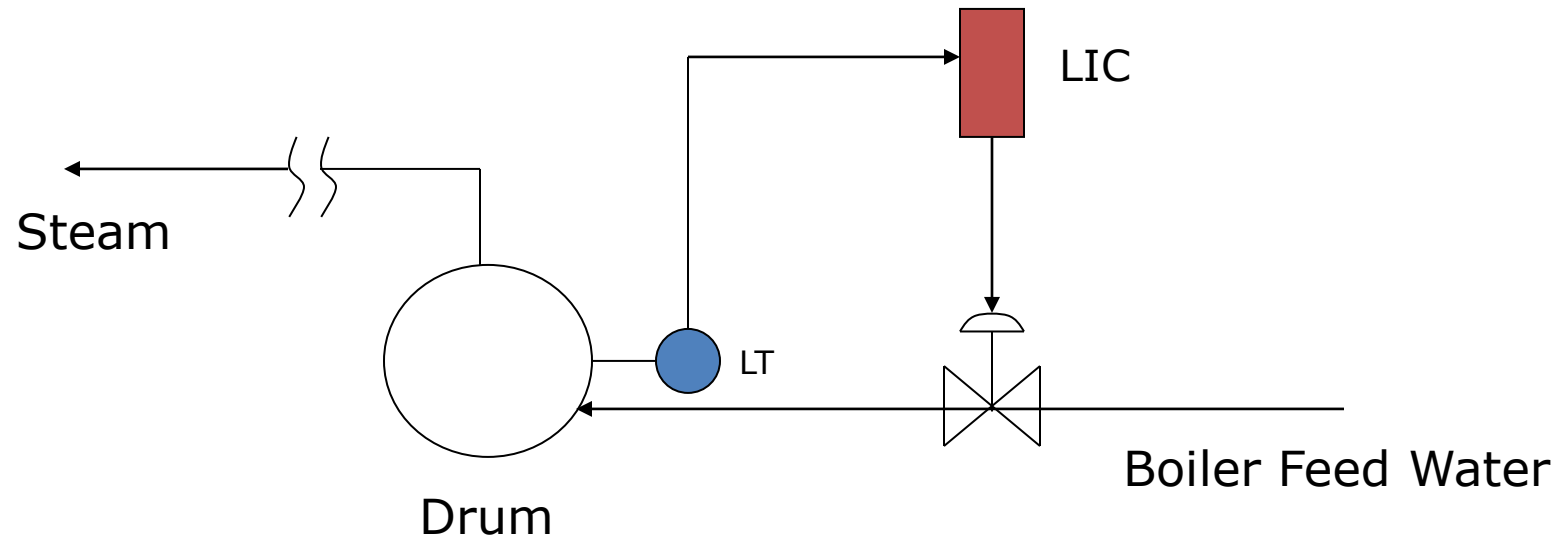
- Speed of response
 - size of header?
 - type of measurement?
- Reliability of measurement
 - Redundancy requirement

Advanced Regulatory Controls

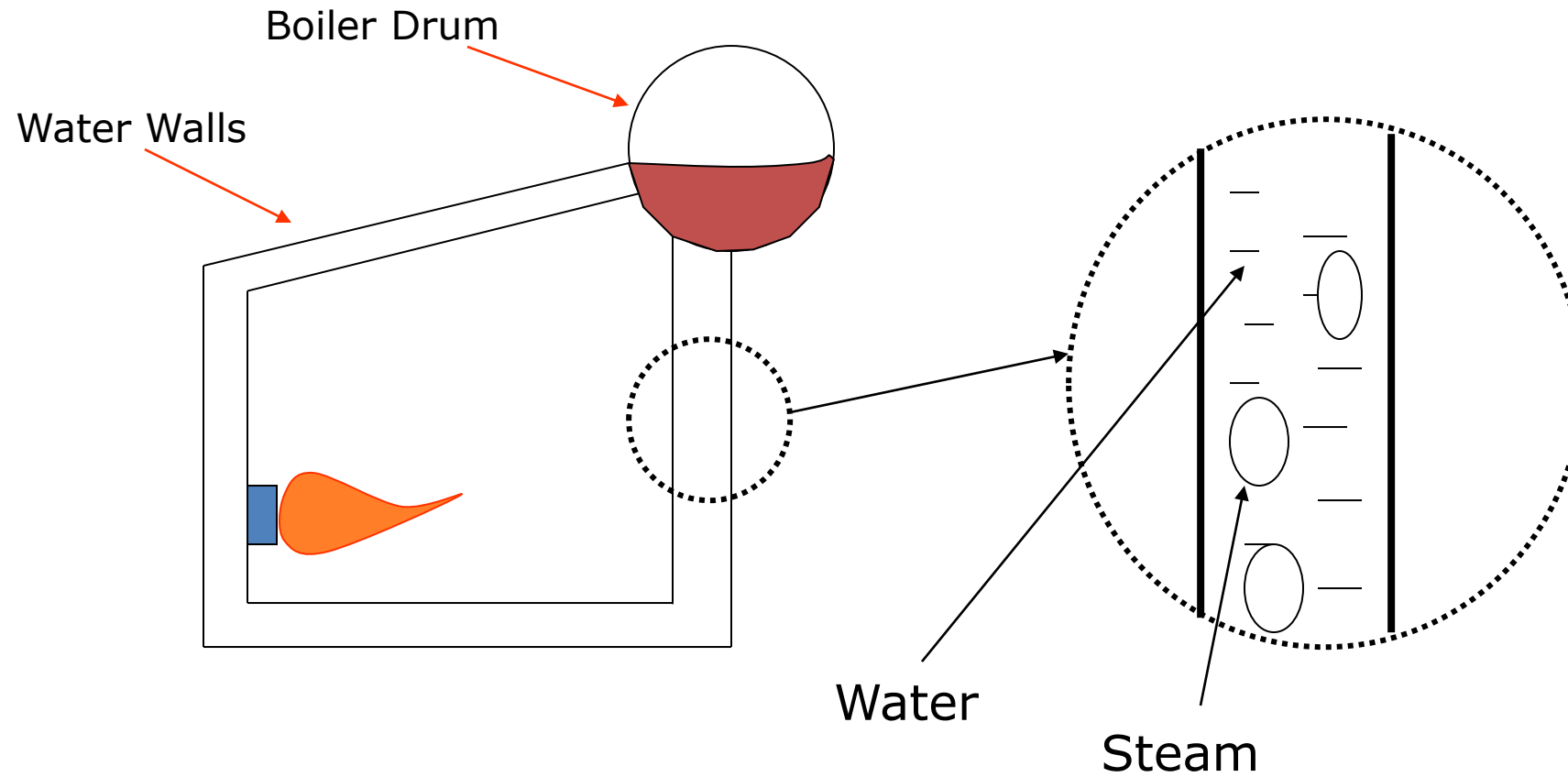


Single Element Control

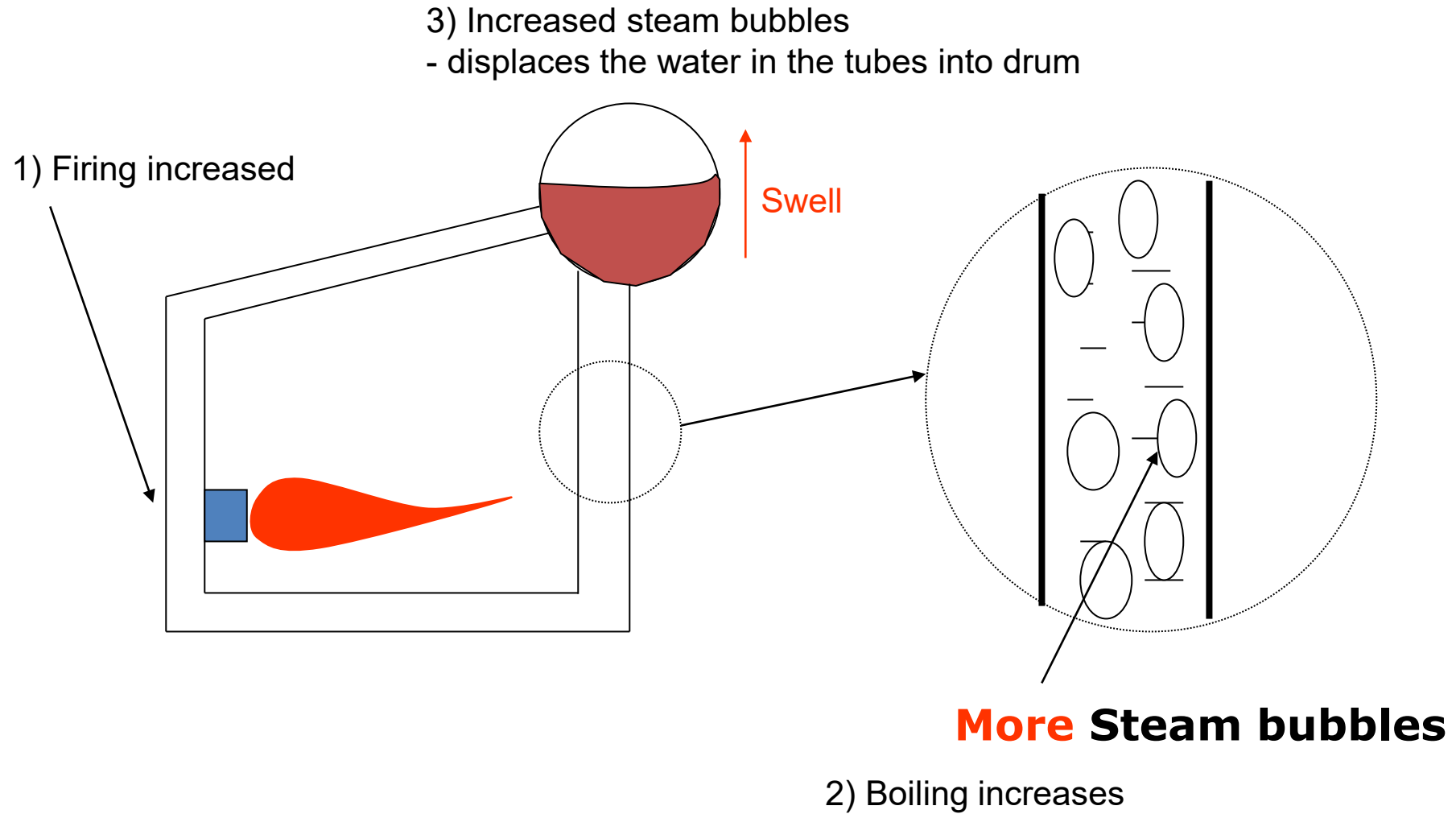
Single measuring 'Element' = Level



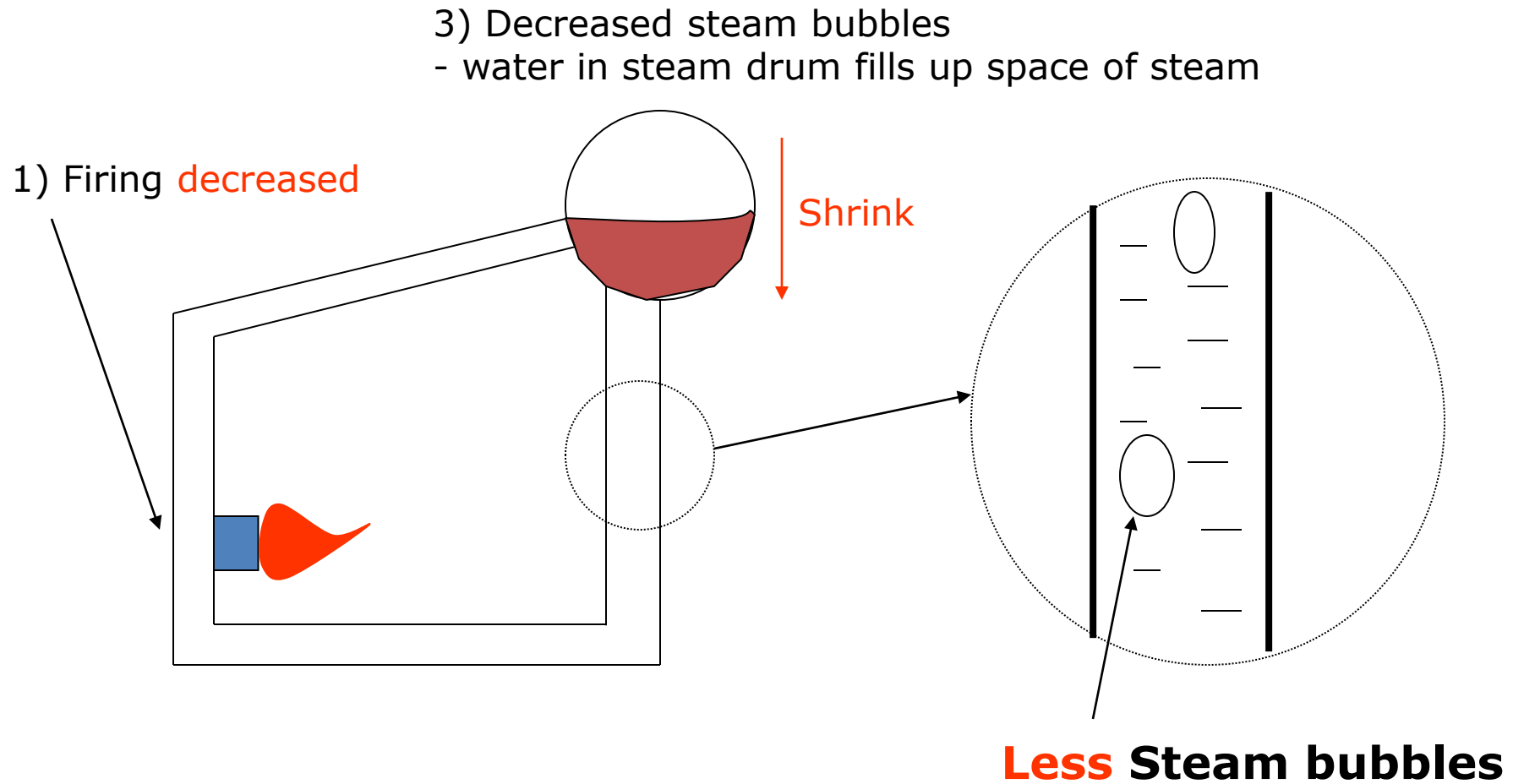
Condition of tubes within a water-tube boiler



Increased Firing & Swell



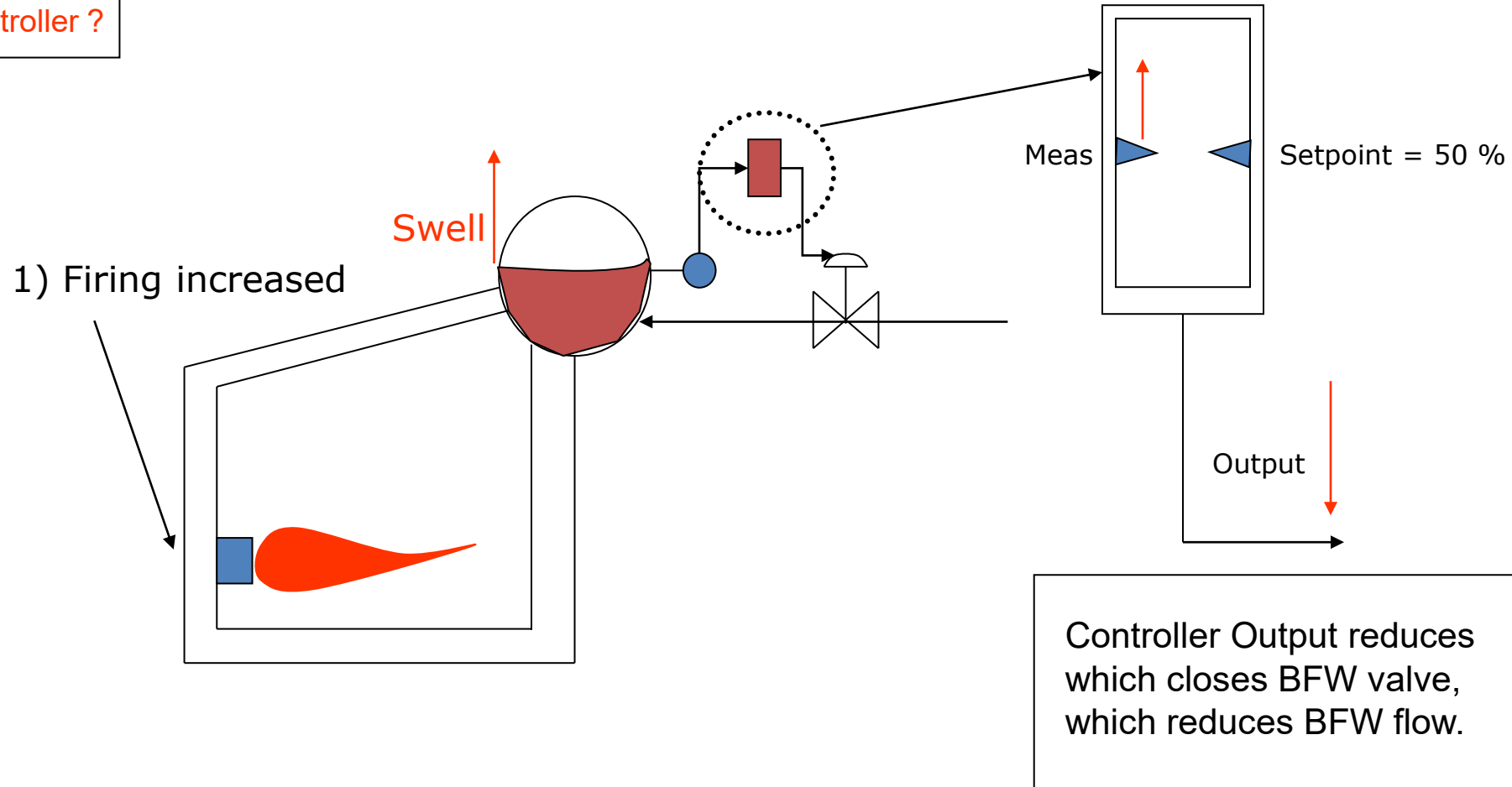
Decreased Firing & Shrink



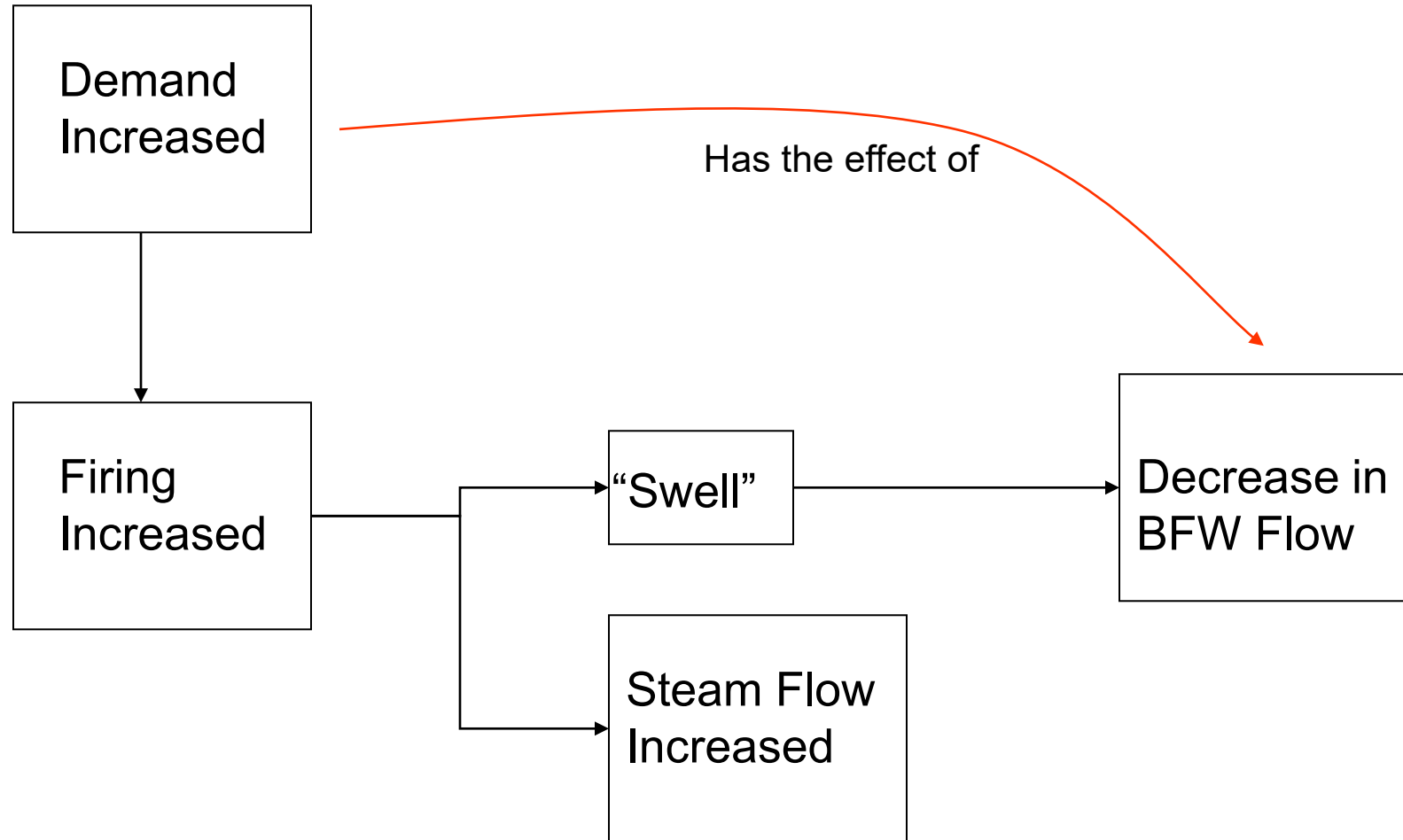
2) Boiling decreases

Increased Firing – effect on Level Controller

Action of Controller ?

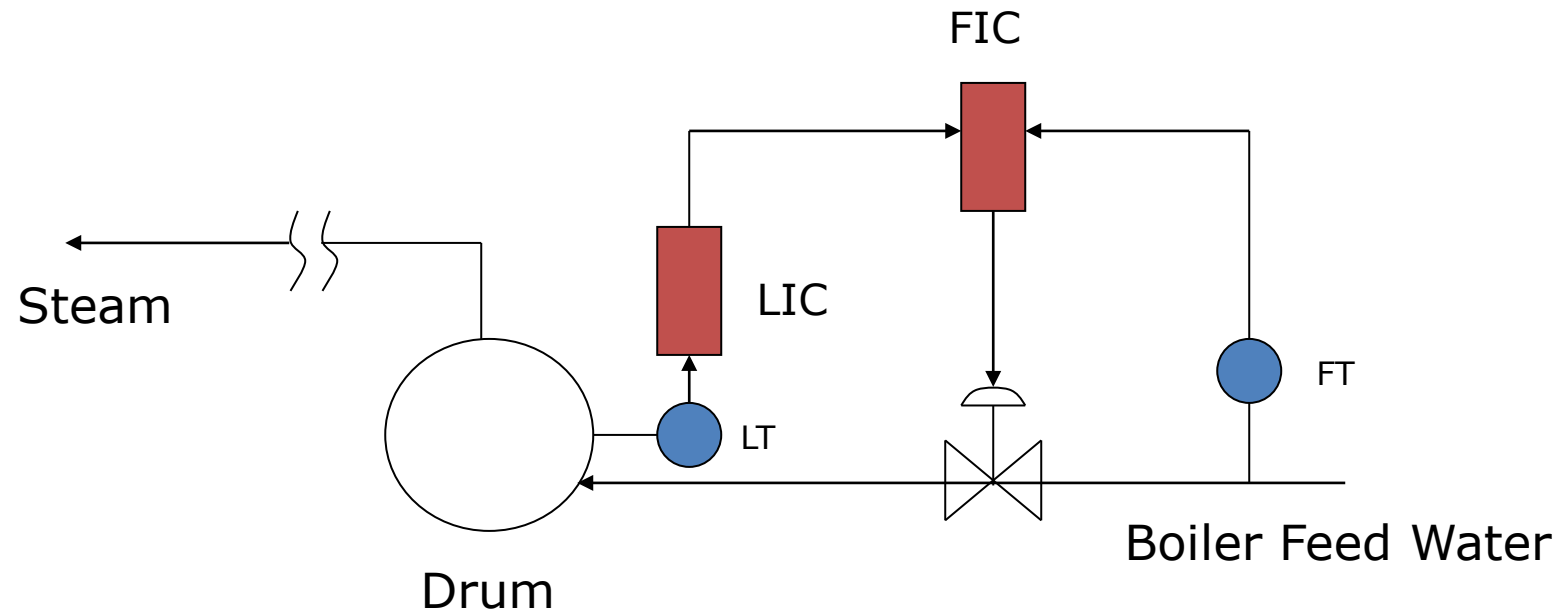


Primary Issue



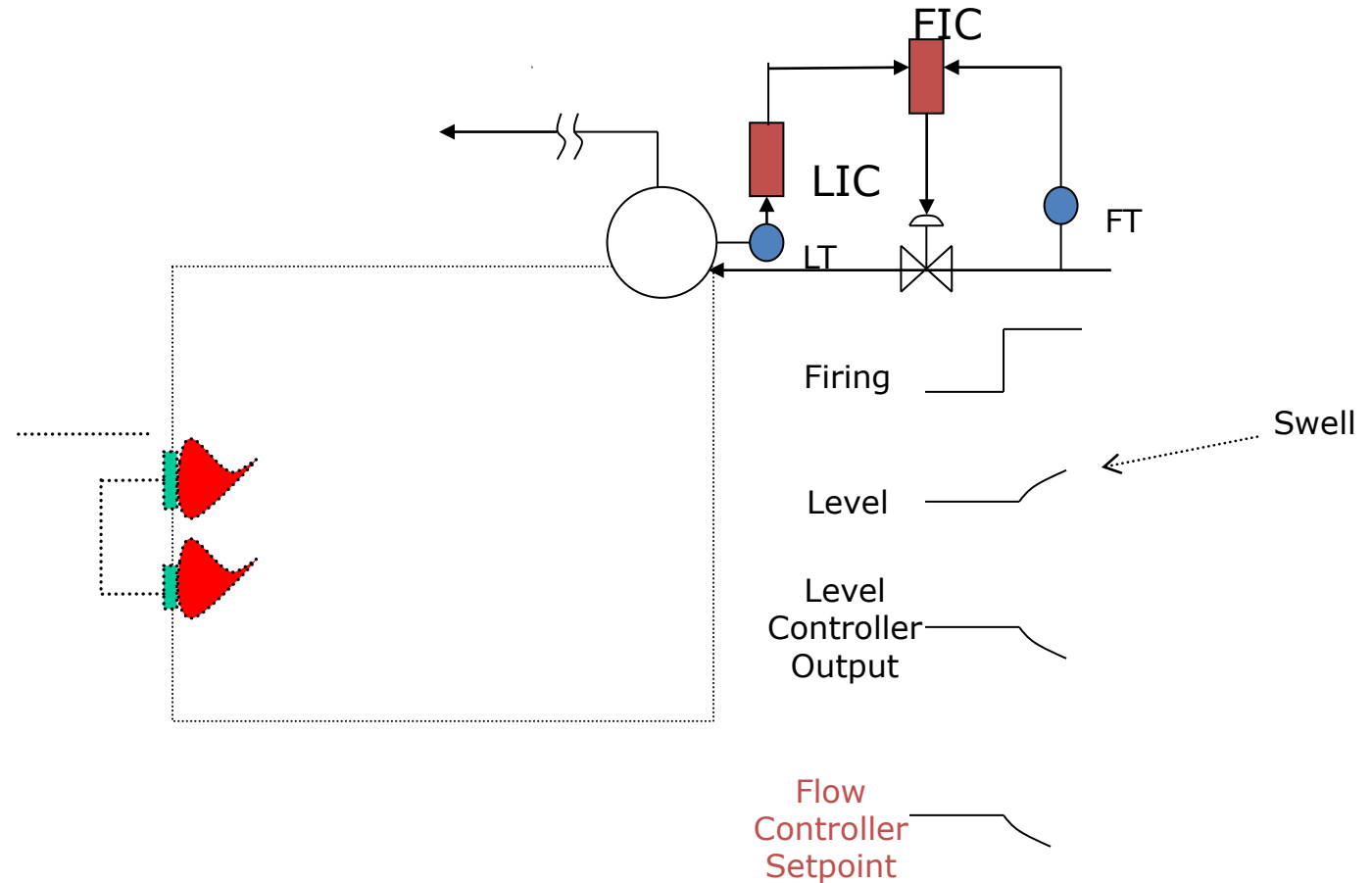
Two Element Control

Two measuring 'Elements' = Level & Flow



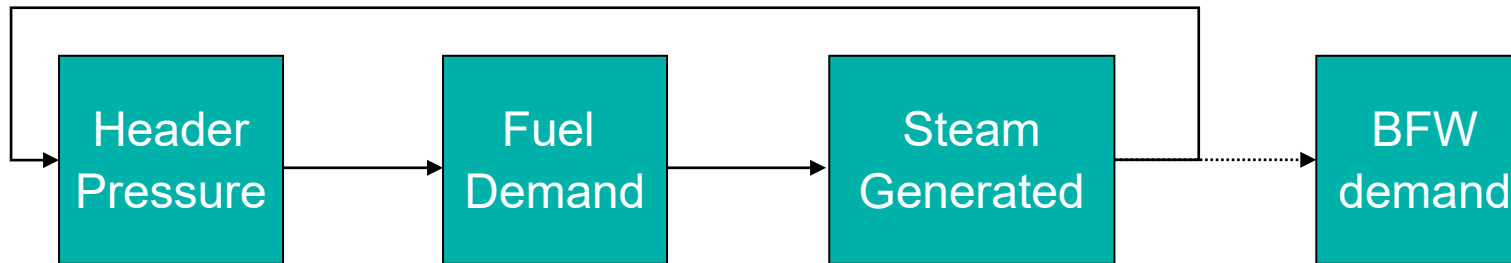
Basic Control with Two Element Loop

- Similar to single element control
- Cannot handle shrink and swell because the MASTER is still the level controller



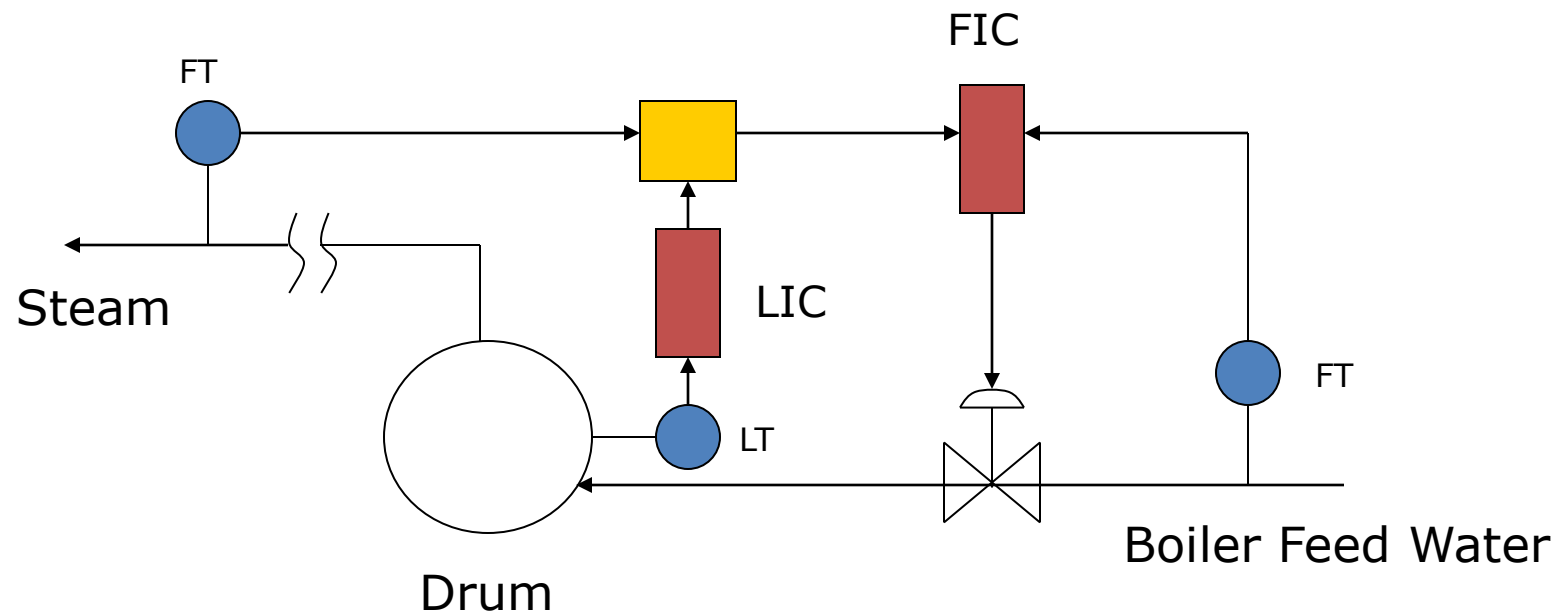
Level control relationship with steam controls

- The steam header pressure controls determines the fuel demand of the boiler.
- This in turn determines the amount of steam generated.
- The amount of steam generated can then determine the amount of BFW utilized.



Standard 3 Element Control

Three measuring 'Elements' = Level & BFW Flow & Steam Flow

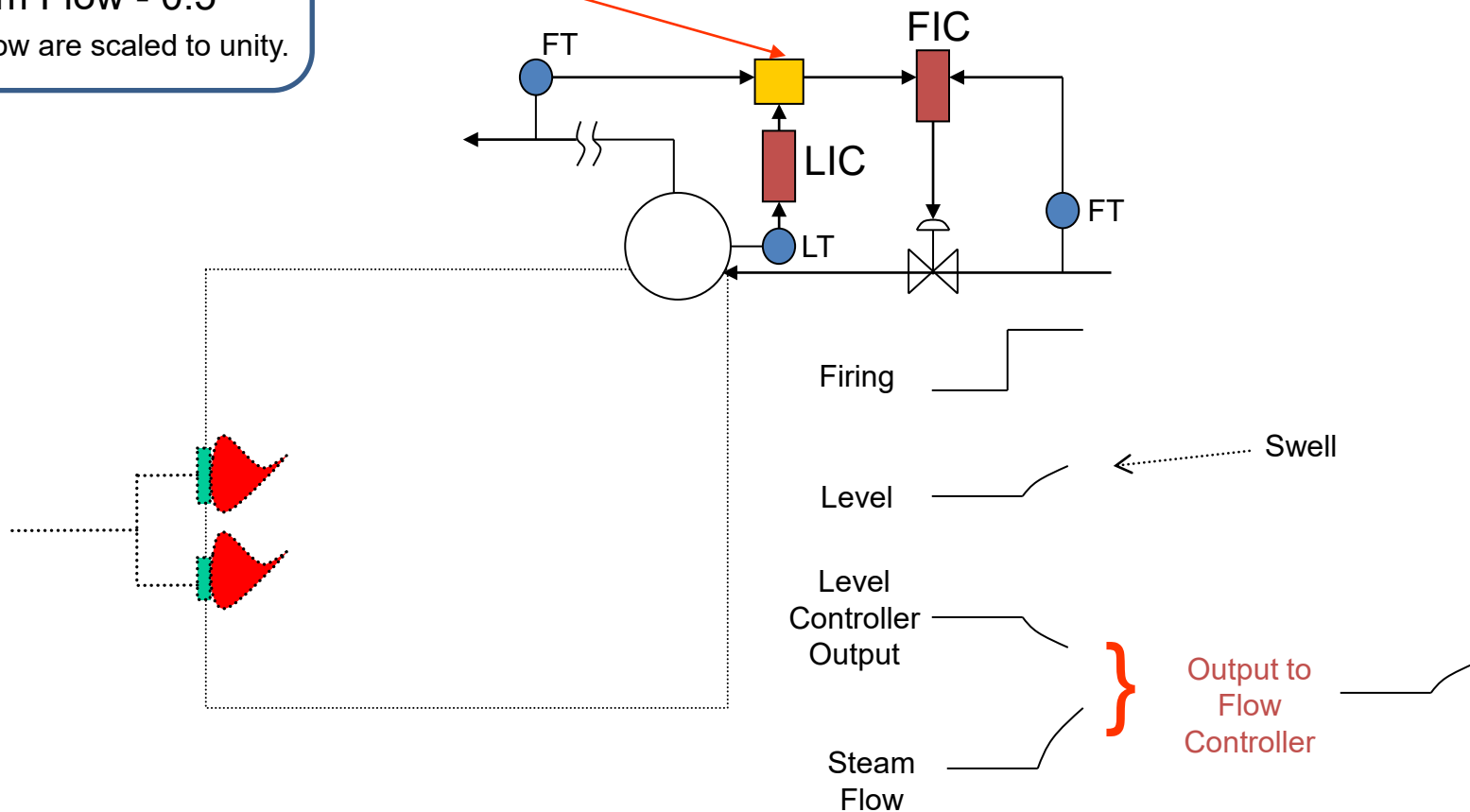


Basic Control with Three Element Loop

Equation

FIC Setpoint = LIC output + Steam Flow - 0.5
Where the LIC Output and the Steam Flow are scaled to unity.

- Can handle shrink and swell as the steam flow counteracts the level controller's intention to go in the wrong direction.



Implication of the equation

- Equation utilized is
 - $\text{FIC Setpoint} = \text{LIC output} + \text{Steam Flow} - 0.5$
- IF Level in the boiler drum is stable and controller is (ideally) giving an output of 50 % (or 0.5)
 - $\text{FIC setpoint} = \text{LIC output} + \text{Steam flow} - 0.5$
 - $\text{FIC setpoint} = 0.5 + \text{Steam flow} - 0.5$
 - $\text{FIC setpoint} = \text{Steam flow}$
- The impact of the equation is that **BFW flow is controlled primarily by steam flow** and 'trimmed' by the level controller.

Retrieval Practice

1. The issue with boiler drum levels stem from a particular phenomena that occurs in the boiler drum. One of the following best describes the phenomena:
 - a) When the firing increases, the drum level drops which then increases the feedwater flow
 - b) When the firing increases, the drum level increases, which then increases the feedwater flow
 - c) When the firing increases, the drum level drops which then decreases the feedwater flow
 - d) When the firing increases, the drum level increases, which then decreases the feedwater flow

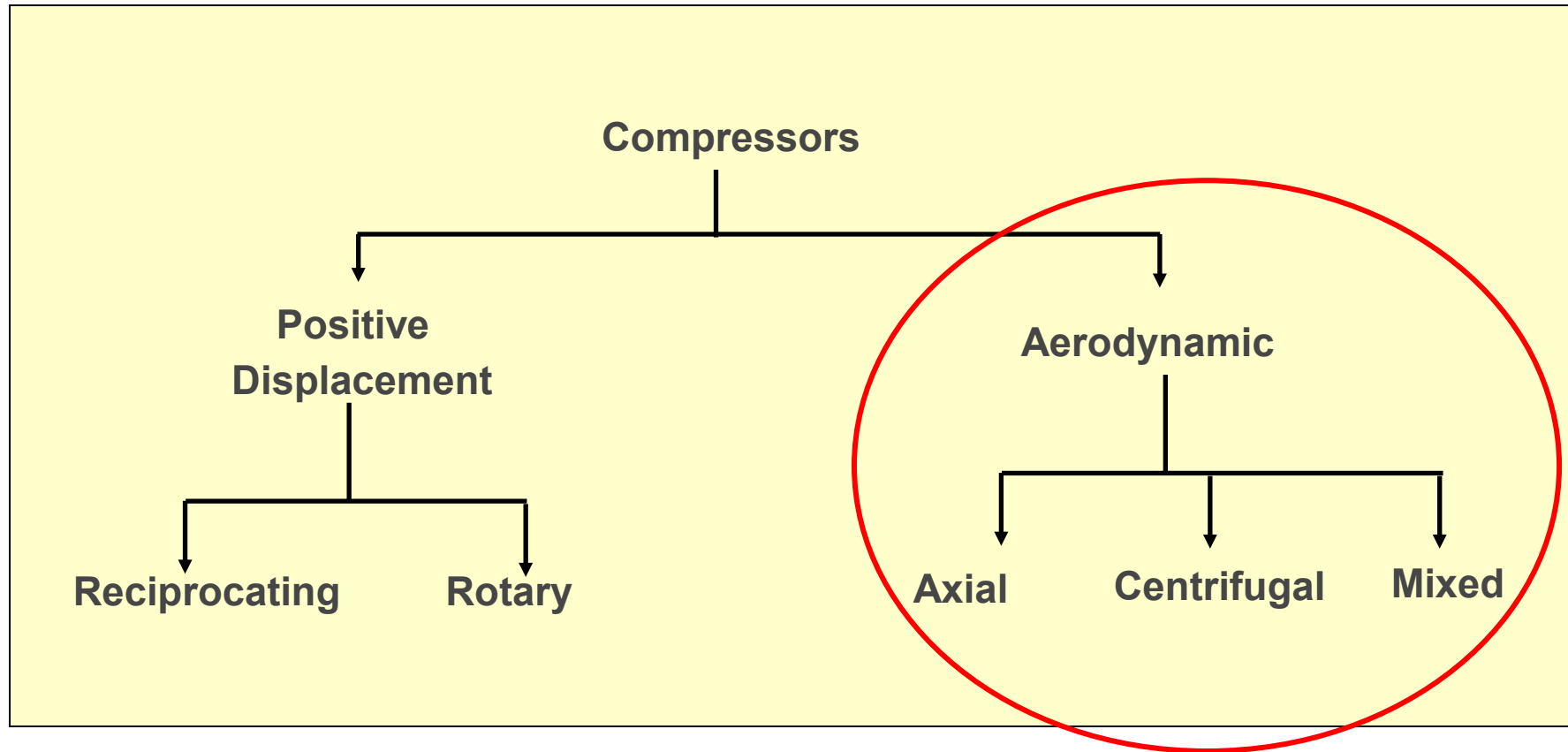
2. The solution for the issue at the boiler drum is to utilize a parameter that is indicative of an increase in demand. What parameter is this:
 - a) The drum level
 - b) The steam flow
 - c) The boiler feedwater flow
 - d) The feedwater temperature

Advanced Regulatory Controls



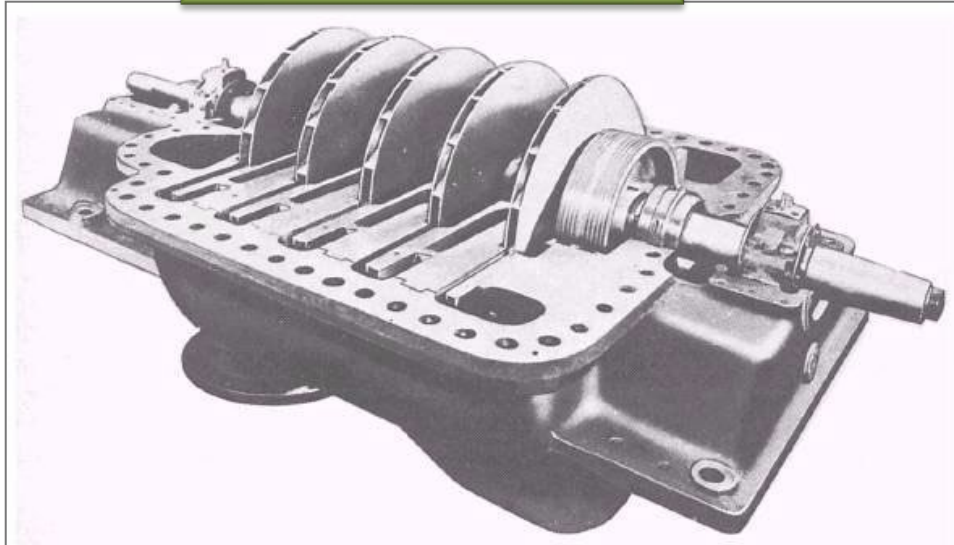
Compressor Antisurge Control

Types of Compressors



Aerodynamic Compressor Types

Centrifugal Compressor



Axial Compressor



Closed Impeller



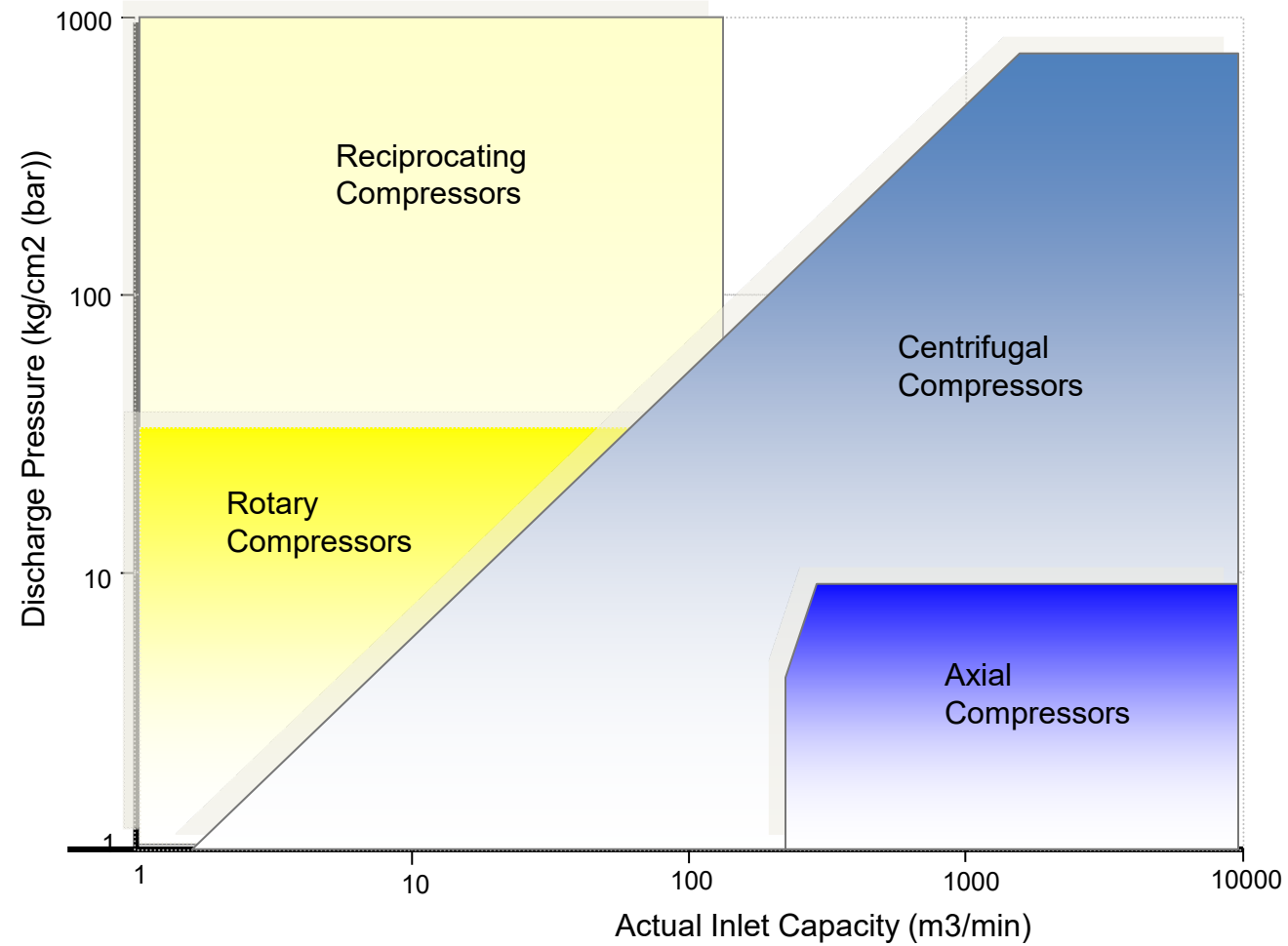
Open Impeller



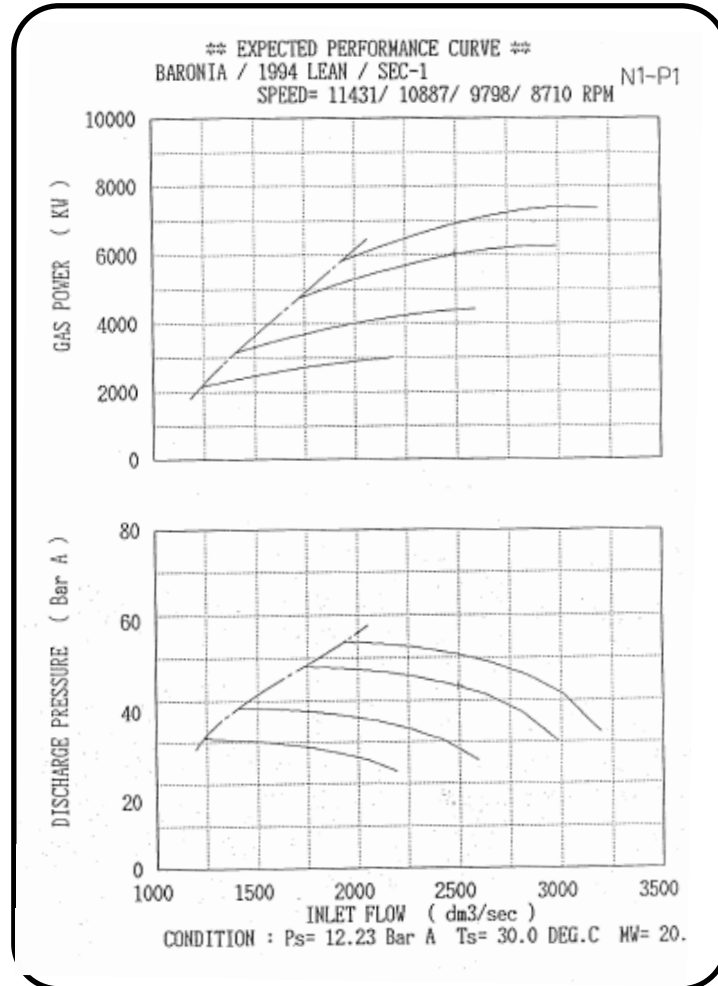
Axial Blades



Application range of compressors



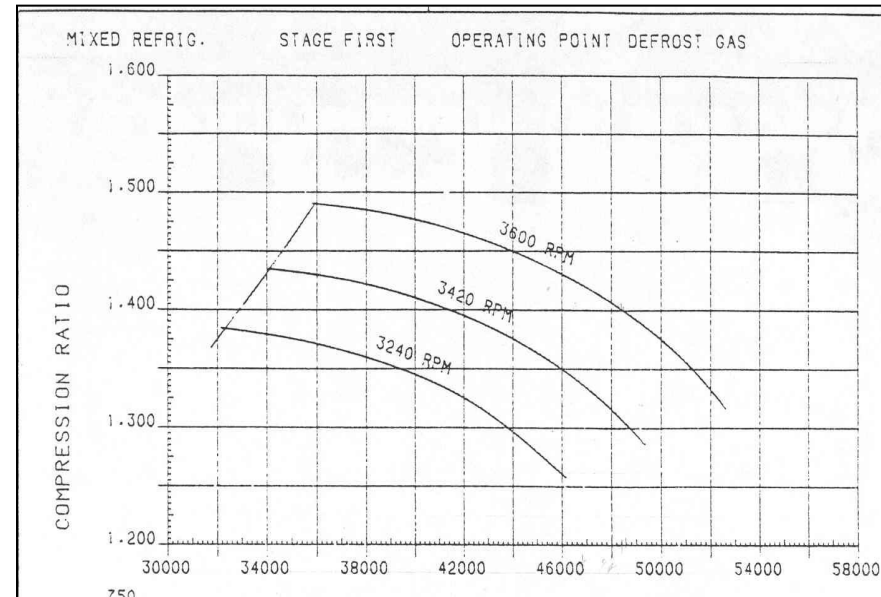
Sample Compressor Maps



5.2.4 Curves

API617

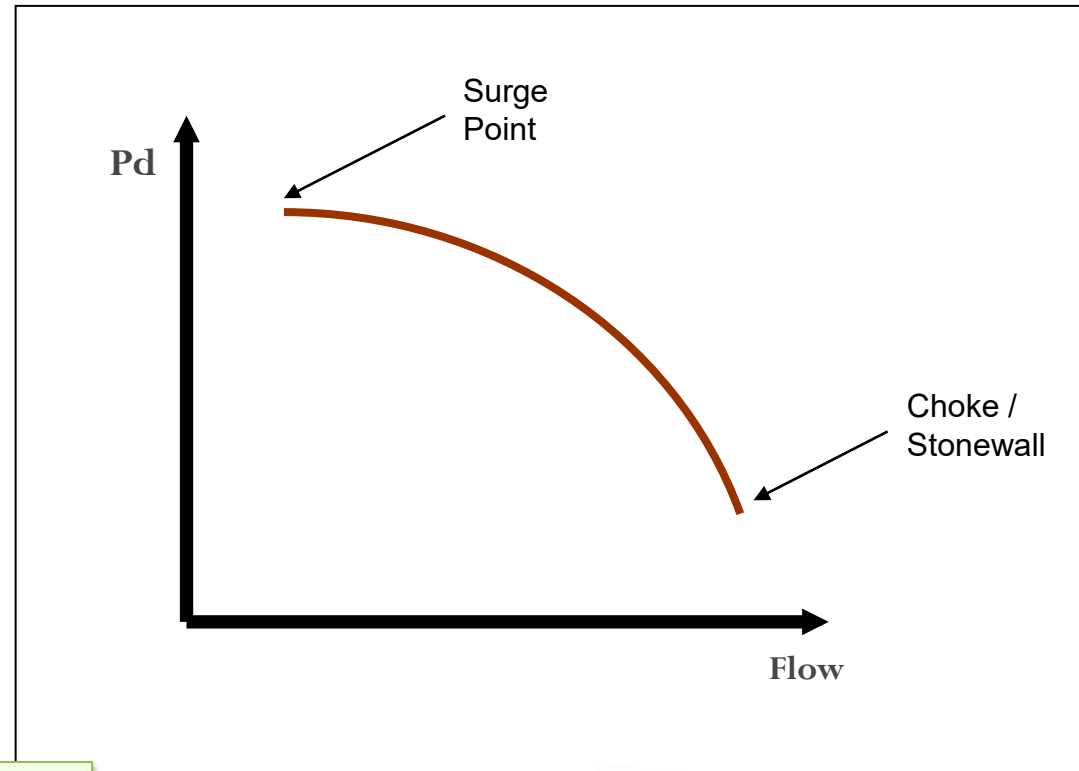
Performance curves shall be submitted for each section (between purchaser's process nozzles) of each casing as well as an overall curve for the train. These curves shall encompass the map of operations, with any limitations indicated thereon. All curves shall be marked "PREDICTED."



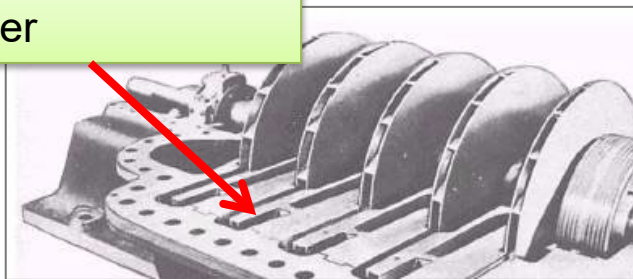
Compressor B

Type : Centrifugal, Variable Speed
 Service : Light Hydrocarbon + N₂

Understanding the Compressor Map

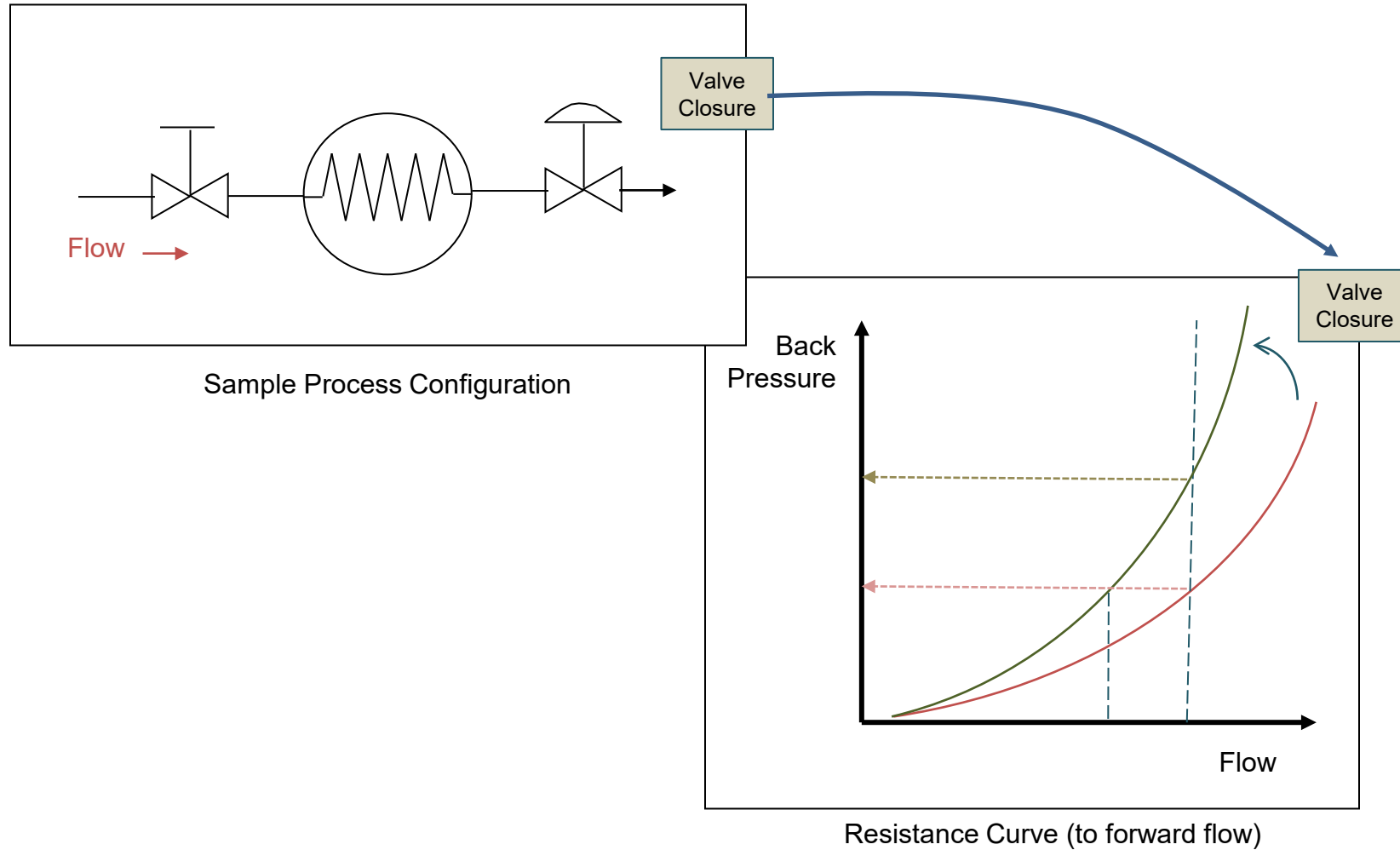


Surge occurs in the diffuser

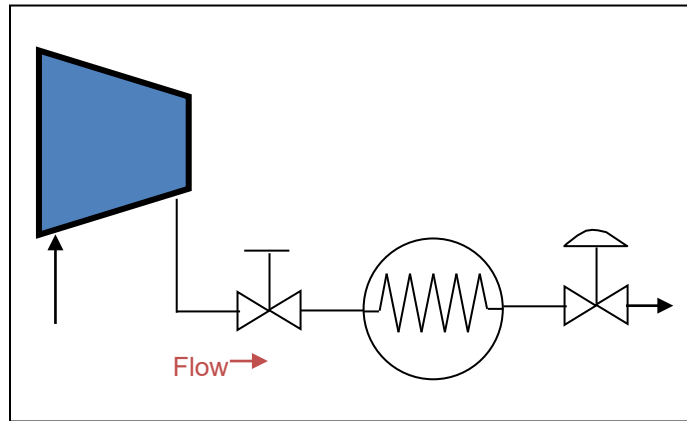


Choke occurs at the impeller

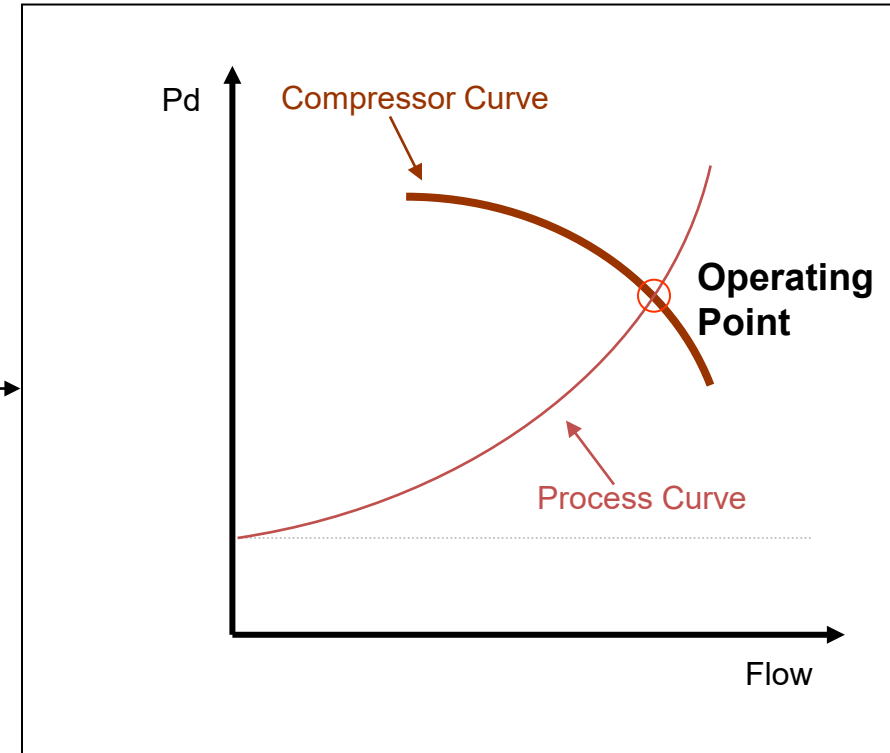
Understanding the Process Resistance Curve



Operating Point

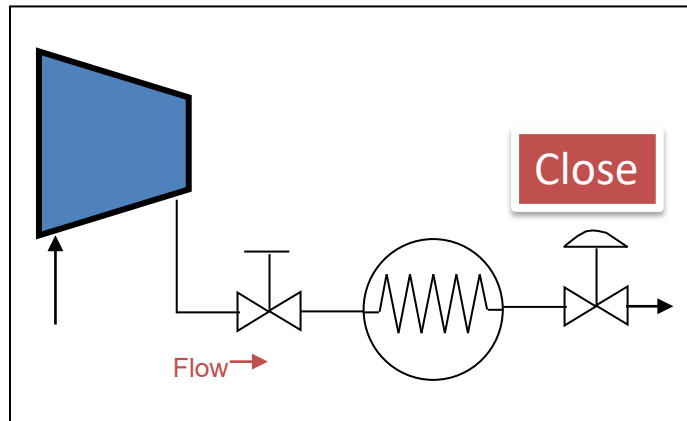


Typical Compression System

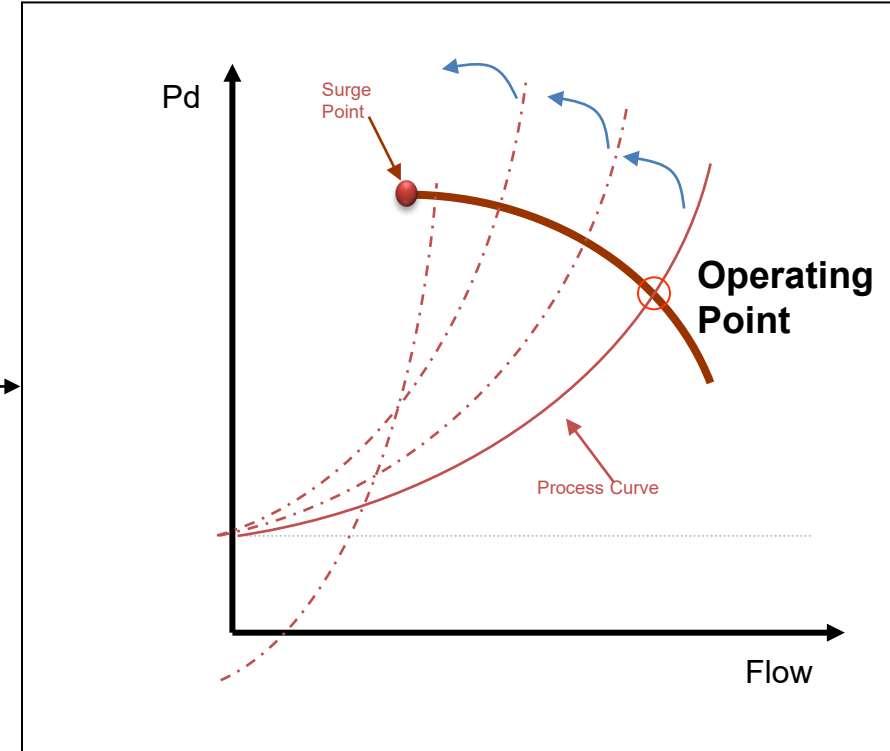


The intersection of the Process curve with the Compressor curve is the Operating Point.

Impact of closing a valve



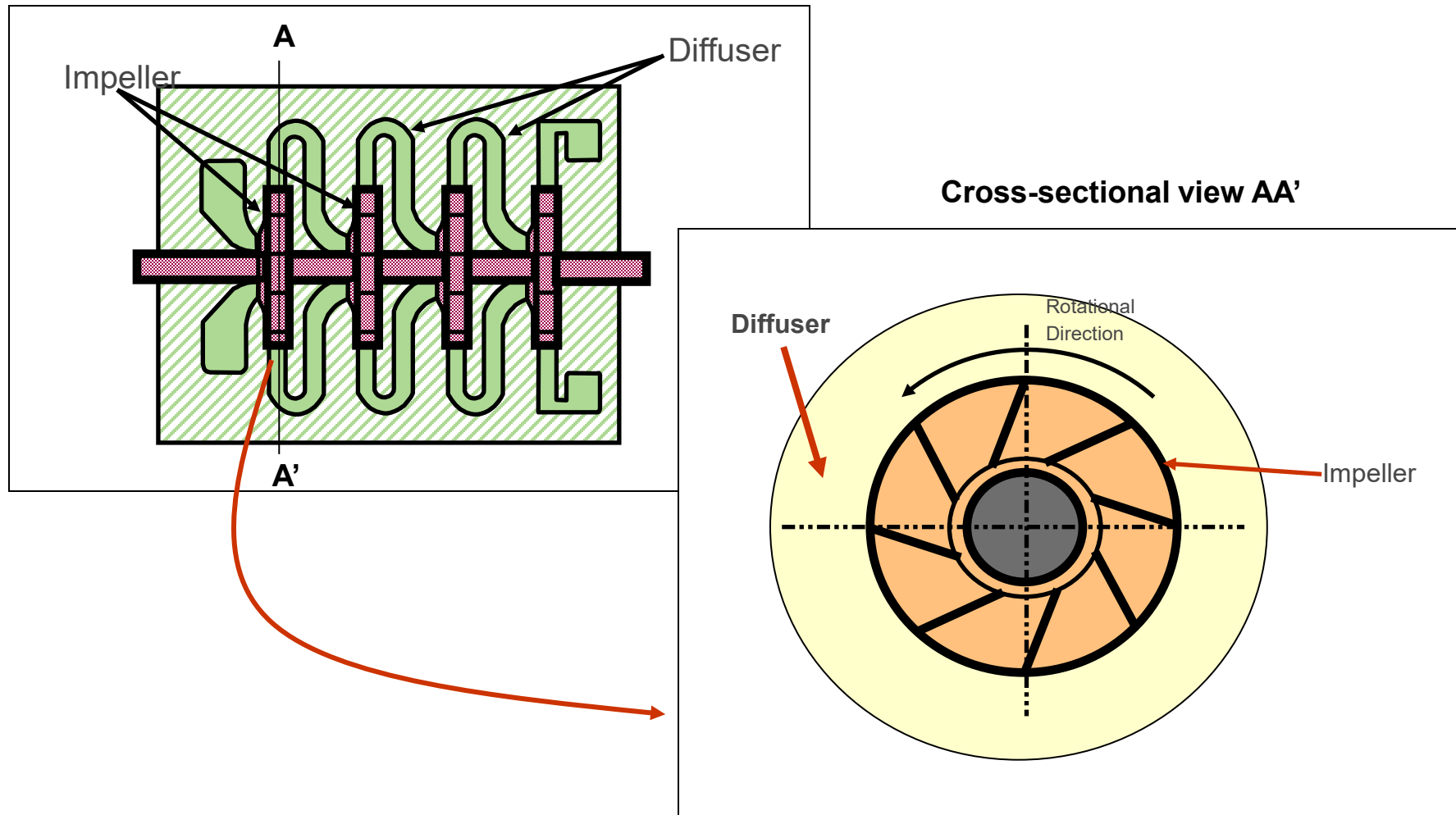
Typical Compression System



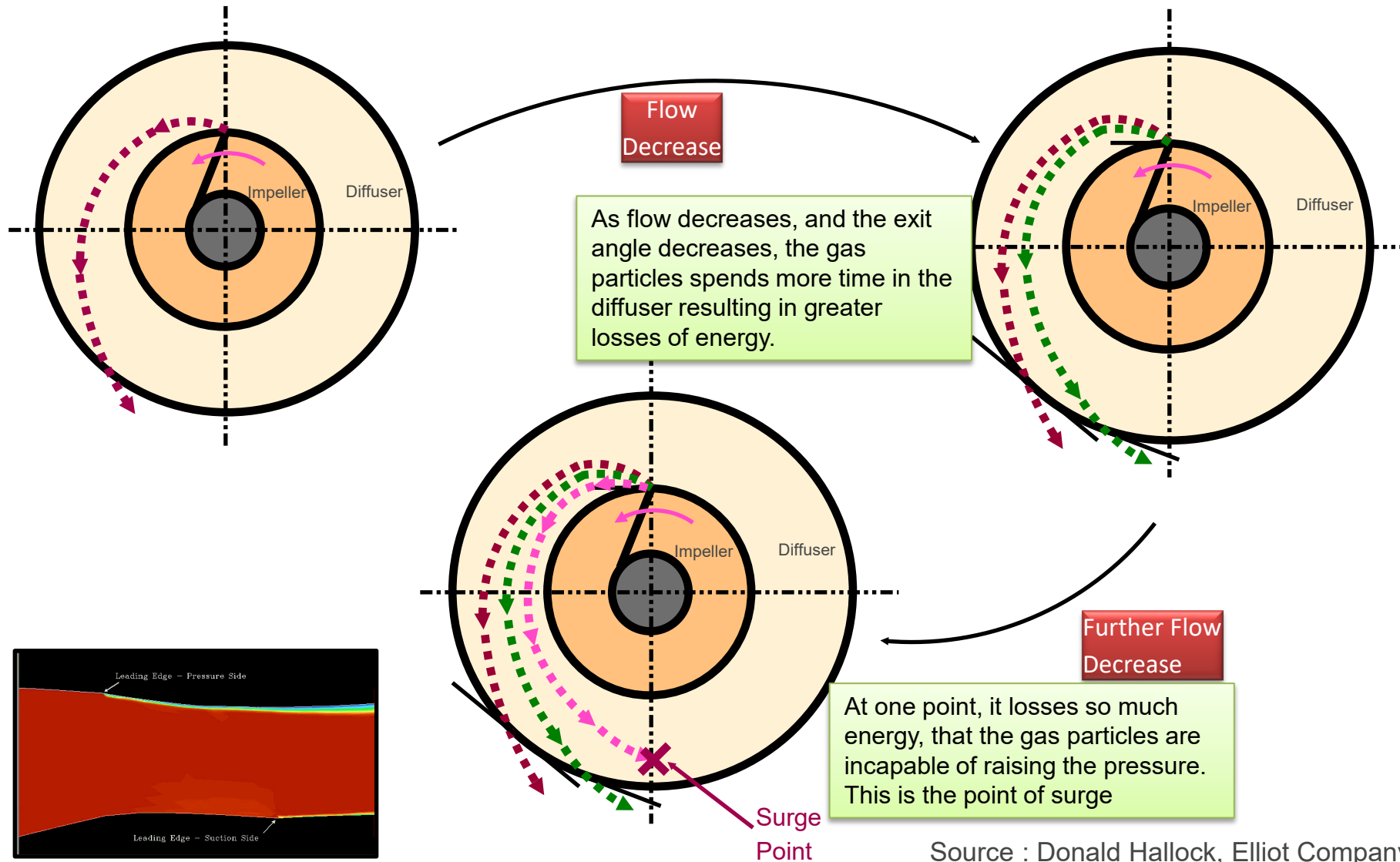
Closure of a valve will cause higher backpressure causing the Process Resistance curve to move to the left.

Surge Location

Centrifugal machine cutaway



The Point of Surge



Process Control Theory and Applications
PETRONAS Skill Group Training

Source : Donald Hallock, Elliot Company

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SKG16.1 Operations Management

Surge Definition

- The Surge Point of an aerodynamic compressor is the point at which the pressure rise capability of a compressor fails and flow reverses through the machine.

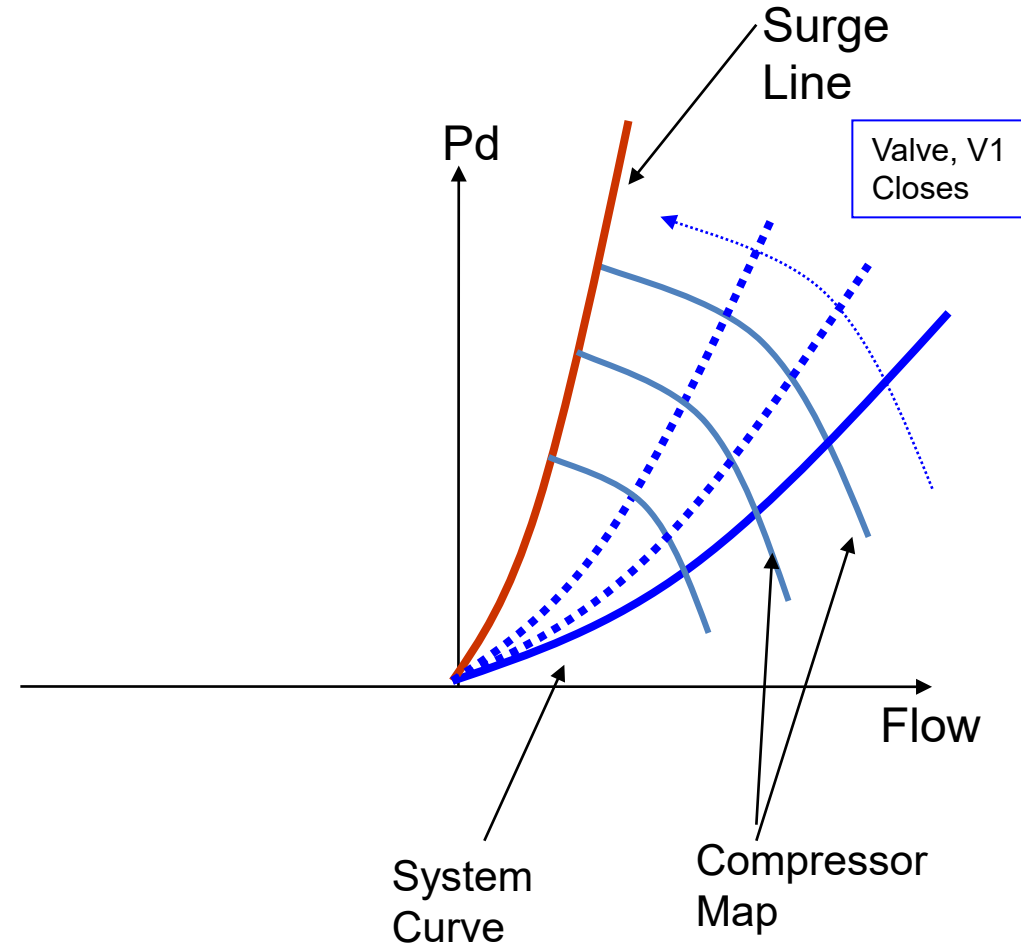
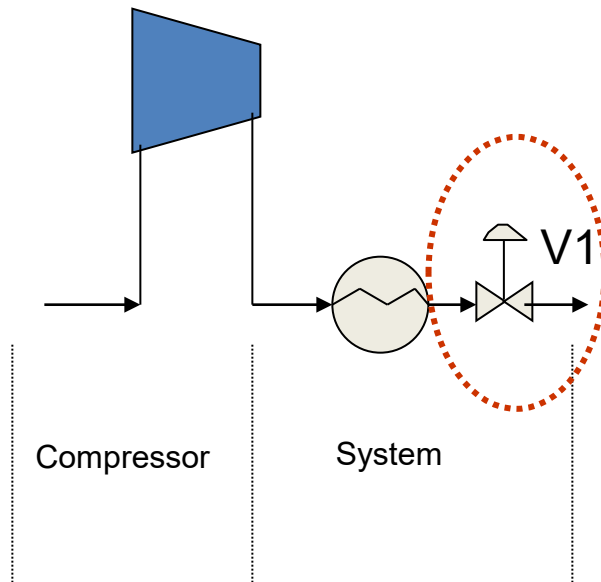
Characteristics

- Rapid flow reversals
- Rapid surge cycles
- Axial and vibration movements on the machine rotor
- Temperature rises

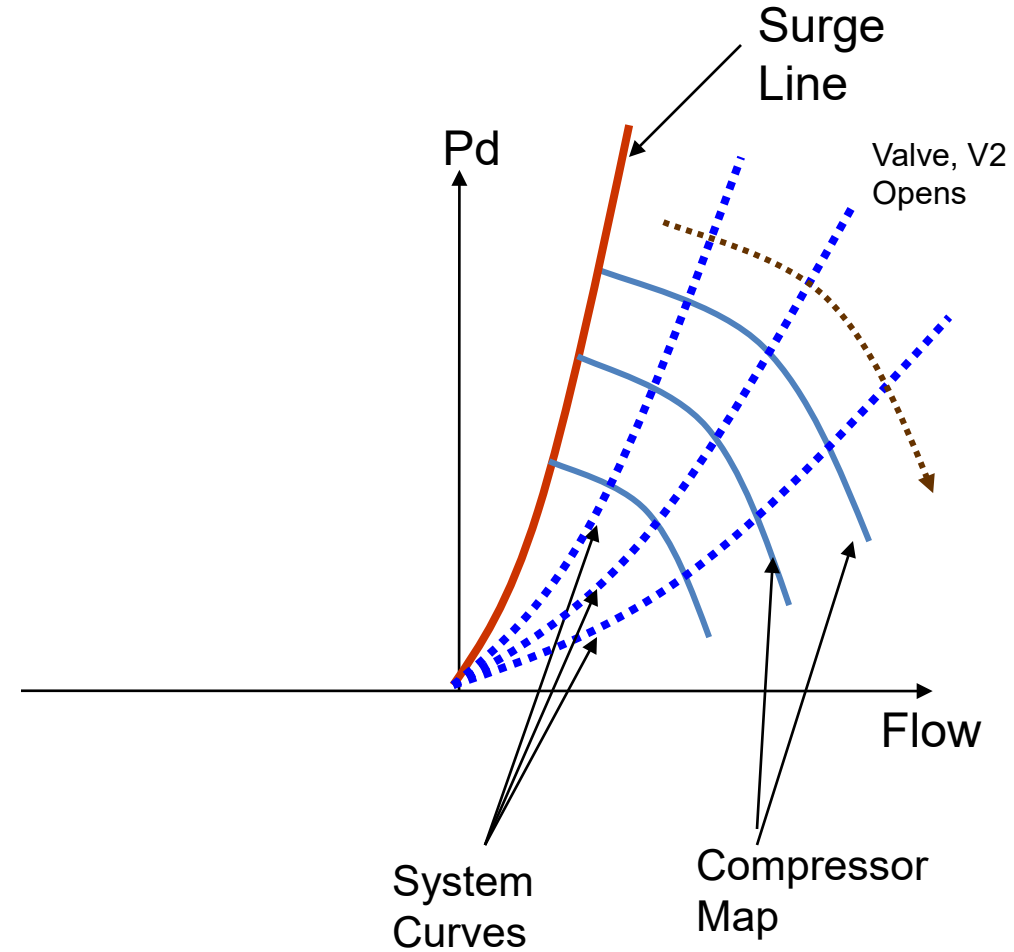
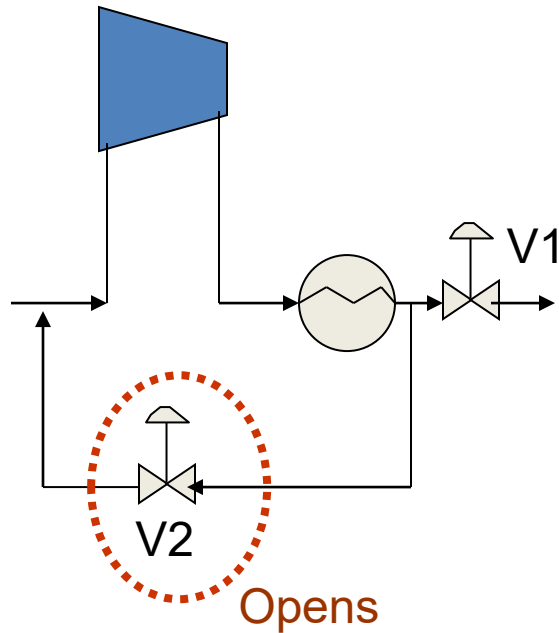
Impact

- Incremental damage to rotor and casing components
- Increased seal clearances
- Lower efficiency
- Prolonged exposure can lead to catastrophic failure

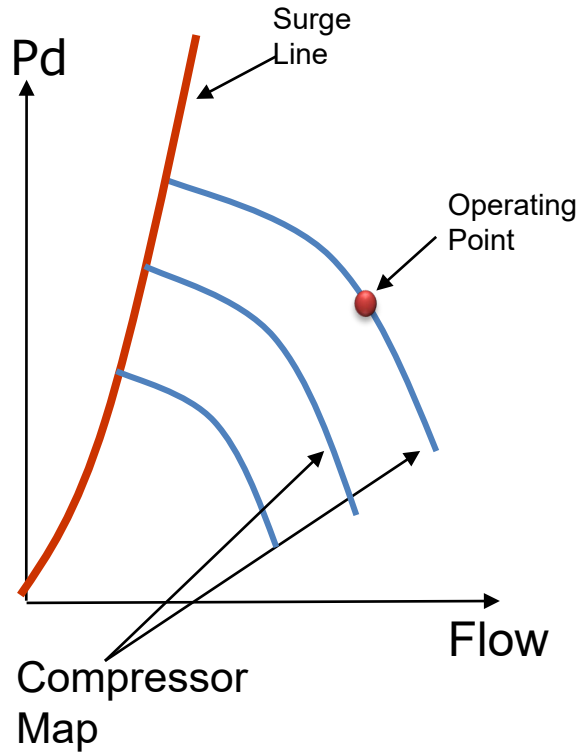
How to move away from Surge...1



How to move away from Surge...2



Measurement of Surge

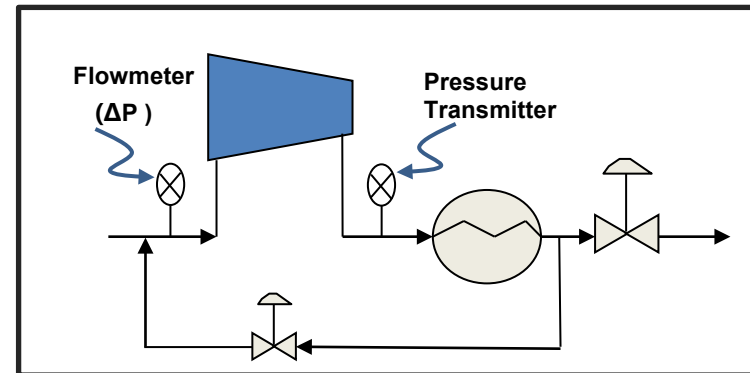


To prevent the machine entering surge, it is important to know where the operating point is

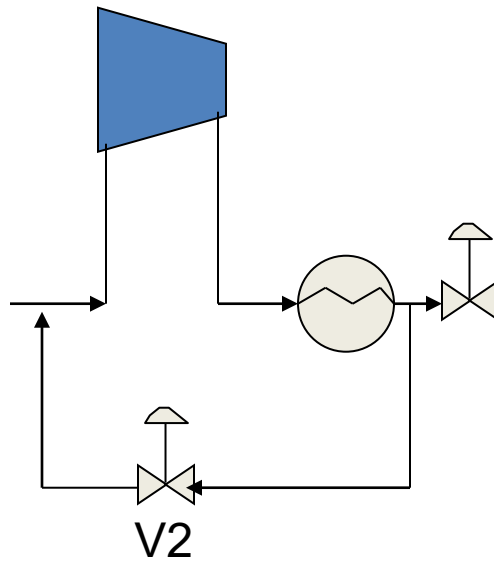


$$S = K \frac{\Delta P}{P_d}$$

Most common form of the surge equation

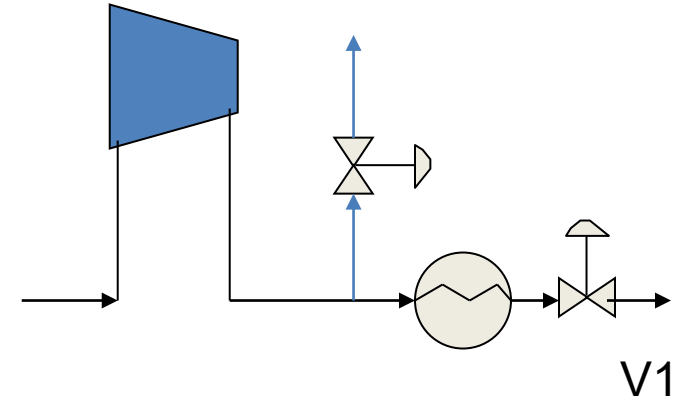


Most Common Handles to move away from Surge



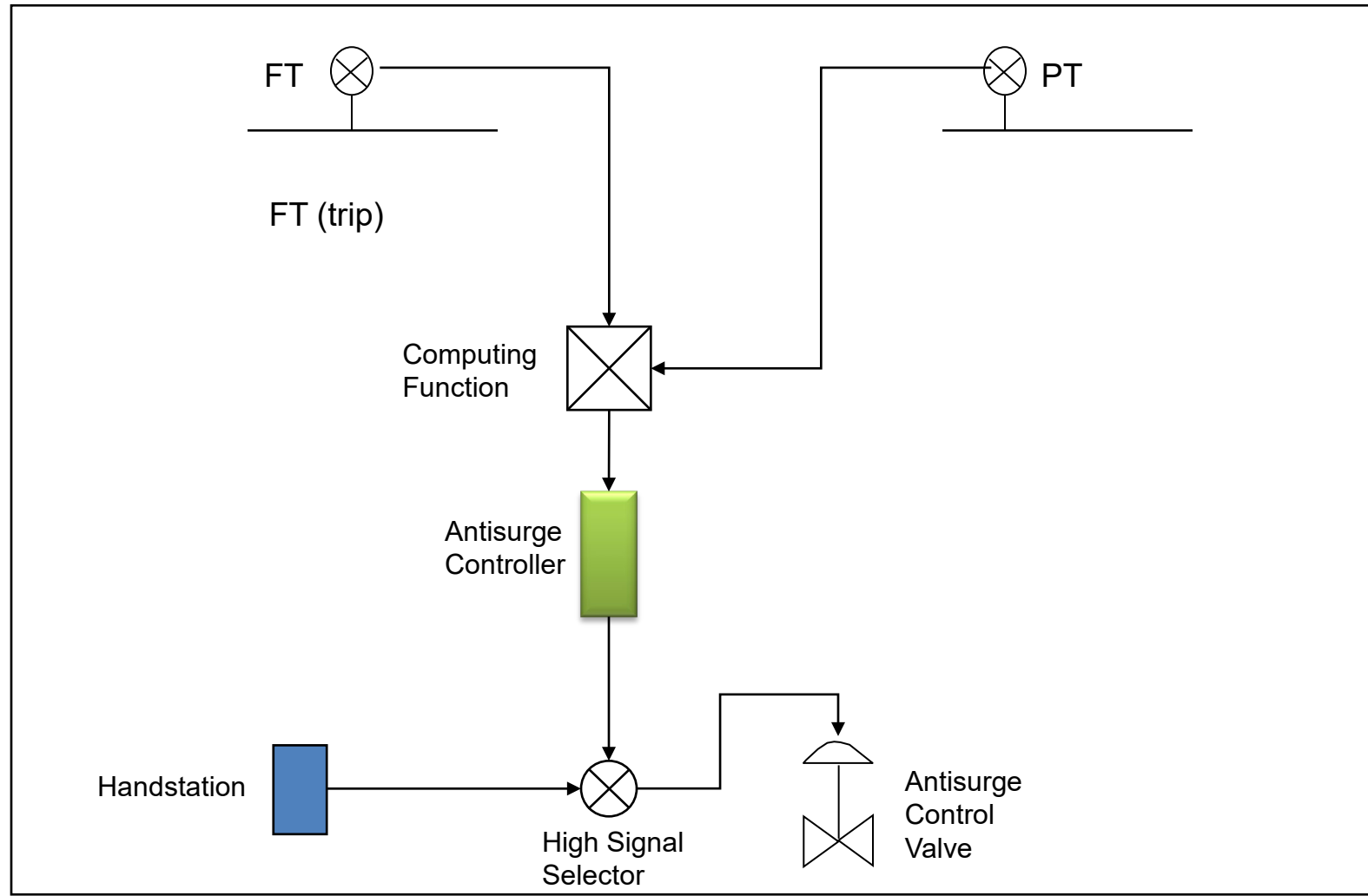
Antisurge
Recycle Valve

V1
→

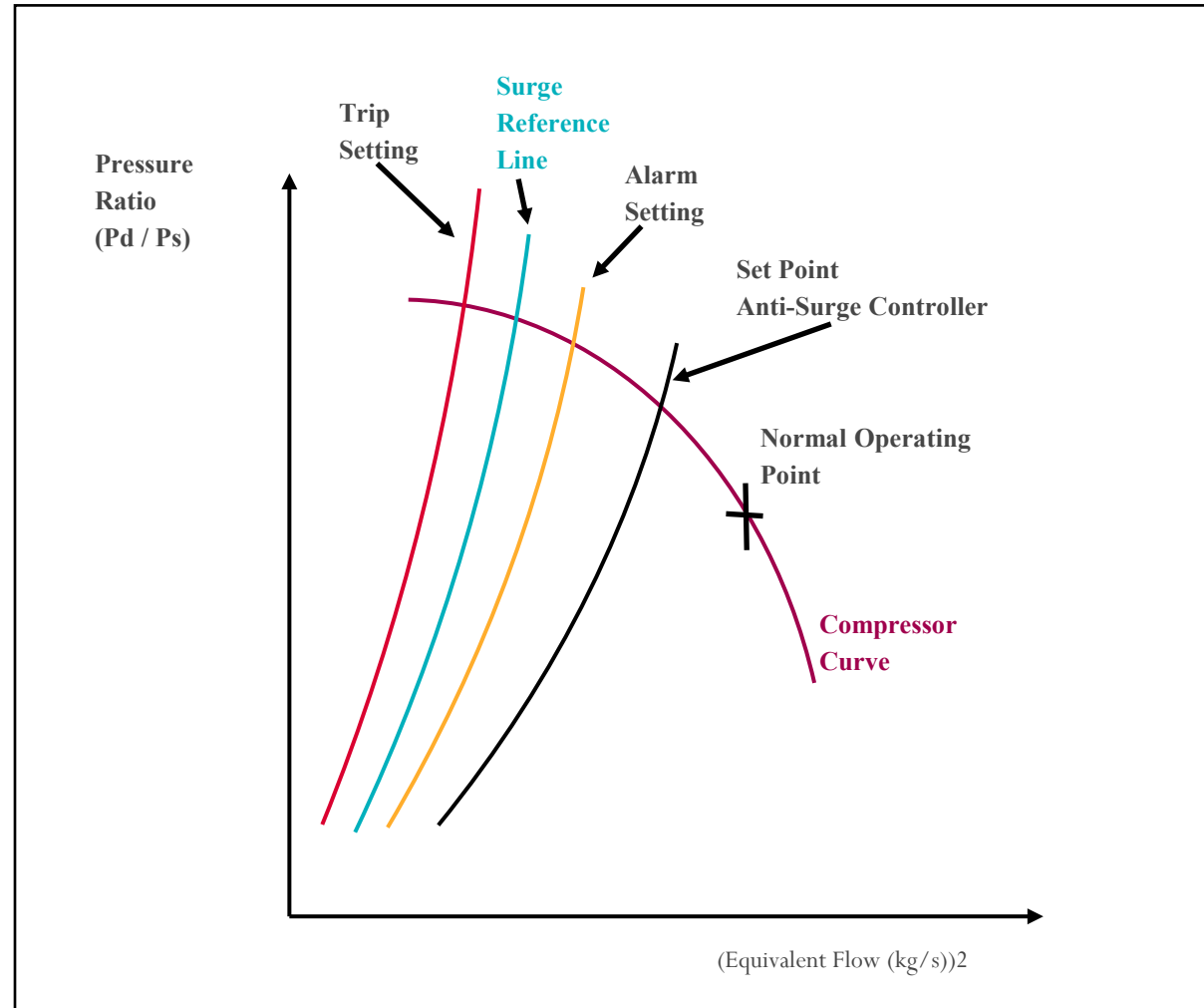


Blow-Off
Valve

Typical Installation (DCS based)

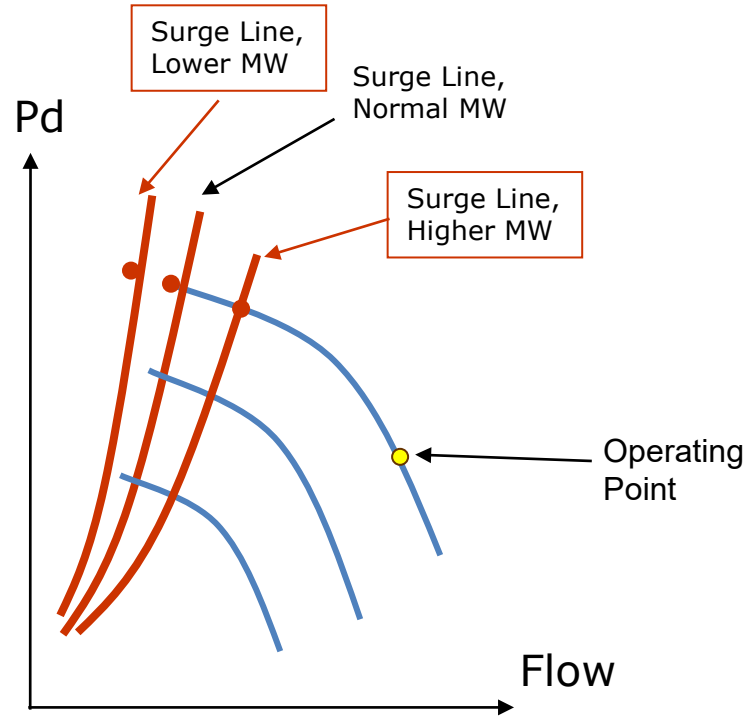


Implementation of Protection Mechanisms

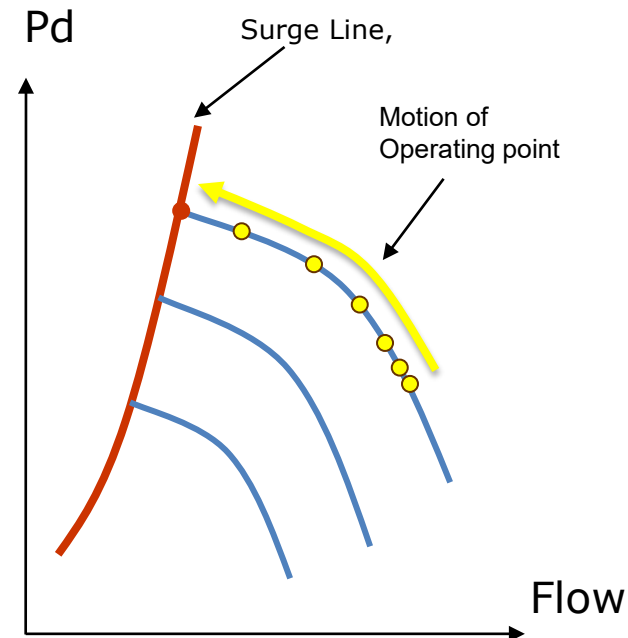


The Complexity of Surge Control

Complexity #1 : The measurement of Surge involves more than 1 parameter, making it unlike the standard control of P, T, F or L.

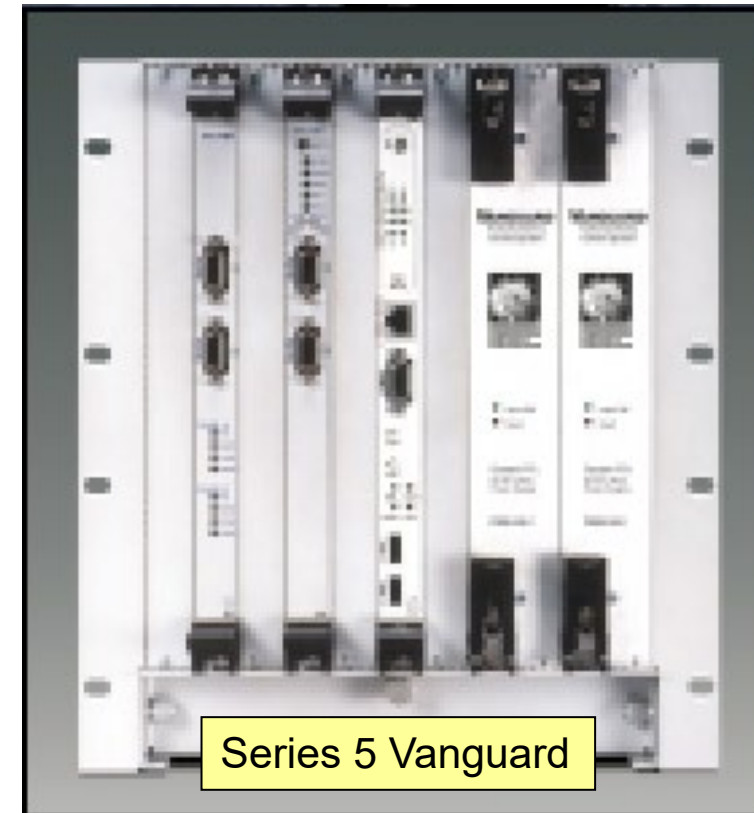


Complexity #2: The Surge Line (Surge Limit) varies with suction operating conditions.
Implication: the Surge Setpoint is not necessarily fixed

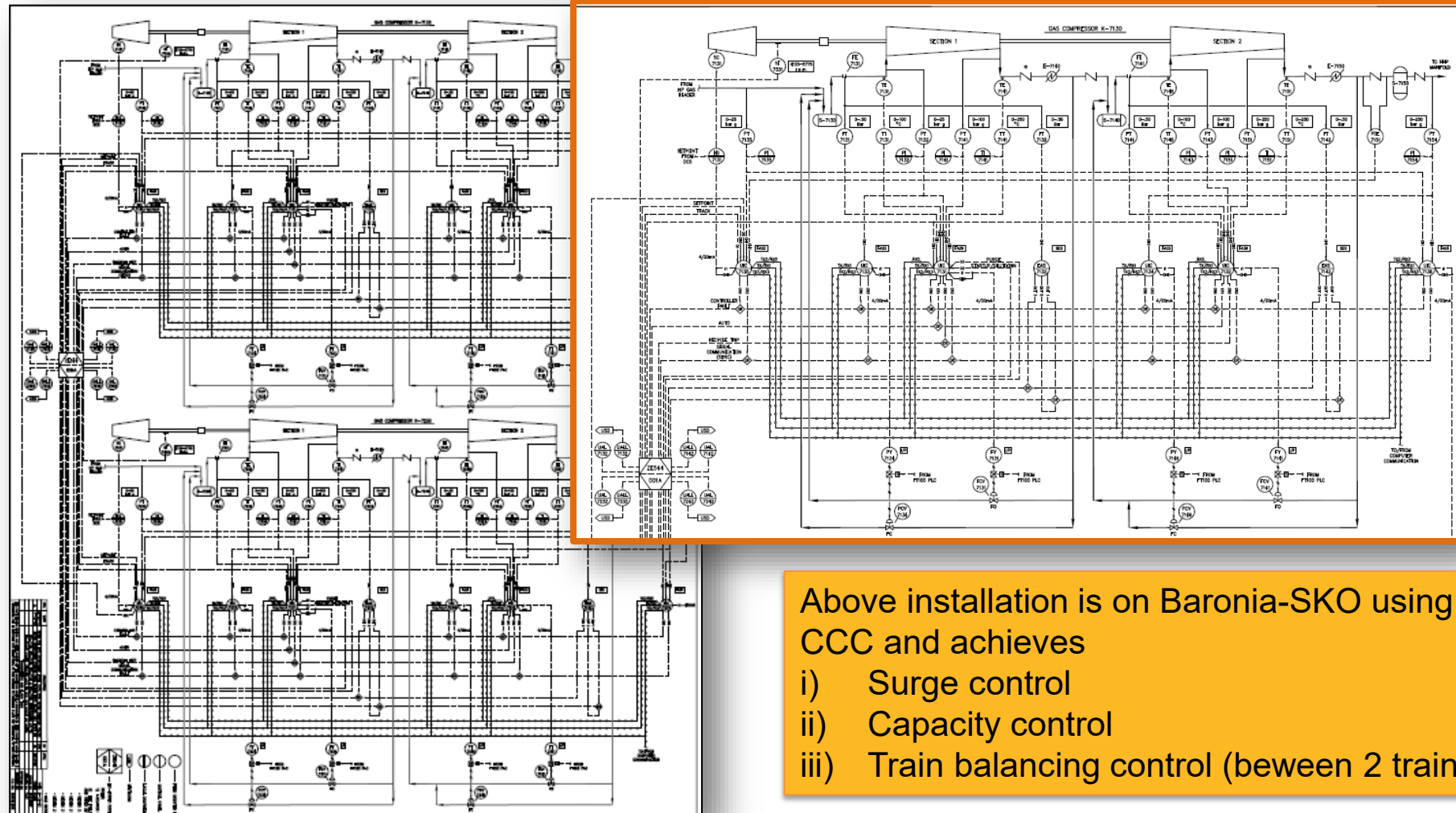


Complexity #3 : The speed by which the operating point can move to the left CAN BE very rapid. In addition, once the machine is in Surge, the surge cycles can be shorter than 2 seconds

CCC Applications



Complex installations in PETRONAS



Above installation is on Baronia-SKO using CCC and achieves

- i) Surge control
- ii) Capacity control
- iii) Train balancing control (between 2 trains)

Other Issues related to Antisurge control

- Multiple compressor trains – in parallel or in series
- Capacity control

Retrieval Practice

1. What conditions promote the move to surge?
 - a) High discharge pressure and high flow
 - b) Low discharge pressure and low flow
 - c) High discharge pressure and low flow
 - d) Low discharge pressure and high flow

2. Flow, temperature, pressure and level are a SINGLE DIMENSION problem. What do you think is the number of dimensions associated with surge control:
 - a) At least one dimension
 - b) At least two dimensions
 - c) At least three dimensions
 - d) At least four dimensions

Advanced Regulatory Controls



Distillation: What is it?

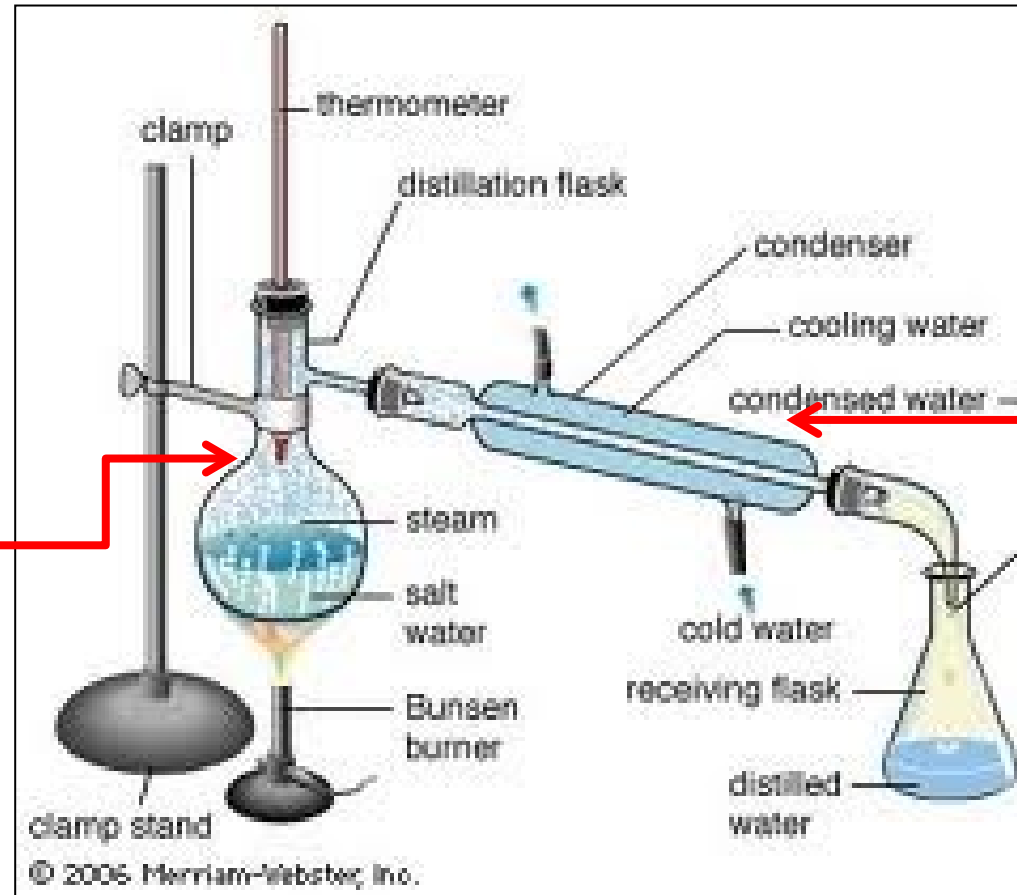
- Distillation is a physical separation process (i.e. and not a chemical reaction process)
- It is a process by which two or more substances are separated into its components based on the **RELATIVE VOLATILITY** or **BOILING POINTS**
- Distillation is achieved by the application and removal of heat.
- The distillation process is extremely energy intensive.

Simple Distillation

Salt Water Distillation (separation of solids from water)

*"The basic process of distillation involves the **boiling** of a liquid mixture and the **condensation** of the vapors"*

Vaporisation

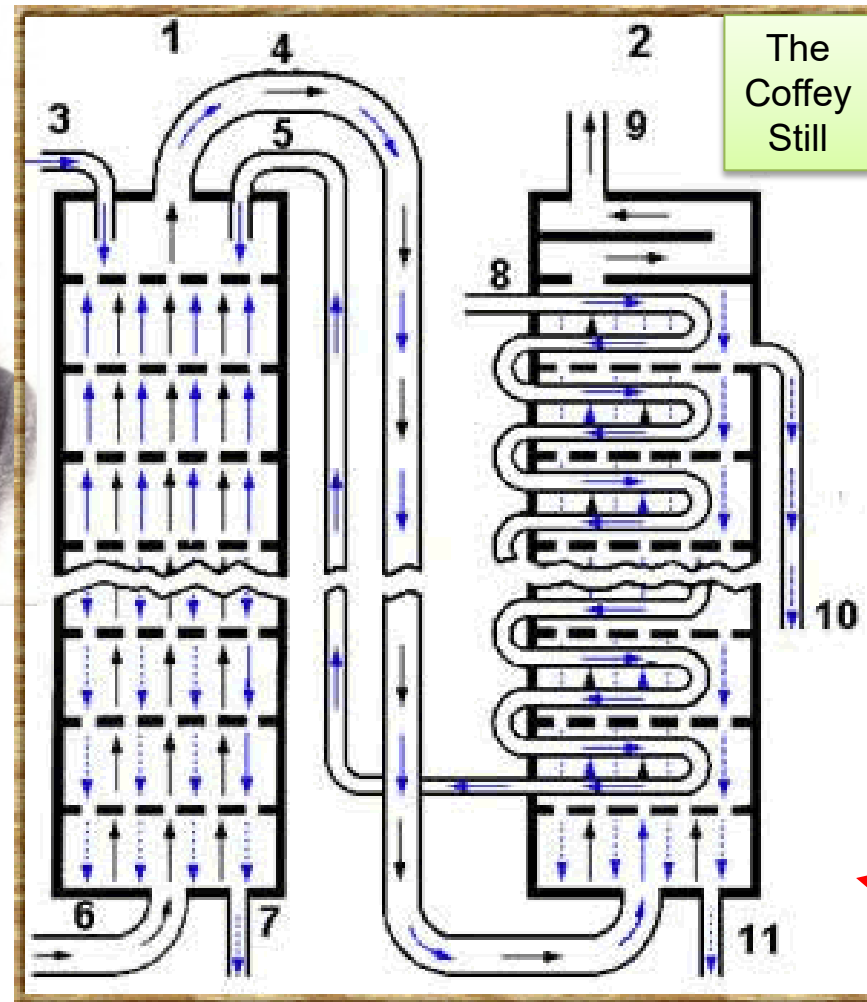


Condensation

Early stages of current distillation technology



Aeneas Coffey
1830



Purpose : Production of Alcohol



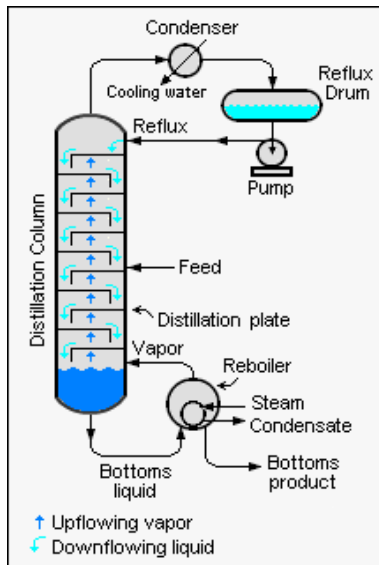
*1830: Starting to look like
classic distillation
equipment*

Later Developments

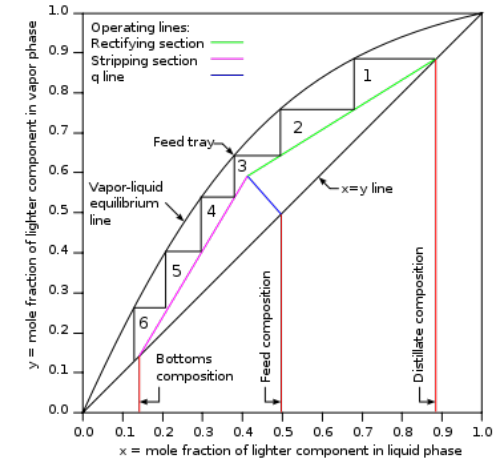
1900's



- Development of Oil and Gas Industry
- Emergence of Chemical Engineering



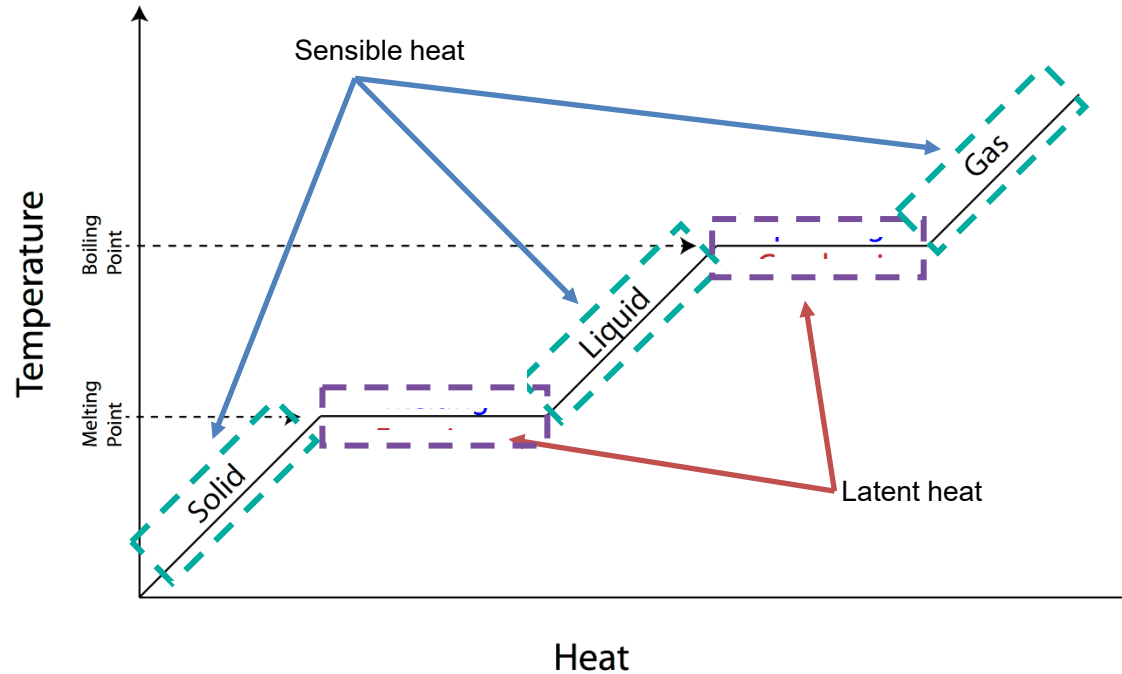
McCabe-Thiele method for the analysis of binary distillation



Fenske's Equation for the calculation of the number of theoretical plates reqd for the separation of a binary feed stream

$$N = \frac{\log \left[\left(\frac{X_d}{1-X_d} \right) \left(\frac{1-X_b}{X_b} \right) \right]}{\log \alpha_{avg}}$$

How does distillation work?



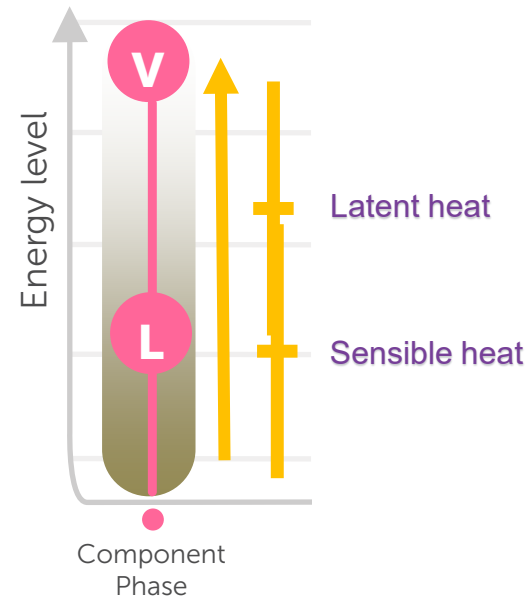
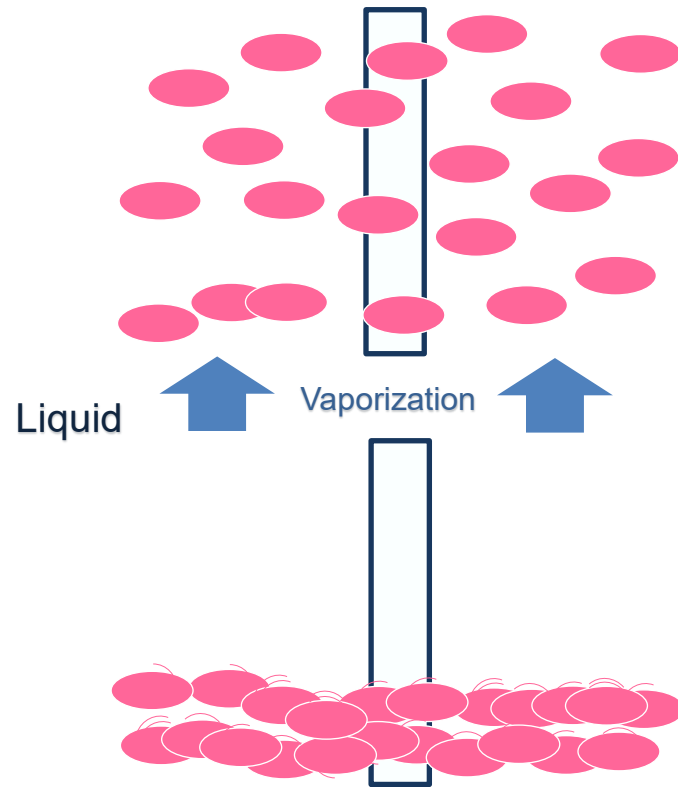
Sensible heat:
Heat that can be felt and measured by a thermometer

Latent heat:
Latent = Hidden
When changing phases, there is no change in temperature, only change in phase.

How does distillation work pt2?

Liquid to Gaseous Transition

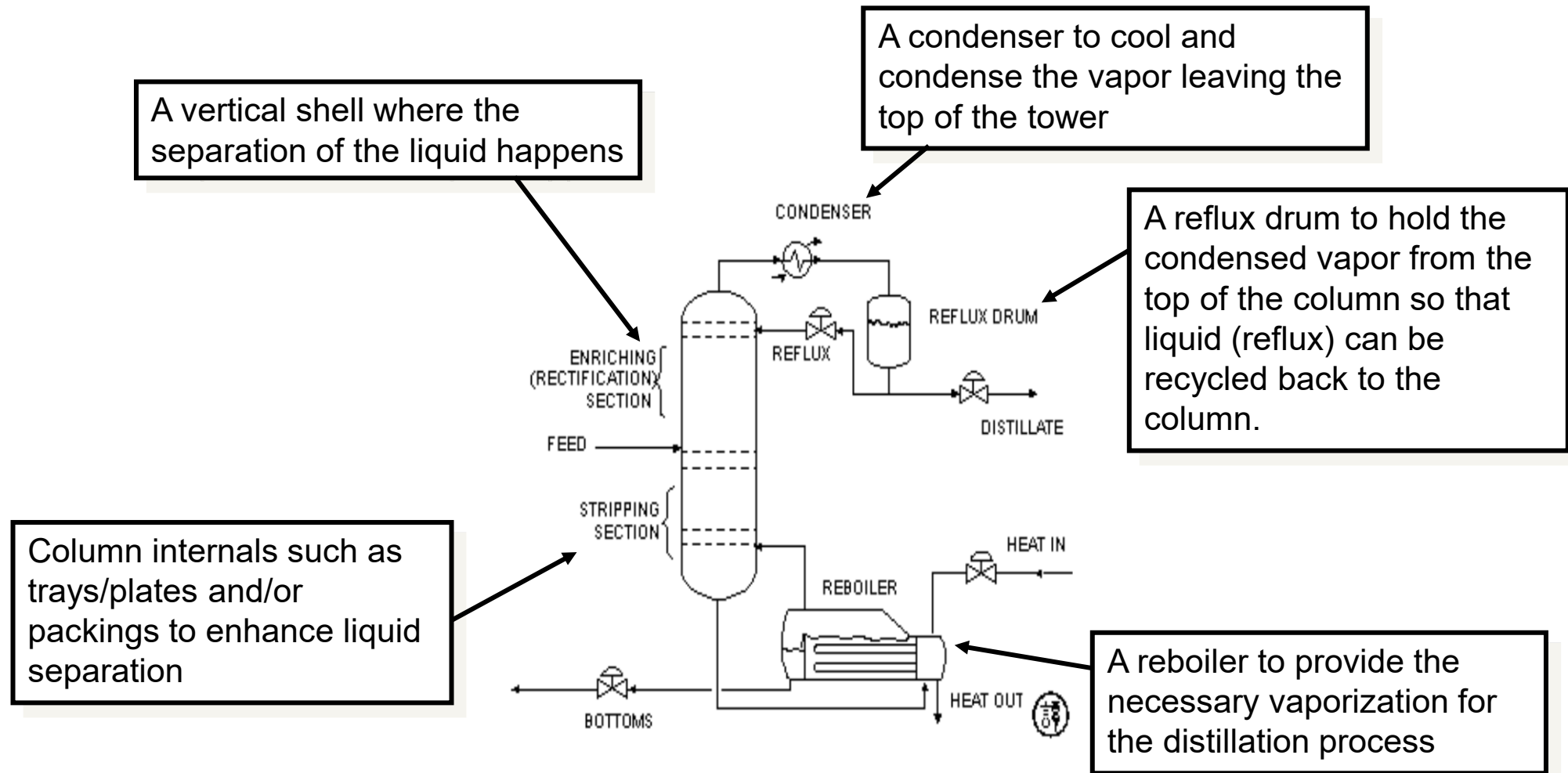
Gaseous



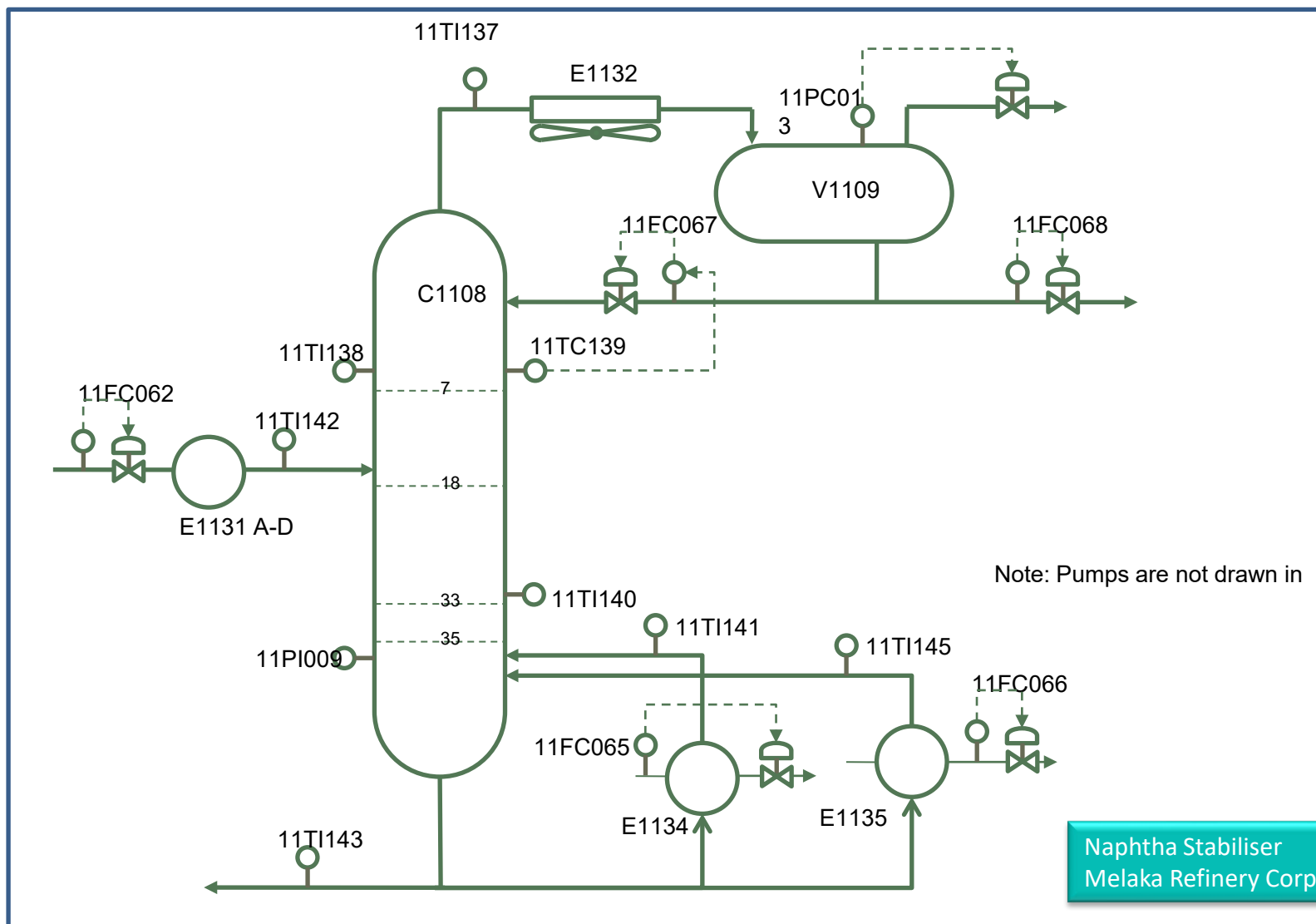
The liquid form of A absorbed heat and there are increase of temperature.

A mixture needs enough latent heat energy for phase change to break the intermolecular forces to vaporize.

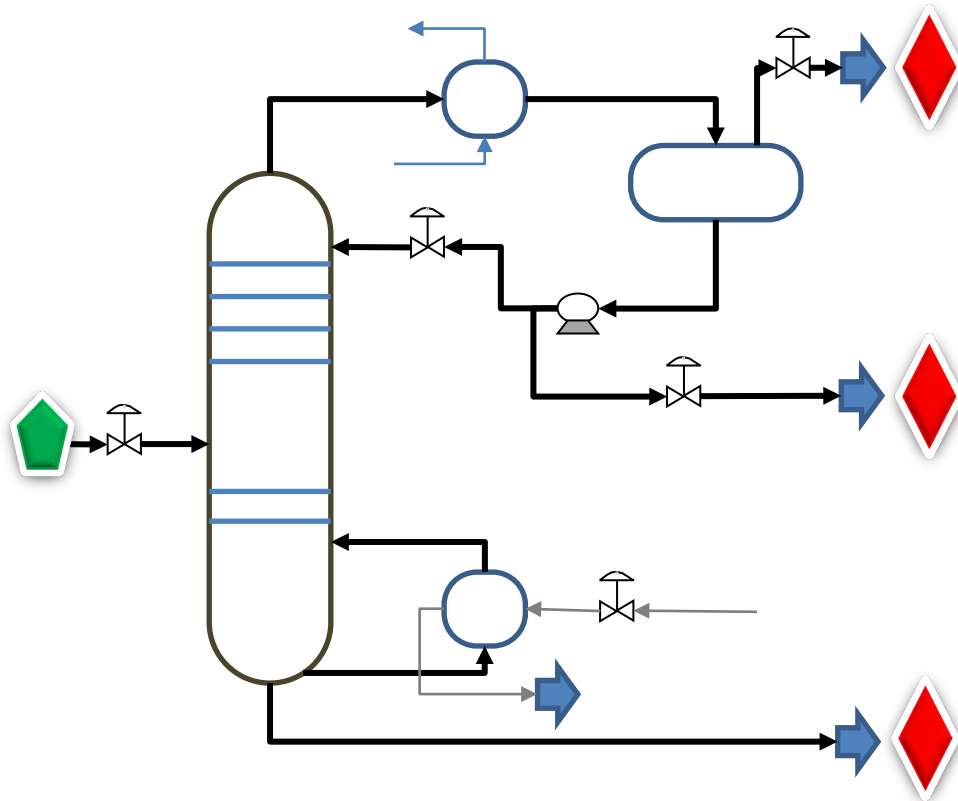
Components of a Distillation tower



Example of Distillation Set-up

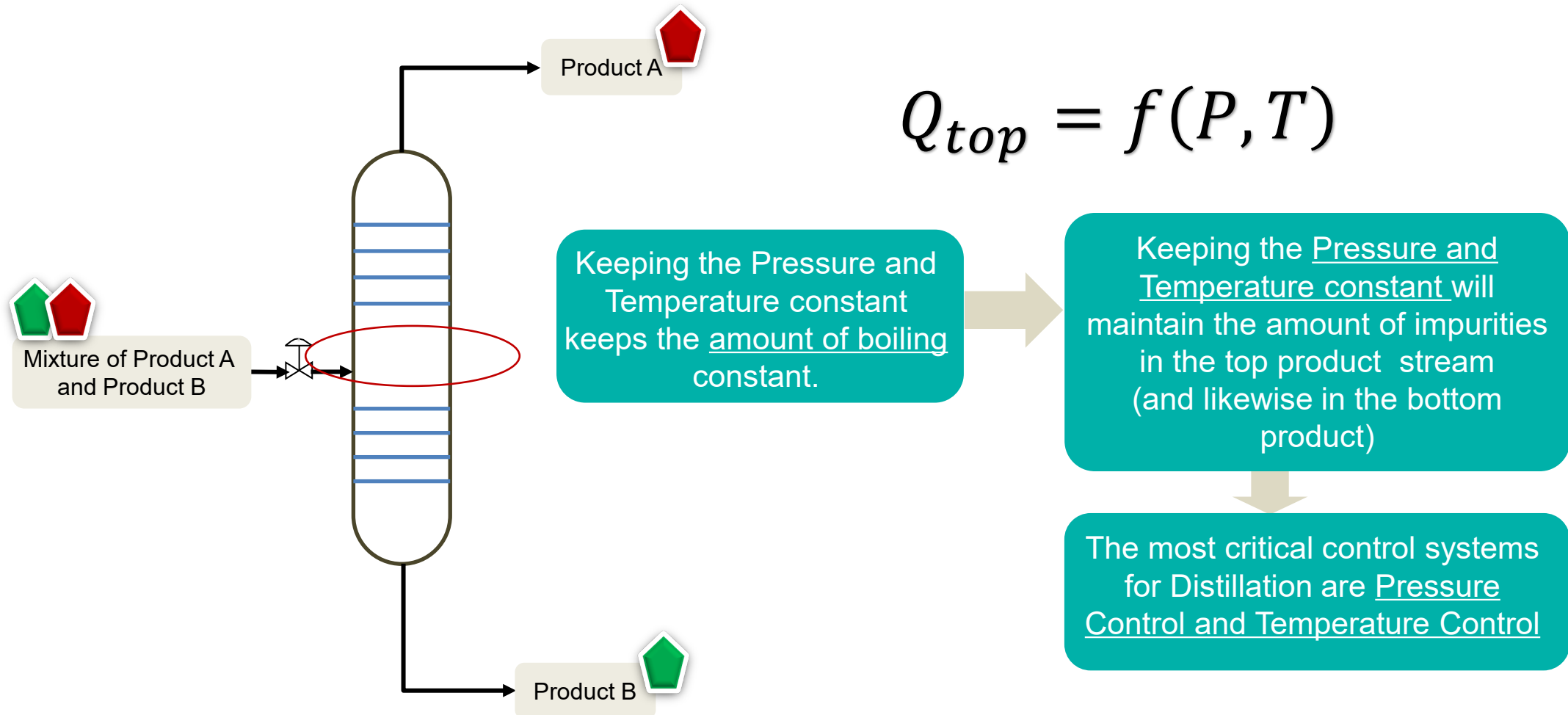


The impact of the operating point on Dynamics



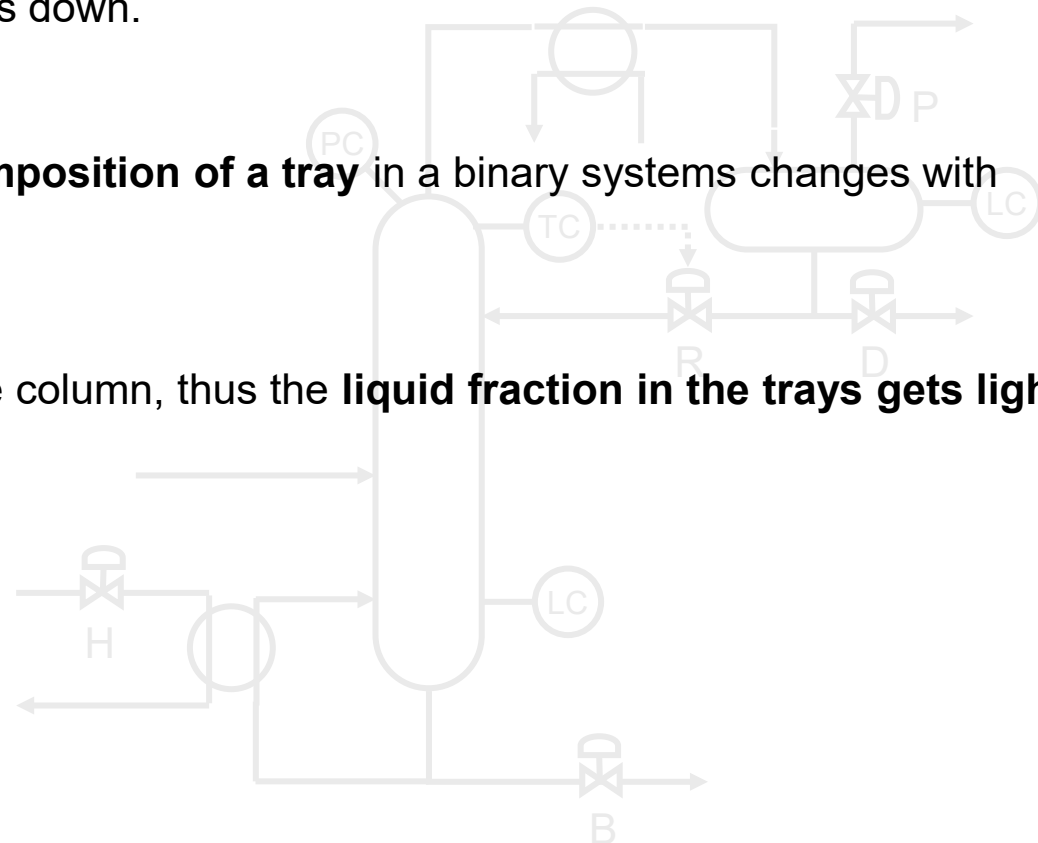
- i. This distillation column has forty trays
- ii. The normal feed to the column is 1000 t/d. The column is currently in turndown mode and operating at 500 t/d. It is meeting both top and bottom product spec
- iii. Assume that as the reflux is scaled 0 ~ 50 t/d and is increased by 5% (of full range). Compare between operation at 500 t/d and operation at 1000 t/d and answer the following. Will the time taken to reach a new equilibrium for tray temperature and composition be longer or shorter?
 - a) 500 t/d will have a faster settling time compared to operating at 1000 t/d
 - b) 500 t/d will have a slower settling time compared to operating at 1000 t/d
- iv. What conclusions can be made with respect to feed conditions and dynamics?

Initial Implications from understanding Boiling



Distillation Control Methods

- Heat is introduced to the column at the bottom and removed at the top.
- Hot vapor rises to the top and cool liquid cascades down.
- It is assumed that at a constant pressure, the **composition of a tray** in a binary systems changes with **temperature**.
- Tray temperature decreases with the height of the column, thus the **liquid fraction in the trays gets lighter**.



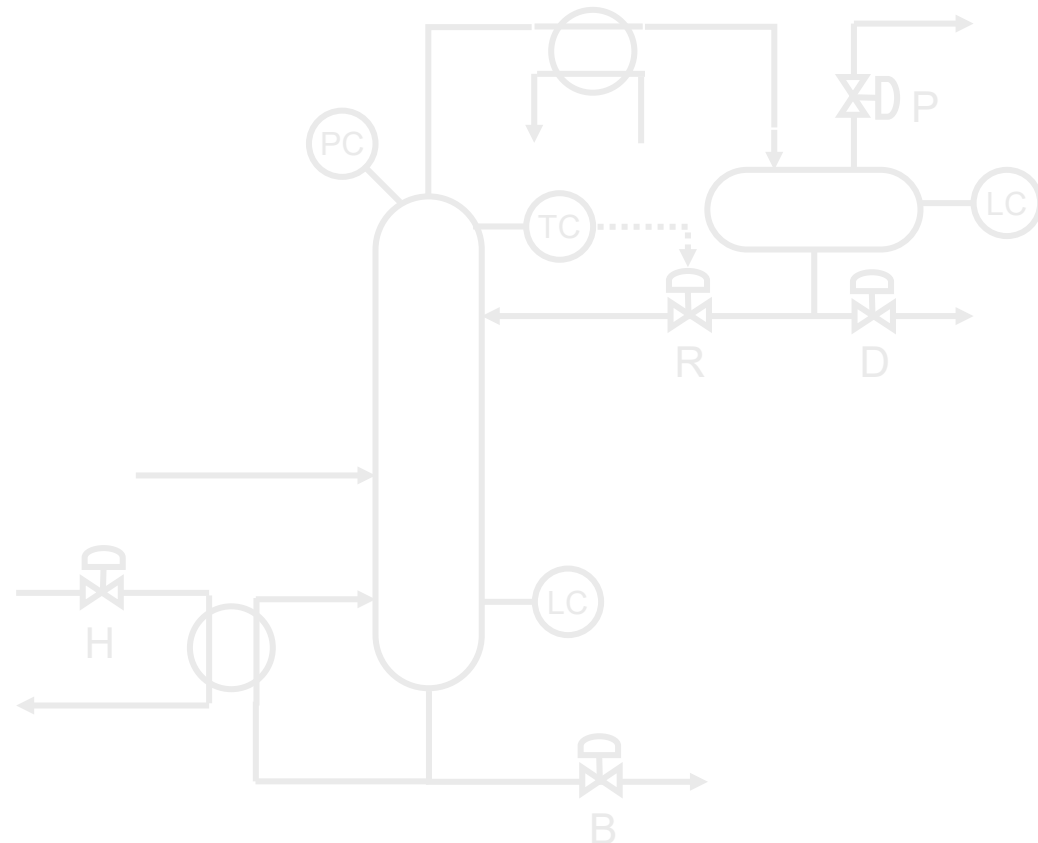
Objectives of Process Control in Distillation

Distillation Process Control Objectives:

1. Product quality:
 - To maintain a constant product quality
 - To control product quantities
2. To maintain liquid levels within limits
3. To maintain the internal column loads within constraints

Distillation Control Methods

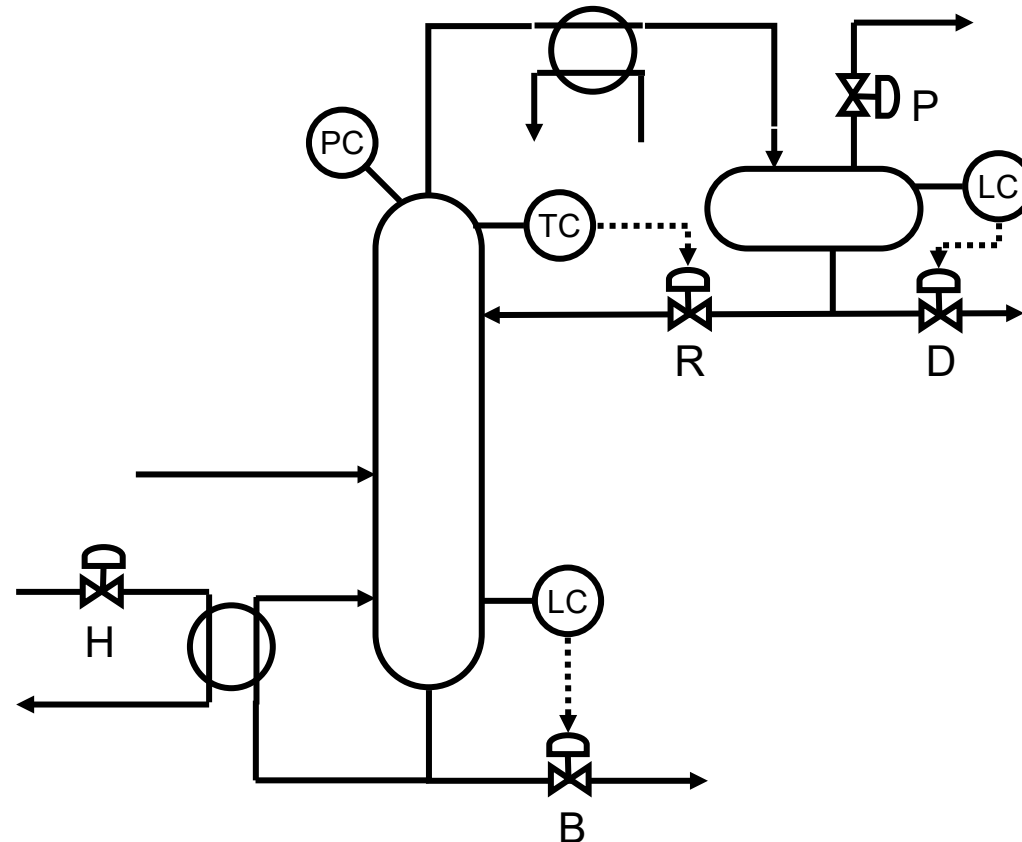
- Heat Balance Control
- Mass Balance Control
- Quality Control (not explained)



Distillation Control Methods

Heat Balance Control

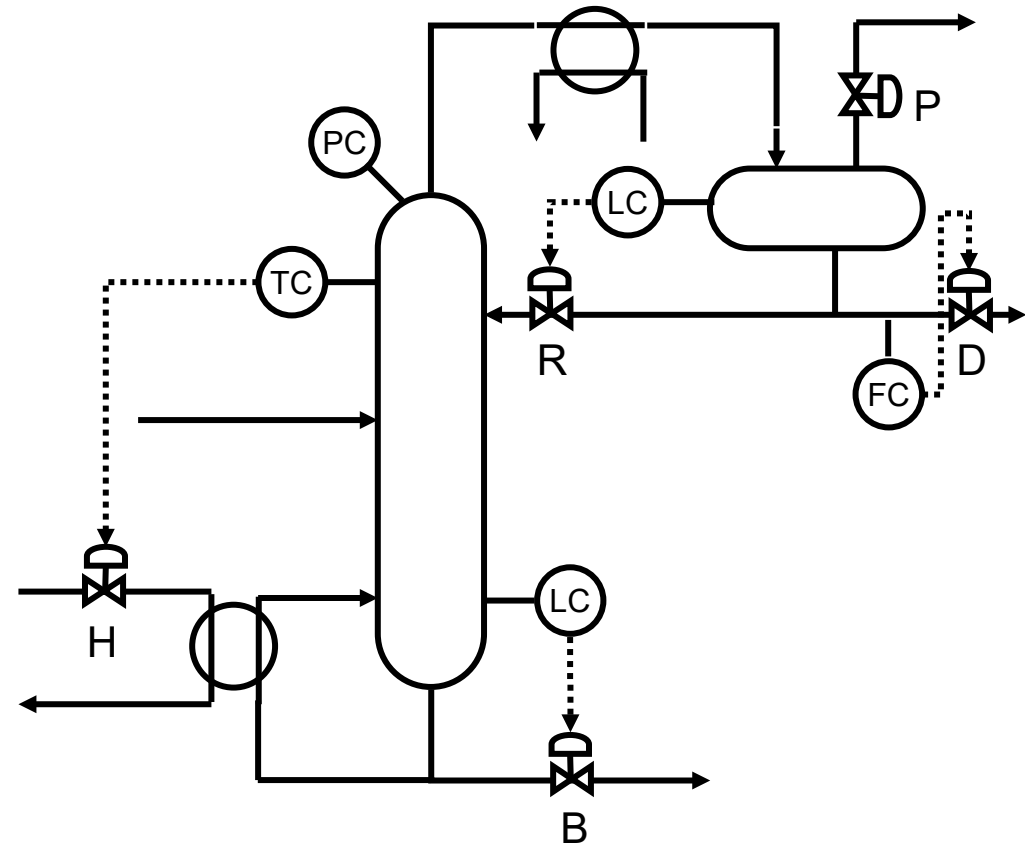
- In Heat Balance Control, the amount of heat in the system is maintained by manipulating the Reflux Flow or Reboiler duty
- Example here is controlling the reflux.



Distillation Control Methods

Mass Balance Control

- In Mass Balance Control, a mass balance of the system is maintained by fixing the yield, either top or bottom product.
- In this configuration, distillate flow / yield is fixed.



Multi-Component Distillation – understanding the refinery crude split



No. of Carbons	Molecular Formula	Name	Acronyms in PETRONAS
1	CH ₄	Methane	C1's
2	C ₂ H ₆	Ethane	C2's
3	C ₃ H ₈	Propane	C3's
4	C ₄ H ₁₀	Butane	C4's
5	C ₅ H ₁₂	Pentane	-
6	C ₆ H ₁₄	Hexane	-
7	C ₇ H ₁₆	Heptane	-
8	C ₈ H ₁₈	Octane	-
9	C ₉ H ₂₀	Nonane	-
10	C ₁₀ H ₂₂	Decane	-
11	C ₁₁ H ₂₄	Undecane	-

C10
C16

C14

C20

Refinery Fuel Gas

LPG

Naphtha

Kerosene

Diesel

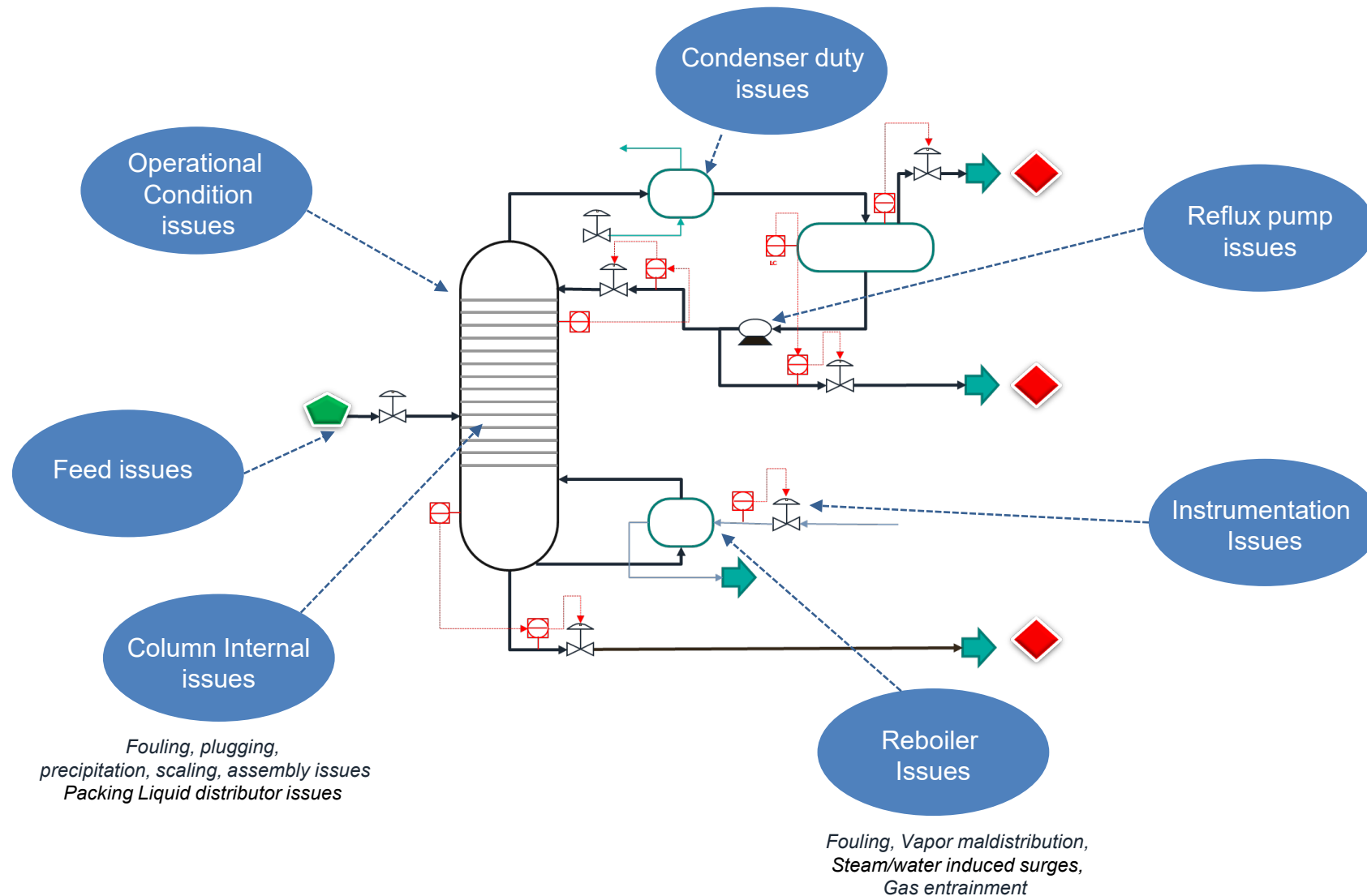
C20
C50
Cxx

Lubricating Oils

Residue

What potential issues impacting distillation columns

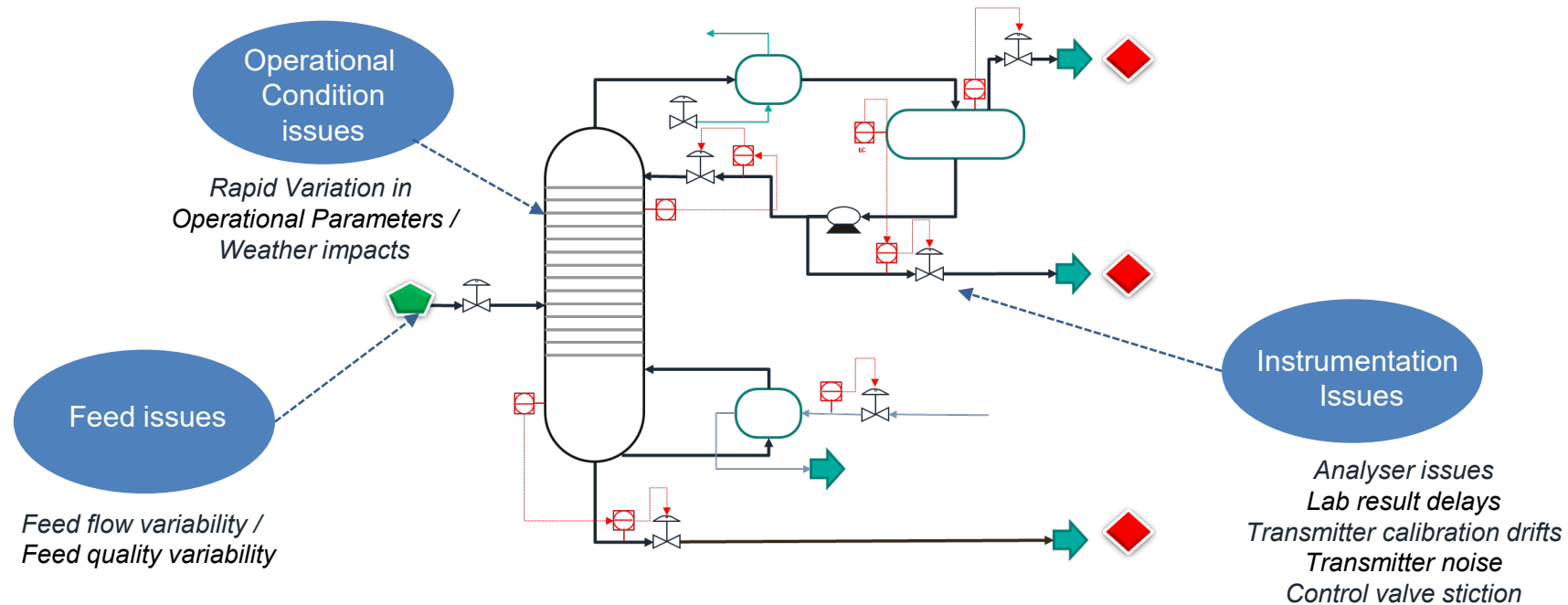
Discussion



What potential dynamic issues impact distillation columns

Primary Conclusion for Dynamics associated with a distillation column

- A Distillation equipment is a LONG DYNAMIC system
- This implies that it takes a long time for the system to stabilize after a change
- IMPLICATION : To stabilize any distillation column, rapid changes to the column should be eliminated / minimised



Summary of Process issues

- Long time constants (reboiler impact to bottom, bottom change impact to top)
- Long deadtimes (change to final product quality)
- Interactivity (impact of top to bottom and vice versa)
- Non-linear (especially for high purity columns)
- Large energy consumer



Long time constant

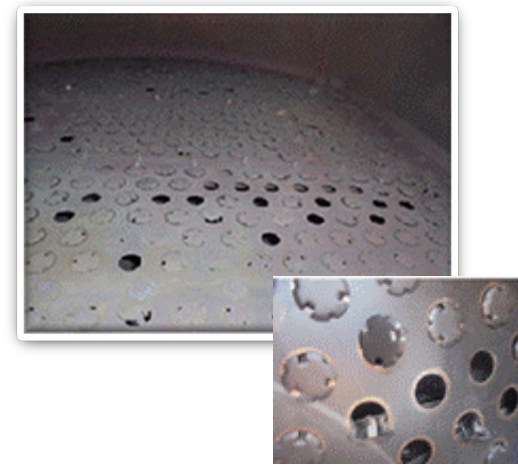
Impacts on Improper Operation



Collapsed
Tray



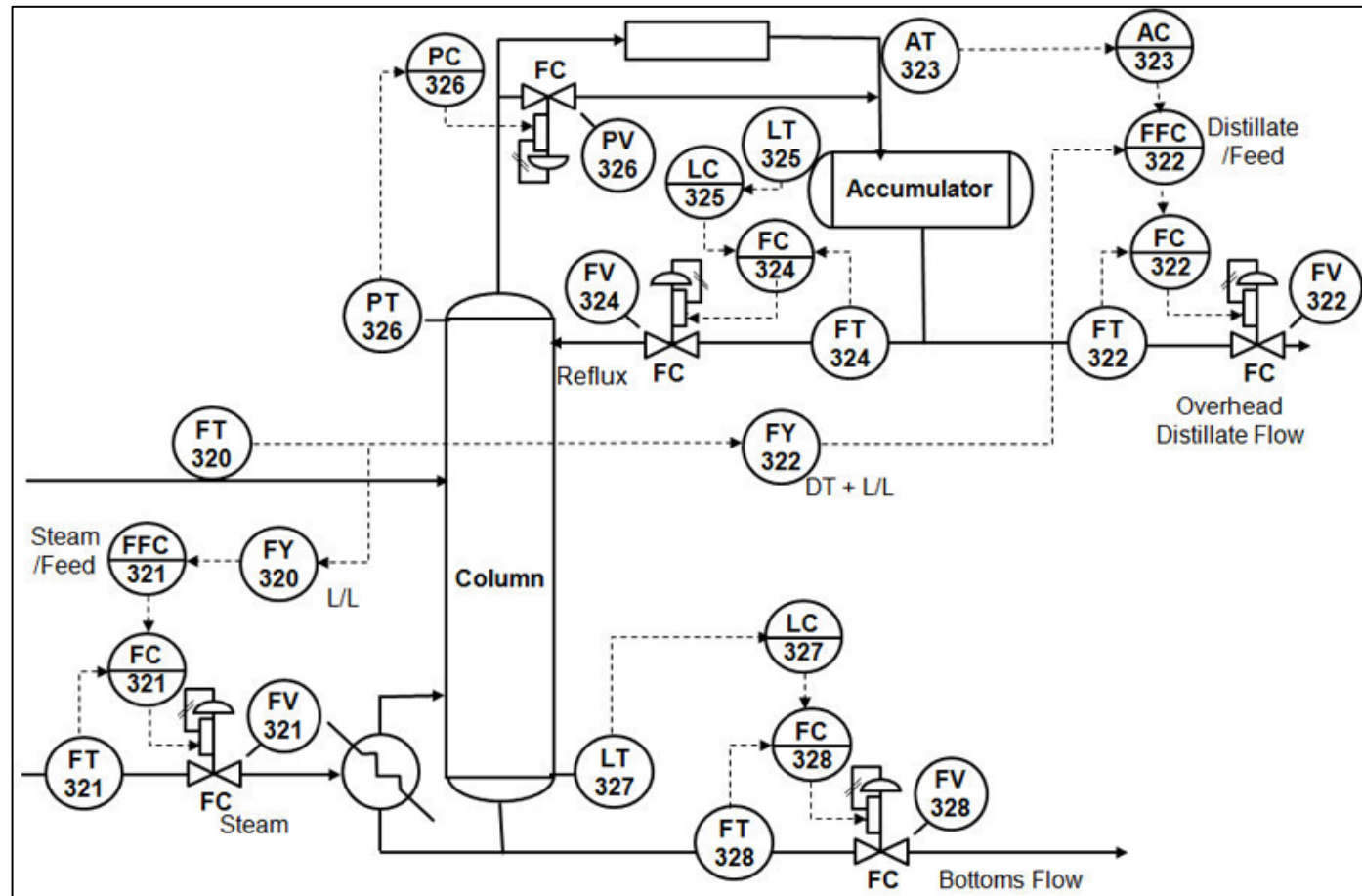
Detached
Manway



Dislodged
Valve

Sample of Feedforward Control

The impact of feed rate on composition control is minimized through the use of feedforward control with dynamic compensation. Heat and composition balances required for stable operation are maintained using heat/feed and distillate/feed ratio control.



Retrieval Practice

1. An instrument engineer approaches you and asks you to explain why distillation columns are so difficult to control. You give the 2 keywords that explains this issue
 - a) _____
 - b) _____

2. The quality of a product is largely dependent on two key process parameters. These are:
 - a) _____
 - b) _____

3. These two process parameters are however sensitive to one or two major disturbance process (a disturbance process is usually not controlled by the unit). These MAJOR disturbance parameters are:
 - a) _____
 - b) _____

Layer 4

Advanced Process Control

Advanced Process Control

This refers to complex control algorithms whose sole purpose is to optimise plant operations through tight control.

Techniques

- Multivariable Predictive Control (MPC)
- Inferential
- Procedural Automation
- Optimisation techniques

Advanced Process Control

Multivariable Predictive Control

Inferentials/Property Estimator

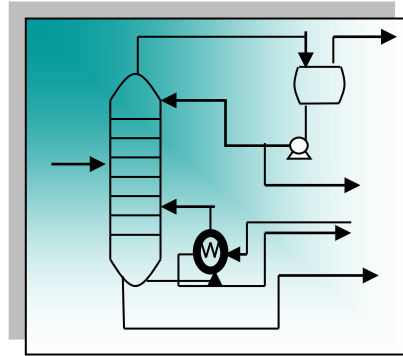
Procedural Automation

Optimisation

APC and Multivariable Control

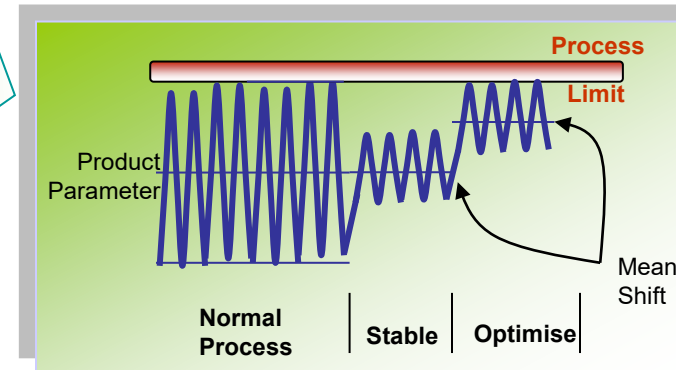
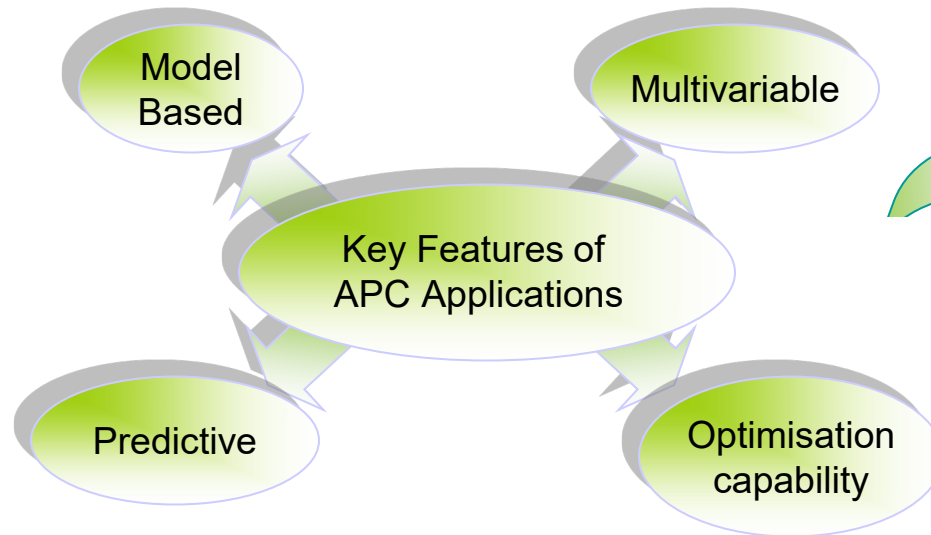
- Advanced Process Control or APC applications are also known by other names:
 - Multivariable Control
 - Multivariable Predictive Control (MPC or MVPC)
- APC is utilized on complex process units that are difficult to control such as distillation and reaction based processes.
 - These processes are difficult primarily due to substantial process interactions and process sensitivities
 - As a result, the Operator has to **monitor a large number** of process parameters and
 - **Manipulate a large number** of handles
- APC is a purpose built (not ready made) Control Application that will automatically stabilise an entire Process Unit.
- By stabilizing a Process Unit, the APC can then target to improve product specifications and yield.

Features of APC

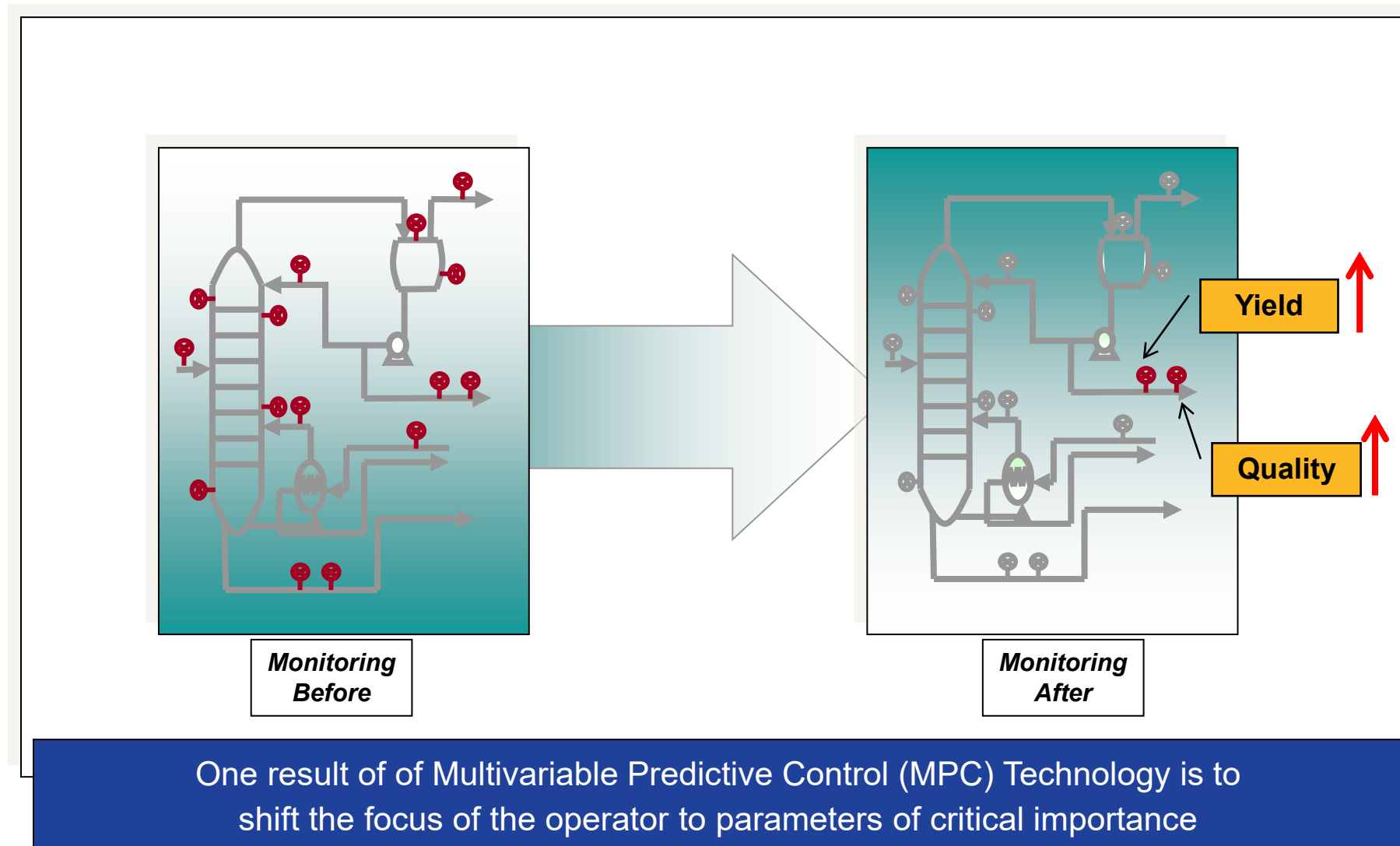


Key Issues :

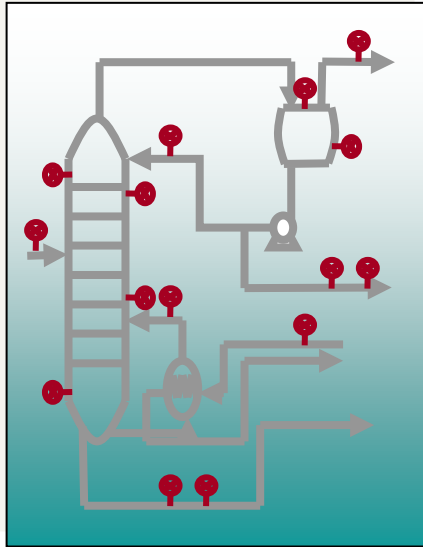
- i) Interactive
- ii) Long time to stability
- iii) Multiple impacts on final quality



Shift of Operational Focus



How to control a complex process?



Key Question:

How to control a Complex Process Unit Automatically using an APC application?

To control a process unit automatically means:

- i) The application **shall be able to handle feed changes** without operator intervention
- i) The application shall **be able to handle external disturbances** (such as utility disturbances and external disturbances such as rainshower)
- ii) The application shall be **able to identify constraints** in the process unit (limiting/fouled heat exchanger, tray flooding, etc) and **keep the process away from constraints**

Step 1

Define relationships in the process:

- i) Top quality (CV) and reflux (MV)
- ii) Top quality (CV) and reboiler (MV)
- iii) Top quality (CV) and pressure (MV)
- iv) Top quality and feed flow
- v) Bottom quality
- vi)

Result

Called
Model
Building

APC
Model

Step 2

Prioritise MV's and CV's

- Which is the most critical CV/MV
- If top quality is in violation, which MV should the application move?

Step 3

Offline Simulation and Testing

Testing with Operations / Technologist personnel

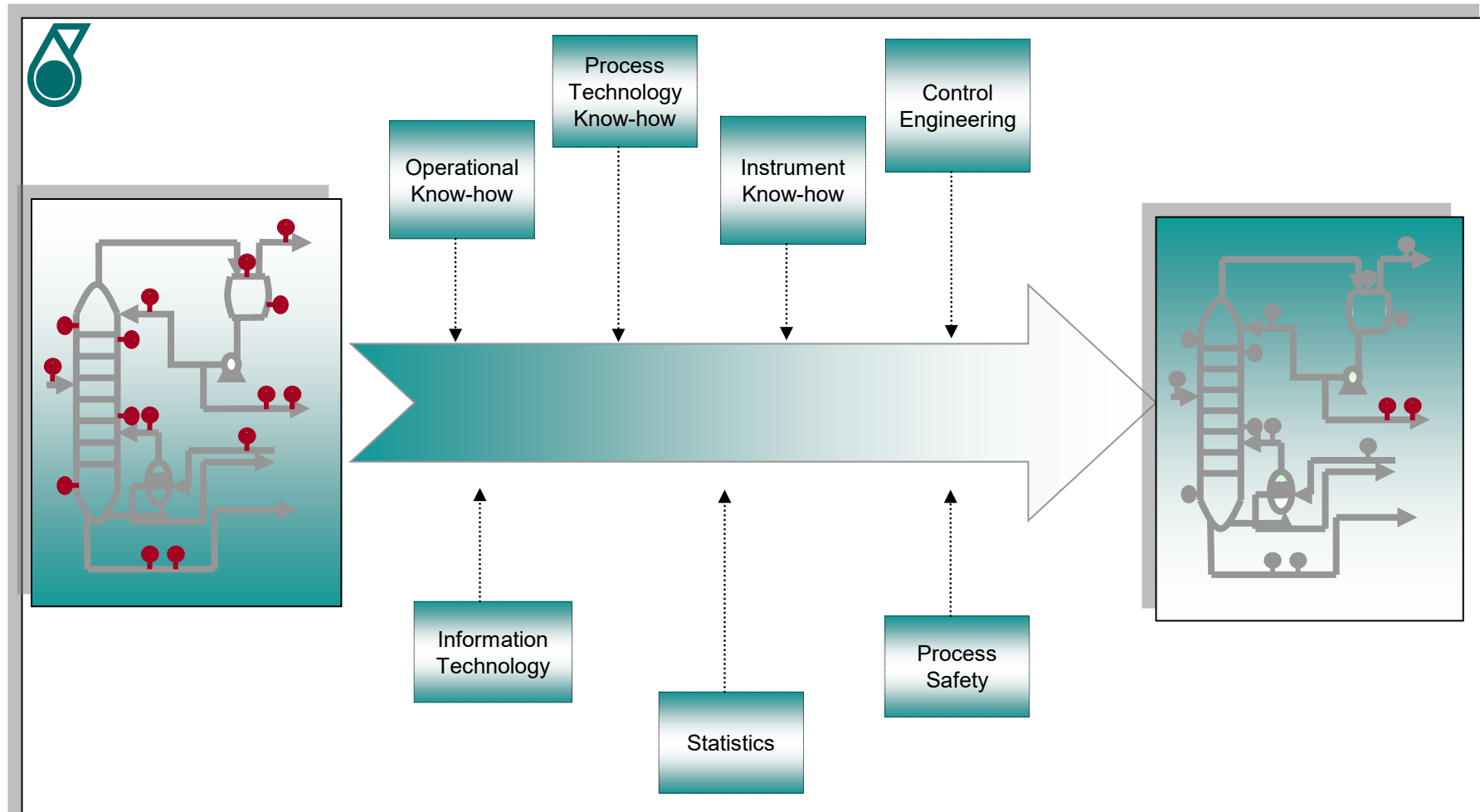
Step 4

Online Installation & Commissioning

Installation at DCS

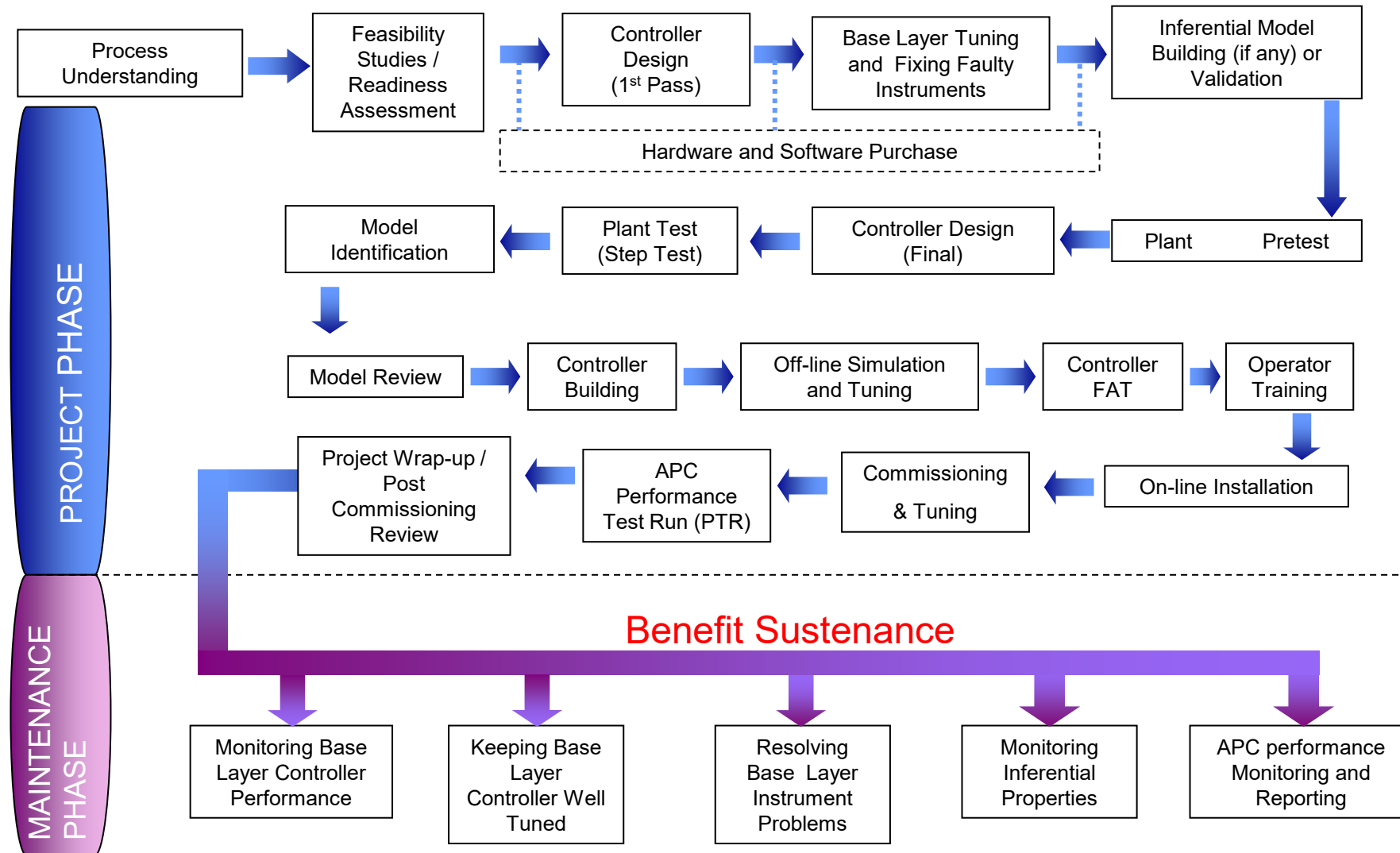
FUNCTIONAL
APC
APPLICATION

Skills Requirement to develop APC



APC represents the convergence of a number technology areas with the objective of stabilizing and optimizing the plant, while modifying the operators focus on to the process variables that matter.

APC Implementation Workflow



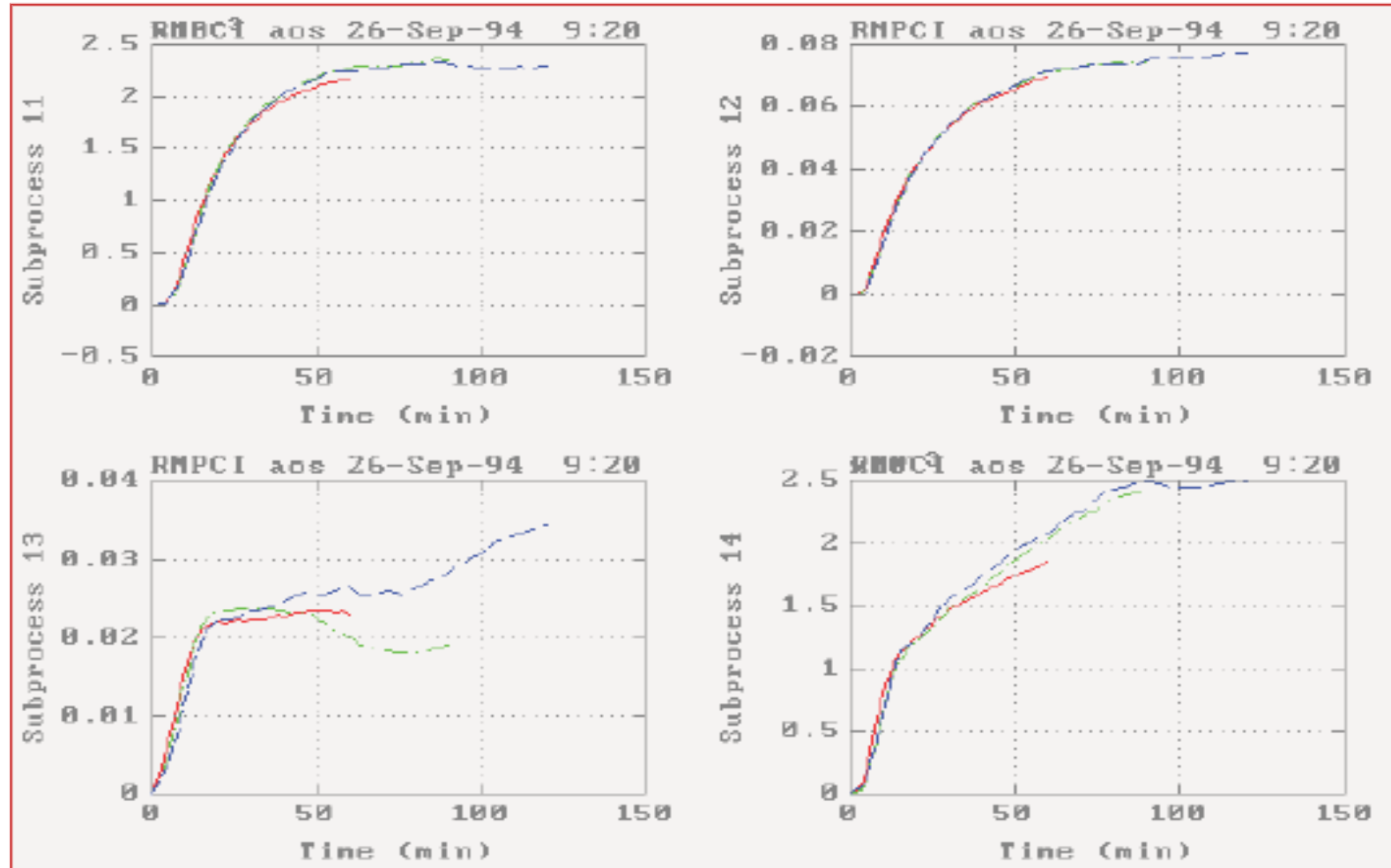
Example of Initial APC matrix

CV #	Description	MV1 PF side draw	MV2 Pass balance flow	MV3 COT	MV4 MF Reflux flow	MV5 HN Prod. flow	MV6 Kero Prod. flow	MV7 Diesel Prod. flow	MV8 Reboiler Temp.	MV9 MF column pressure
1	HN T5				--	+				--
2	HN T95	-			--	++				--
3	Kero Flash	--		+	-	++	++		++	--
4	Kero Freeze	--		+	-	+	++			--
5	Diesel Pour			+				++		--
6	Diesel T95			+				++		--
7	Preflash column level		-							
8	Furnace O2		-	-						
9	Furnace tube skin temperature		-	+						
10	Furnace Fuelgas pressure		+	+						

PF: Preflash
MF: Main Fractionator

COT: Coil Outlet Temperature
HSRN: Heavy Naphtha

Sample of APC models (from Honeywell RMPCT)



Typical APC DCS Display: PACE PCFK (on APC server)

APC HMI - Platform for Advanced Control and Estimation - Run Time

Save APC_ENGINEER Shed All Controllers Help

Runtime Message

5/31/2019 12:30:00 PM Low Live NH3 11PIC02.MV Processor NH3: POV 11PIC02.MV predicted value was clamped.
 5/31/2019 12:29:00 PM Low Live NH3 13AI01.PV Processor NH3: POV 13AI01.PV predicted value was clamped.
 5/31/2019 12:29:00 PM Low Live NH3 15PIC17.DV Processor NH3: POV 15PIC17.DV predicted value was clamped.

Navigation Pane

Application Management

Variable Set Name Status Actual Mode Request Mode KPI Tuning Status

CO2_Feed Control Active Active 94 % Ok

Control Estimation Standard Trends Custom Trends Measurements Parameter Group Operating

MV

Name	Description	Unit	Loop PV	Output	SS Value	LL	HL	Move	Status	Act M...	Req Mode
15PIC09.SV	CO2...	Bar	0.501	0.502	0.502	0.495	0.55	0	PredO...	Inac...	Inactive
31-PICA-6...	CHL...	Bar	6.916	6.915	6.687	5	9	-0.0005	Control	Active	Active
31SIC01-P...	K-3...	RPM	9320.5...	9350.7...	9389.1...	9200	9550	0.1772	Control	Active	Active

CV

Name	Description	Unit	Current Val...	SS Value	LL	Setpoint	HL	Status	Act M...	Req Mode
15PV09A.MV	CO2 VENT O...	%	30.122	20	20			Control	Active	Active
R31_V1OP.PV	K-31 MG...	%	84.628	85.014			90	Control	Active	Active
31TI09.PV	K-31 COMP...	DegC	21.202	20.543	15		30	Control	Active	Active
31PI10.PV	K-31 COMP...	Bar	79.857	79.8			79.8	Control	Active	Active
33CAL02.CPV	UR PLANT L...	%	103.571	104.629			105	Control	Active	Active

APC HMI - Platform for Advanced Control and Estimation - Run Time

Save APC_ENGINEER Shed All Controllers Help

Runtime Message

2019 12:29:00 PM Low Live NH3 15PIC17.DV Processor NH3: POV 15PIC17.DV predicted value was clamped.
 2019 12:29:00 PM Low Live NH3 11PIC02.MV Processor NH3: POV 11PIC02.MV predicted value was clamped.
 2019 12:28:00 PM Low Live NH3 13AI01.PV Processor NH3: POV 13AI01.PV predicted value was clamped.

Navigation Pane

Application Management

Controller Name Status Actual Mode Request Mode KPI Tuning Status

UREA Control Active Active 100 % Ok

Detail Model Matrix Auto Step Overview Shed Logic

Variable Set

Name	Description	Status	Act Mode	Req Mode	Essential?	KPI (%)
CO2_Feed	CO2 Feed	Control	Active	Active	False	94
Urea_Synt...	Urea Syn...	Control	Active	Active	False	100
Recovery	Recovery	Control	Active	Active	False	100

Economic Function

Name	Description	Status	Act Mode	Req Mode	Value	St Priority
EF	EF for Feed and Synthe...	Optimize	Active	Active	-26064.8...	9000
EF_Rec	EF for Recovery	Optimize	Active	Active	18869.991	9000

Typical APC DCS display: AspenDMC PCESB (on DCS screen)



Typical APC DCS Display: RMPCT MRCSB

24MC100 (24MC100) on Local - Honeywell Profit Viewer									
Honeywell Profit Controller (RMPCT) OPTIMIZING									
CONTROLLER DETAIL		ON	OFF	WARM	APP MENU	OPTIONS	STATUS MSGS	REPORTS	
CV SUMMARY	CV DETAIL	CV OPTIMIZE	CV CONTROL	CV PROCESS	CV ADV TUNING	GAIN DELAY			
MV SUMMARY	MV DETAIL	MV OPTIMIZE	MV CONTROL	MV PROCESS	DV SUMMARY	DV DETAIL			
				MOVE DOWN	MOVE UP	MOVE TO TOP	BY MV#		
MV#	MV DESCRIPTION	STATUS	VALUE	MOVE	FUTURE	SS VALUE	LO LIMIT	HI LIMIT	MODE
1	24FC016.SP	ON	55.00	0	55.00	55.00	23.00	55.00	RMPC
2	24TC018.SP	ON	82.26	-0.00364	82.26	82.00	82.00	86.00	RMPC
3	24TC036.SP	ON	88.00	0	88.00	88.00	80.00	88.00	RMPC
4	24UC004.SP	ON	19.76	0	19.76	19.76	19.76	25.00	RMPC
5	24PC014.SP	ON	8.100	0	8.100	8.100	7.900	8.100	RMPC
6	24HC005.OP	ON	13.03	-0.0301	13.03	10.84	0	50.00	RMPC
7	27FC001.SP	ON	40.80	0.0269	40.80	42.76	28.00	43.00	RMPC
8	24FC033.OP	ON	35.00	0	35.00	35.00	28.00	40.00	RMPC
9	24PC018.SP	ON	2.500	0	2.500	2.500	2.000	2.500	RMPC
10	24HC015.OP	ON	19.60	0.00391	19.60	19.88	0	50.00	RMPC
11	24PDC016.OP	ON	0	0	0	0	0	50.00	RMPC

APC Applications in PETRONAS

Refining	PCG		Gas & Energy	
MRC SB	PCESB	PCASB	MLNG-1	GPK
PP(T)SB	PCMTBE	PCFK	MLNG-2	GPS
ENG EN	PCOGD	ABF	MLNG-3	
PRefChem-R	PCLDPE	PCMSB	Tr 9	
	PCARO	PRefChem-PC		
Total APC	Total APC		Total APC	
56	87		72	

Total APC applications in PETRONAS

215

Legend

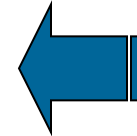
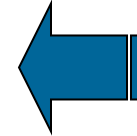
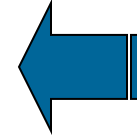
Completed

Ongoing

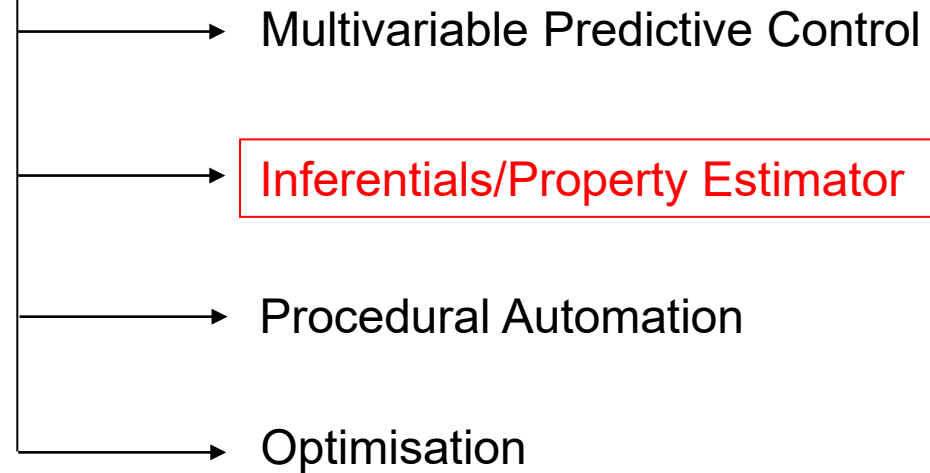
Future?

Common MPC vendors

- RMPCT
 - Robust Multivariable Predictive Control Technology
- DMCplus
 - Dynamic Matrix Control
- PACE
 - Platform for Advanced Control & Estimation
- There are sites using other than above due to licensor or integrator requirements



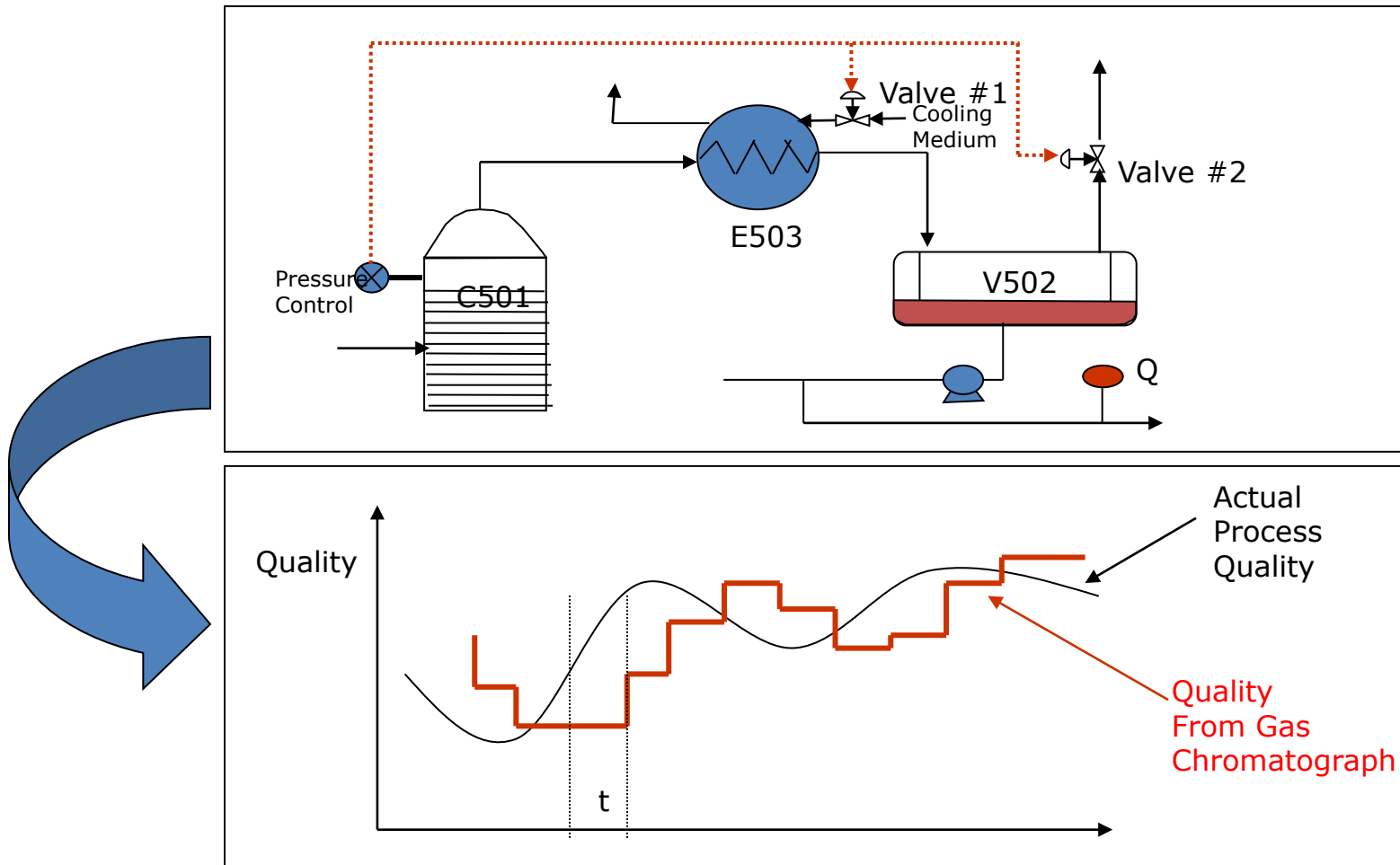
Advanced Process Control



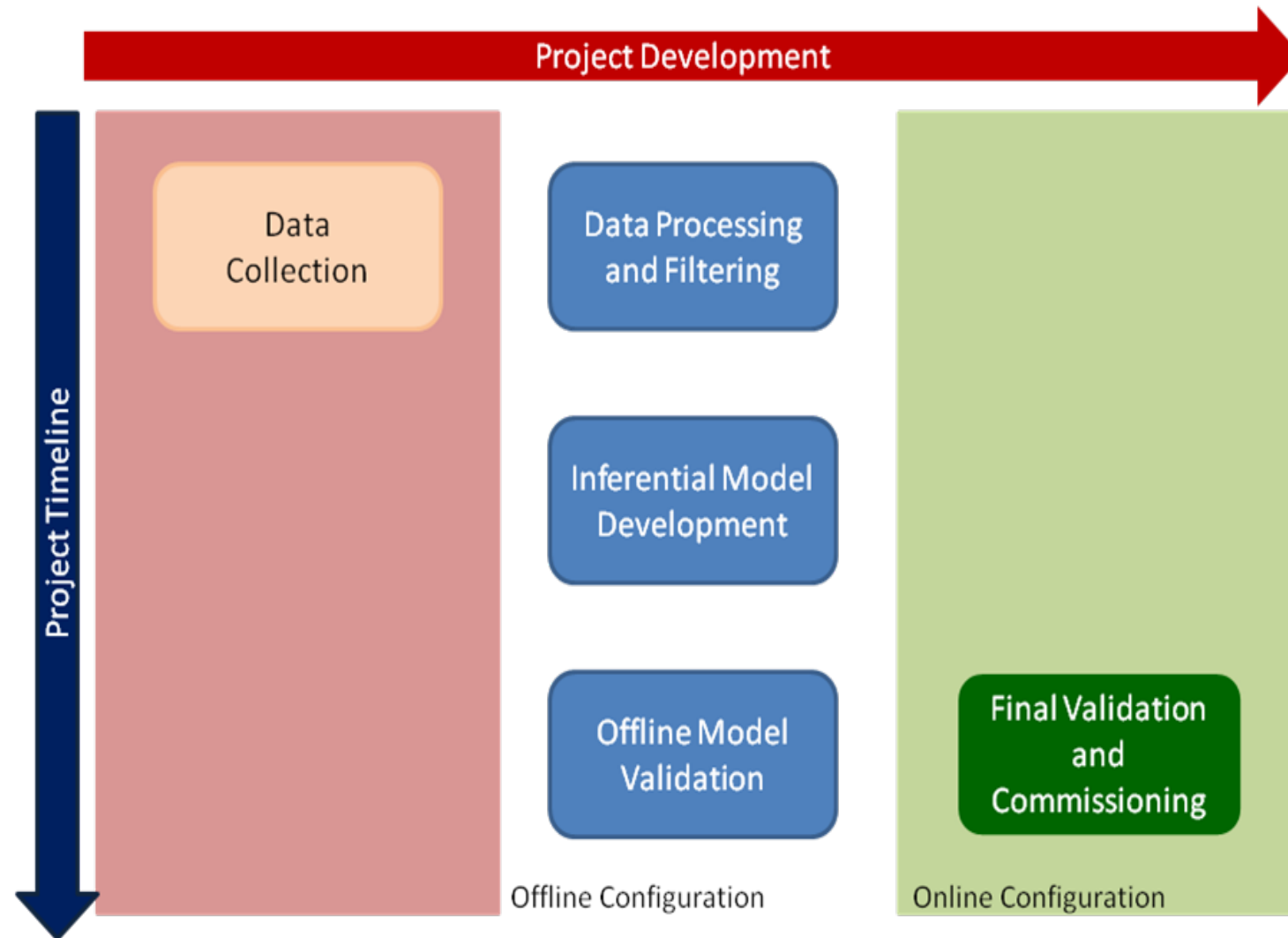
Inferentials (Property Estimation)

- Its key objective
 - to utilise the secondary variables in a process to 'infer' the value or 'property' of the primary or key variable.
- Why this technique is favourable:
 - Provides an instantaneous/real-time value of the key variable, eliminating transport lags and measurement delays.
 - Provides a measurement of key variable when it is unavailable (either no analysis made available, or analyser is down for maintenance)
- Note if the inferred value is sufficiently accurate then it can even be utilised in closed loop control.

Example



Inferential Development Workflow



Inferential Development

- 2 types of inferential exist:
 - Statistical/Regression based
 - First Principles
- Most APC vendors supply in addition to the APC package an additional package for Quality Estimation.
- Requirement for updating the Inferential to ensure estimation is still valid for purpose.

Sample of Statistical Regression Tool - *RQEPro*



Retrieval Practice

1. The key advantages of using APC controllers is when it is applied to processes that are difficult to control. What are the key characteristics of certain “difficult” processes for which APC applications are suited for:
 - a) _____
 - b) _____

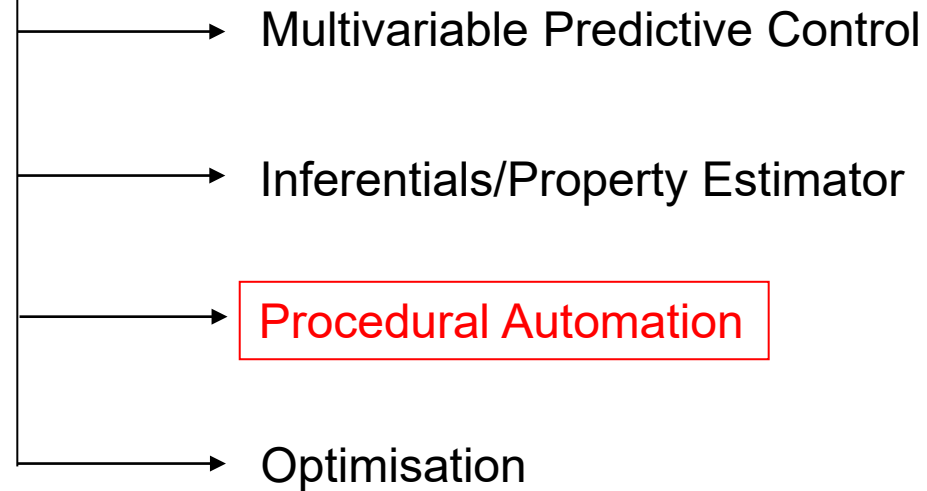
2. What is the one key driver for developing an APC controller? _____

3. Since APC Controllers are built from models of the process, why are simulation models (such as ICON models) not utilized for developing APC controllers?
 - a) _____
 - b) _____

4. One of the key parameters to monitor in the performance of a distillation column is the level of impurity in the product stream. This is usually measured by an analyser. Why does the analyser add a level of complexity to the control of the column?

- _____

Advanced Process Control



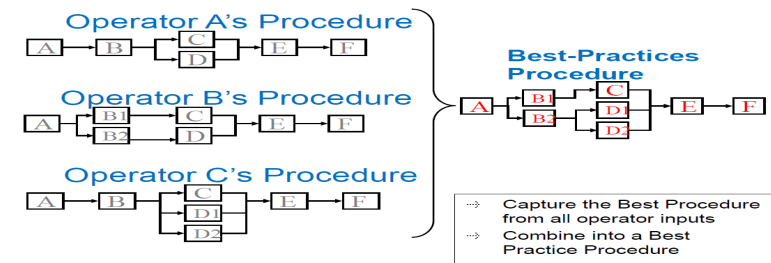
About Procedural Automation (PA)

What is Procedural Automation?

- Procedural Automation utilises event programming to automate manual processes and procedures
- It suits sequential operation i.e., start up/ shutdown, batch process, continuous routine process
- It captures the best practices from operators

It is important to overcome key issues such as:

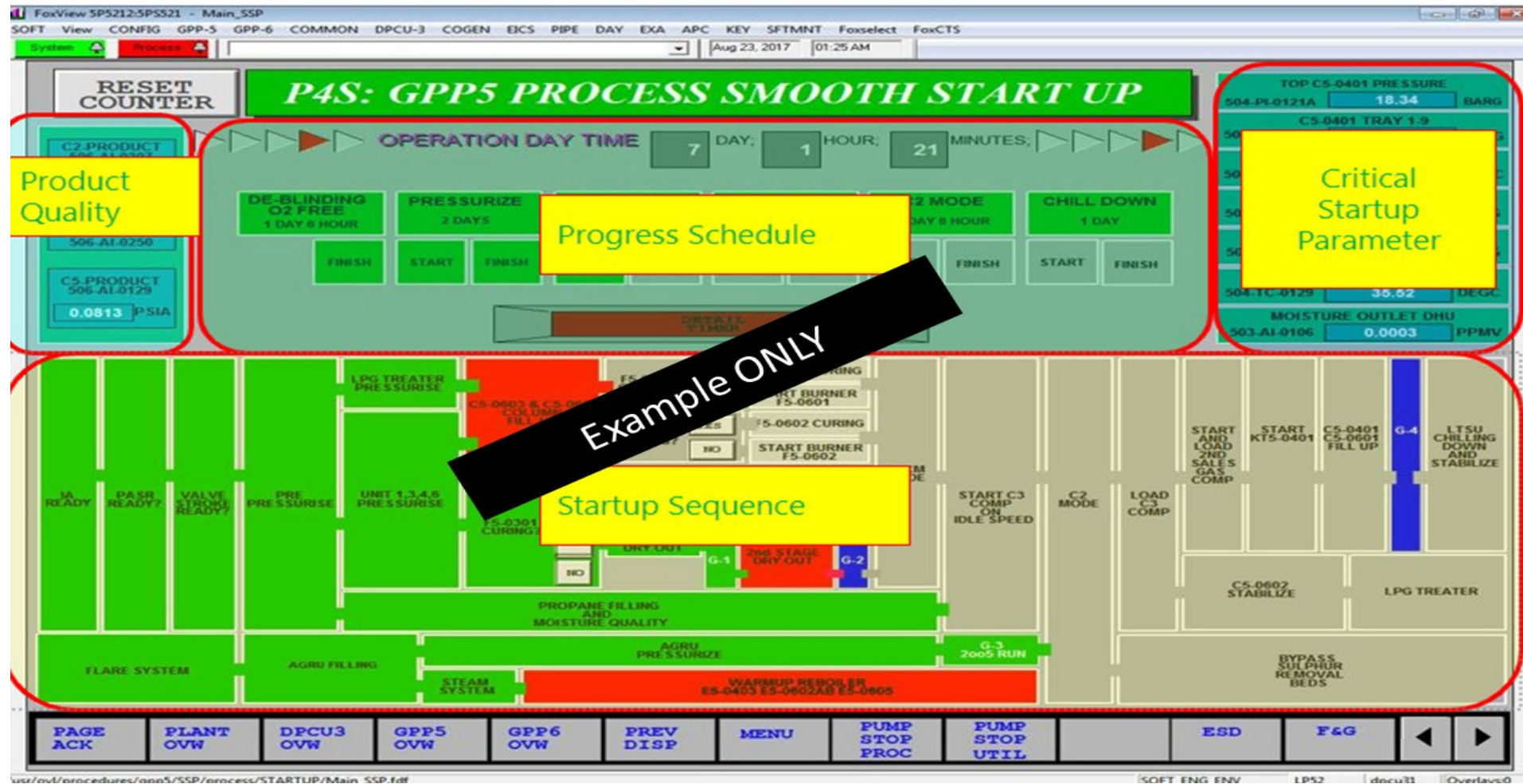
- Experienced operator retiring
- Poorly written procedures
- Non-compliance to procedure
- Inconsistency in operation
- Late detection of process anomaly



Aka Expert Systems

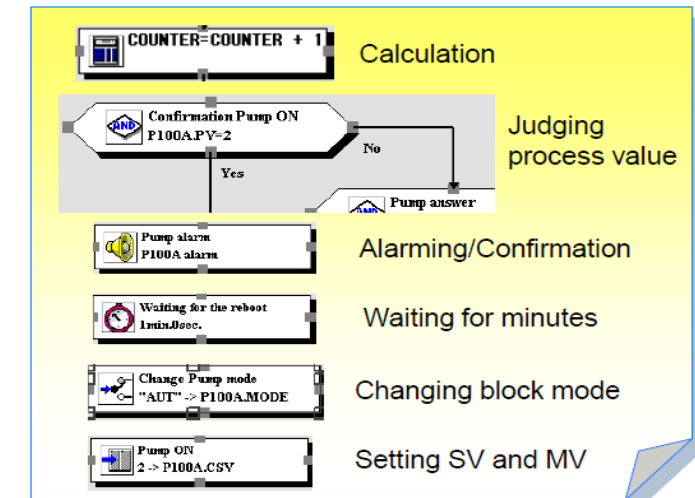
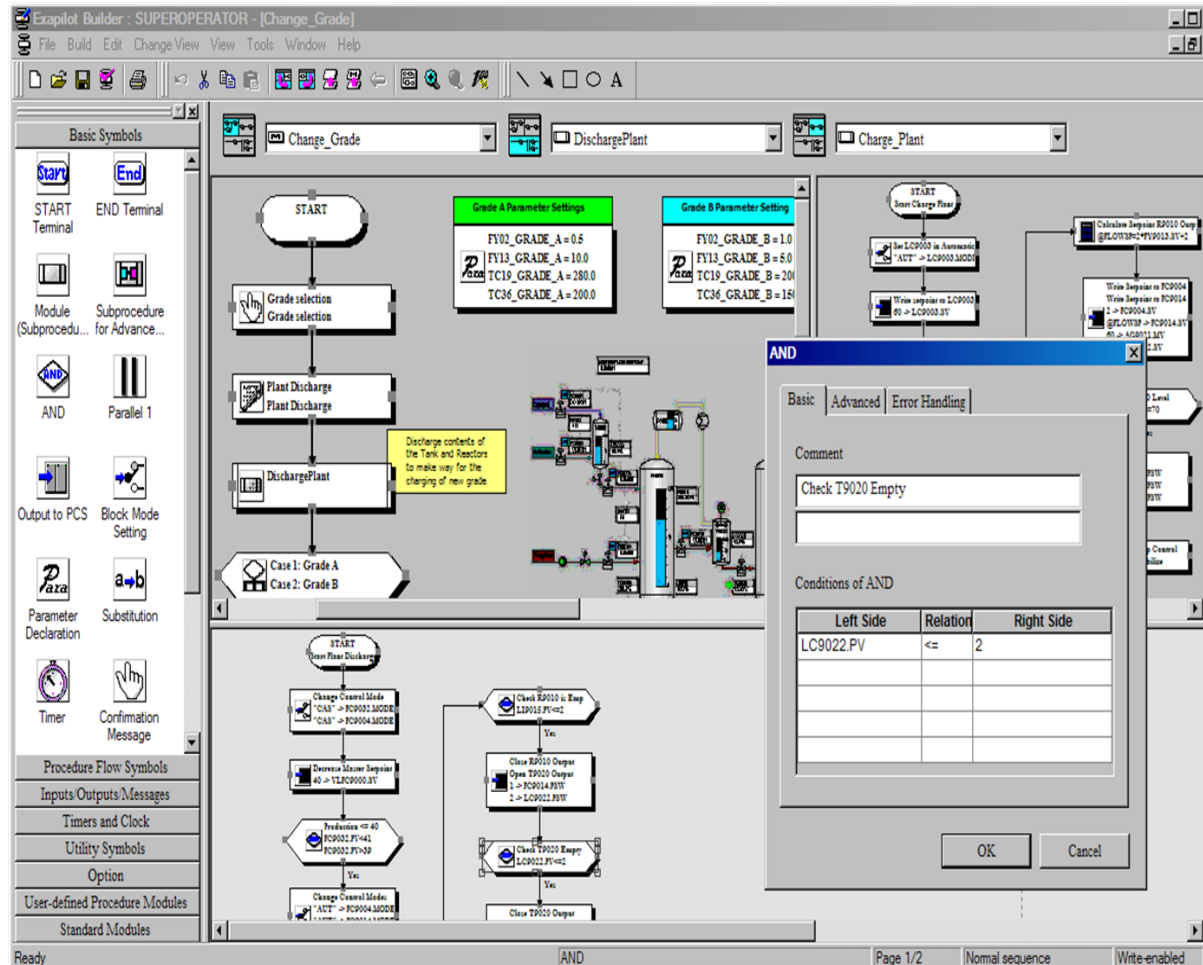
Example advisory dashboard display at DCS

Typical display of advisory dashboard procedural automation which allows transparency in monitoring



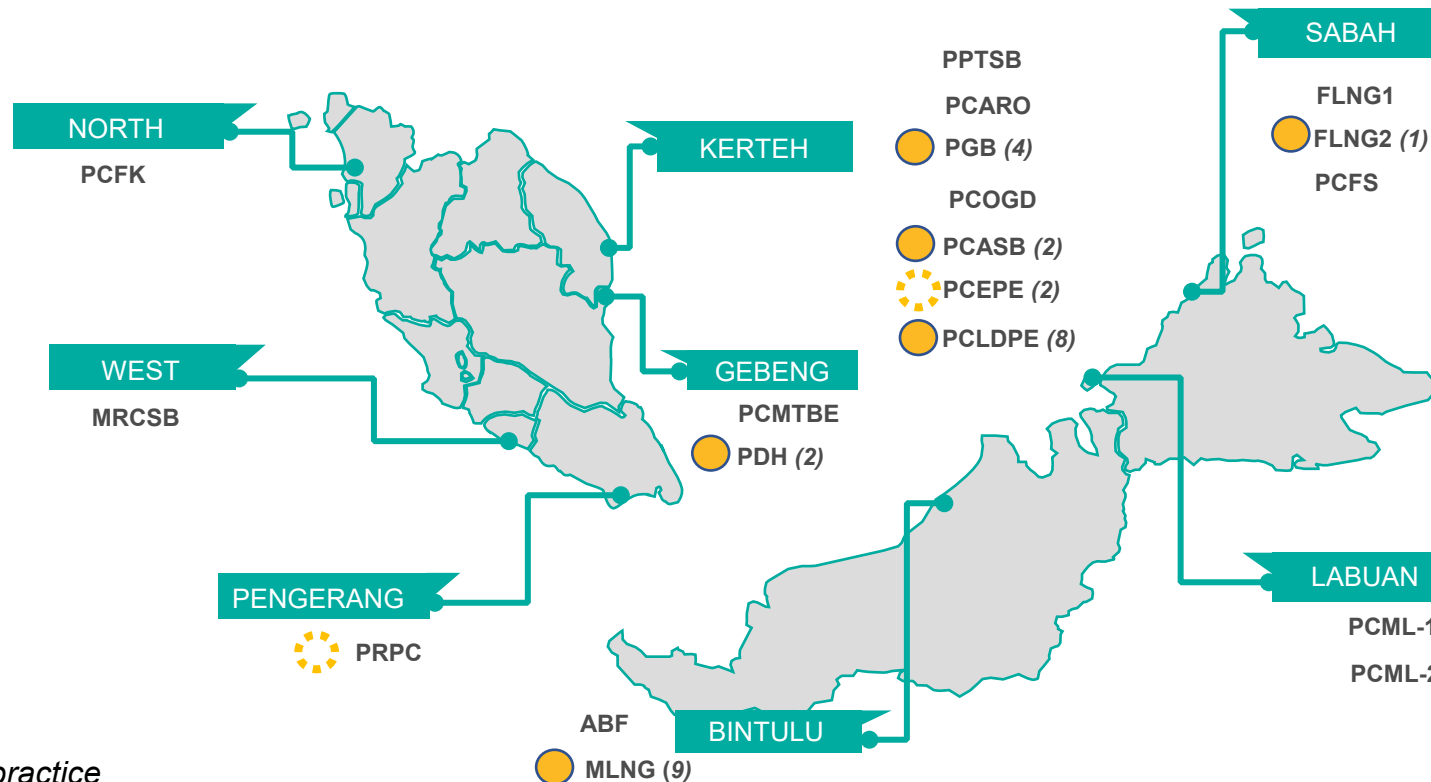
Example of 3rd party Procedural Automation programme (ExaPilot)

Typical display of 3rd party procedural automation (e.g., EXAPILOT Yokogawa)



Sites with Procedural Automation

PA was expanded from batch automation (implemented at DCS system) to 3rd party automation system



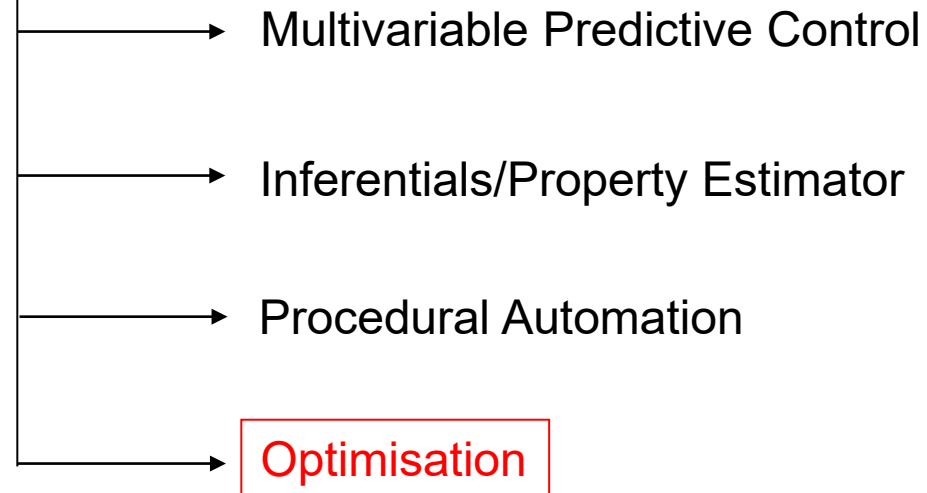
Benefits:

- Consistent Start up practice
- Efficiency and sustainability
- Yield maximization
- Safety monitoring
- Reliability improvement

No of applications
28

- Implemented
- Ongoing

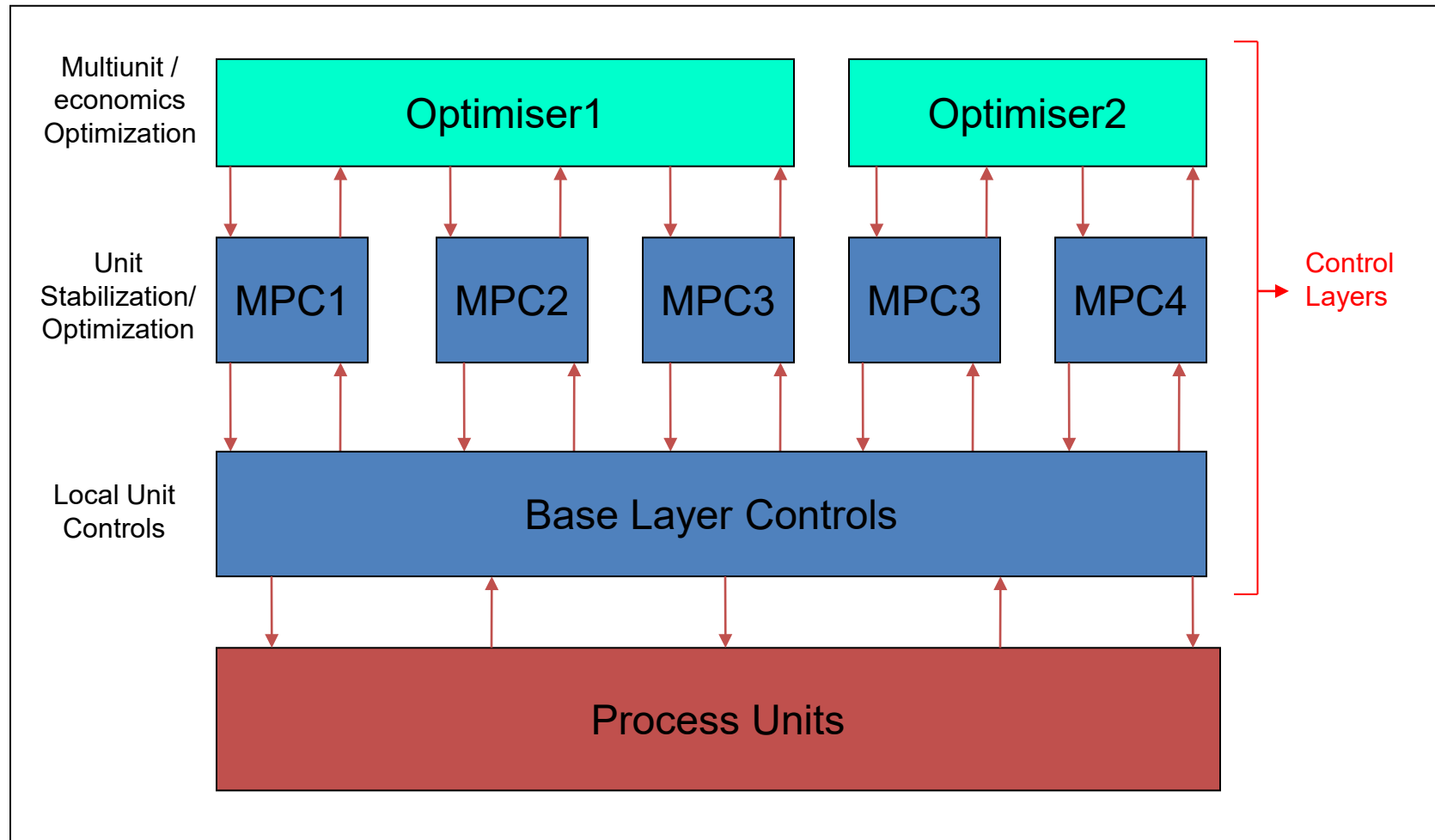
Advanced Process Control



Optimisers

- APC (MPC) Applications
 - MPC applications tends to be utilised to stabilise and optimise the operations of a single process unit
 - In addition, the objective of an MPC controller (while economic in nature) is defined upfront by configuring which handle to move.
 - MPC applications typically run at 1-minute intervals
- Optimisers (Real Time Optimisation - RTO)
 - RTOs are typically utilised to control multiple units or even plants
 - Real Time Optimisers are configured with ECONOMICS. As such, the output of the RTO will decide the most economic point of a unit or series of units
 - RTO output is typically sent to an APC application to modulate the unit
 - RTO's typically run every 6 hours to once a day
 - A new type of optimiser can be using Machine-Learning programming

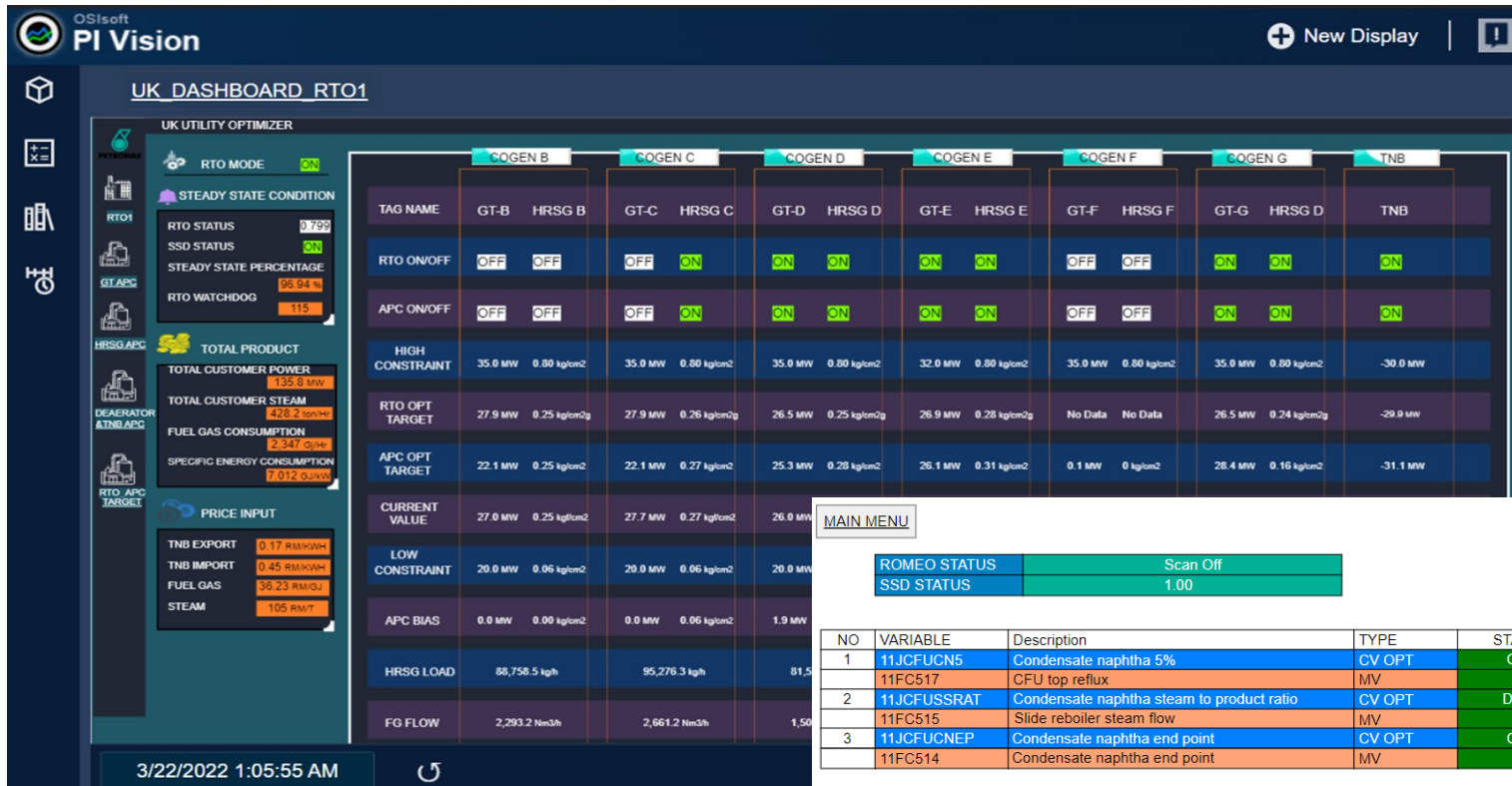
Optimiser Implementation



Similarity/Differences in using off-the-shelf Optimiser with ML

Description	RTO	ML
Typical use	Closed loop application	Advisory mostly, rarely for closed loop
Modelling methods	Model is based on 1st principles techniques.	Models is based on statistical methods, using big data.
Application location	Resides in business layer	Resides in cloud layer
Resources for modelling	Mainly process engineer who will develop the models with vendor	Model is developed by Data Scientist with process understanding provided by process engineer
Resources for integration	APC/I&C engineers with vendor	APC/I&C engineers, Solution Architect, UI/UX engineer
Visualisation	Not readily available, requires end-user configuration that can be from a separate dashboard application	
Programme Protective Features for direct process modulation	Readily available to be configured by end-user	Not readily available, needs to be engineered in
Maintenance	Mostly by APC engineer	Mostly Data Scientist, Solution Architect

Examples of RTO display using PI Vision



CRUDE FRACTIONATOR COLUMN (CDU) PSR1

MAIN MENU

ROMEO STATUS	Scan Off
SSD STATUS	1.00

APC CDU PSR1 STATUS	OFF
---------------------	-----

CV OPTIMIZATION

NO	VARIABLE	Description	TYPE	STATUS	HIGH LIMIT	LOW LIMIT	OPT TARGET	SP	PV	OPT STATUS
1	11JCFUCN5	Condensate naphtha 5%	CV OPT	CRIT	102.00	80.00	Scan Off	NA	23.54	Scan Off
	11FC517	CFU top reflux	MV		88.00	86.00	NA	20.00	20.00	NA
2	11JCFUSSRAT	Condensate naphtha steam to product ratio	CV OPT	DROP	80.00	20.00	Scan Off	NA	Bad	Scan Off
	11FC515	Slide reboiler steam flow	MV		76.00	74.00	NA	0.00	0.00	NA
3	11JCFUCNEP	Condensate naphtha end point	CV OPT	CRIT	190.00	165.00	Scan Off	NA	260.00	Scan Off
	11FC514	Condensate naphtha end point	MV		2.20	2.20	NA	0.00	Bad	NA

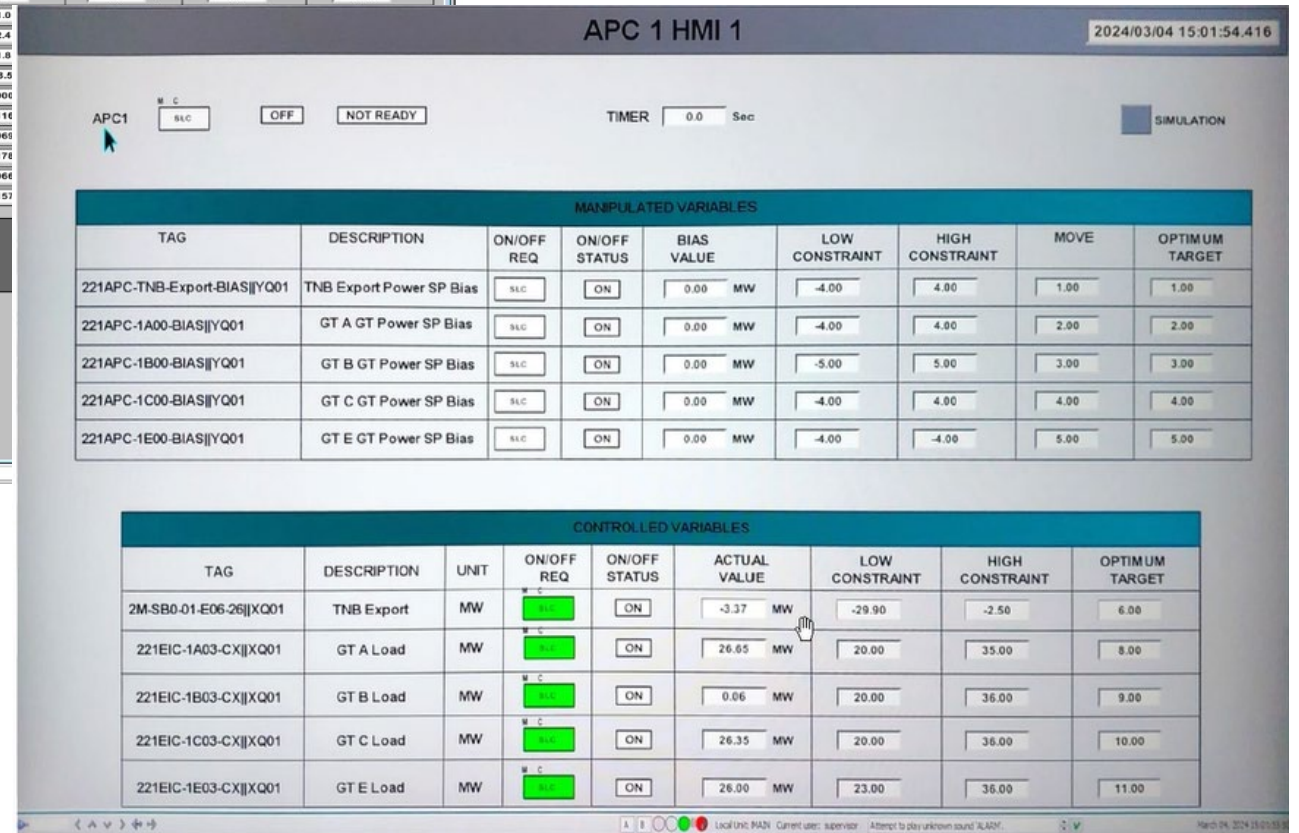
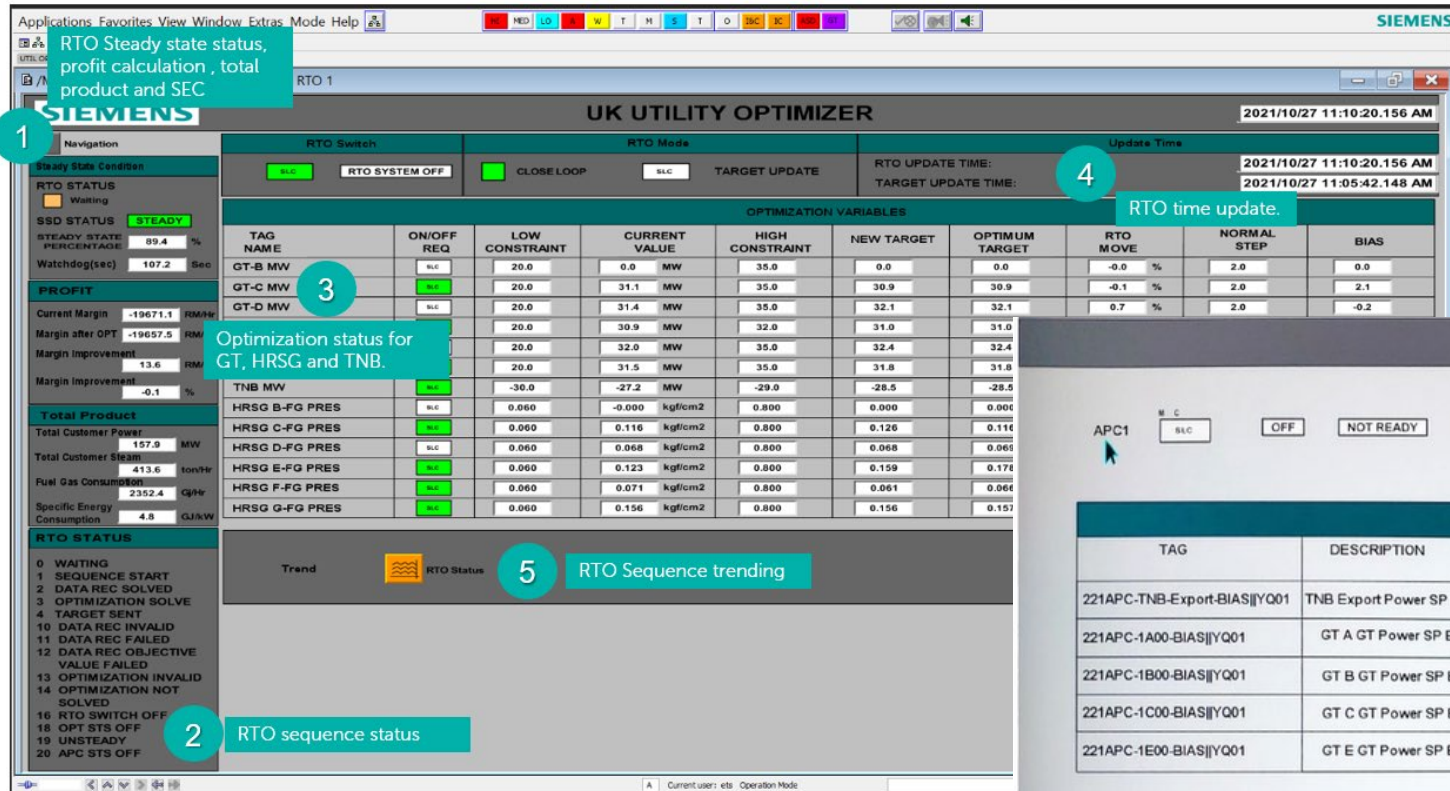
MV OPTIMIZATION

NO	VARIABLE	Description	TYPE	STATUS	HIGH LIMIT	LOW LIMIT	OPT TARGET	SP	PV	OPT STATUS
1	11TC512	Feed temperature	MV OPT	8.16	73.00	72.00	Scan Off	200.00	Bad	Scan Off
2	11TC530	Furnace reboiler COT	MV OPT	0.00	31.00	27.00	Scan Off	200.00	Bad	Scan Off

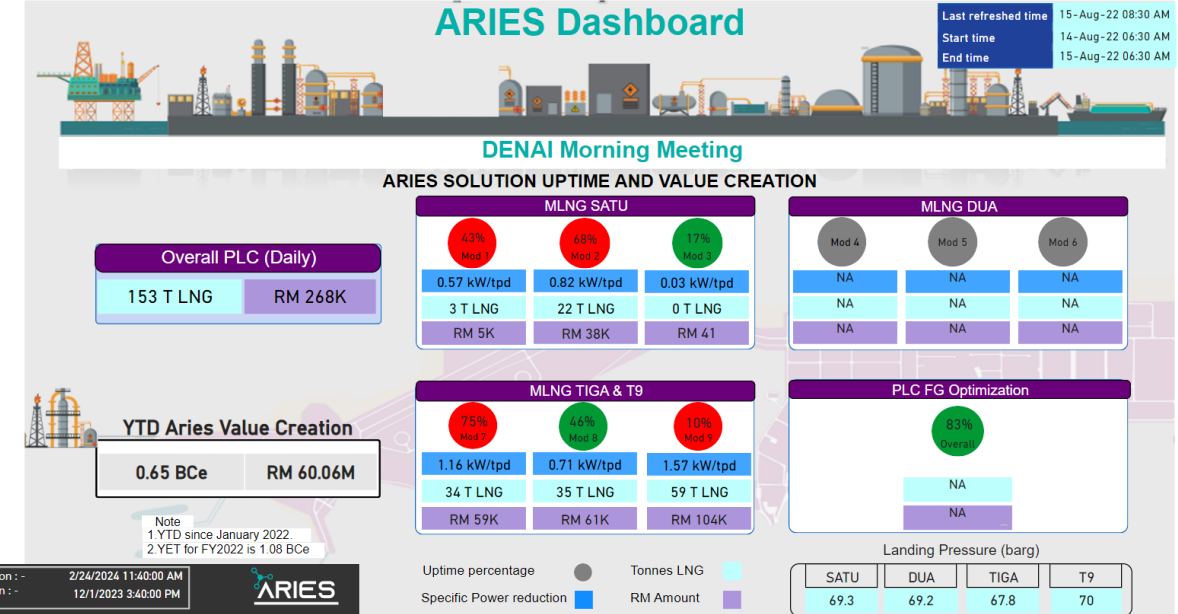
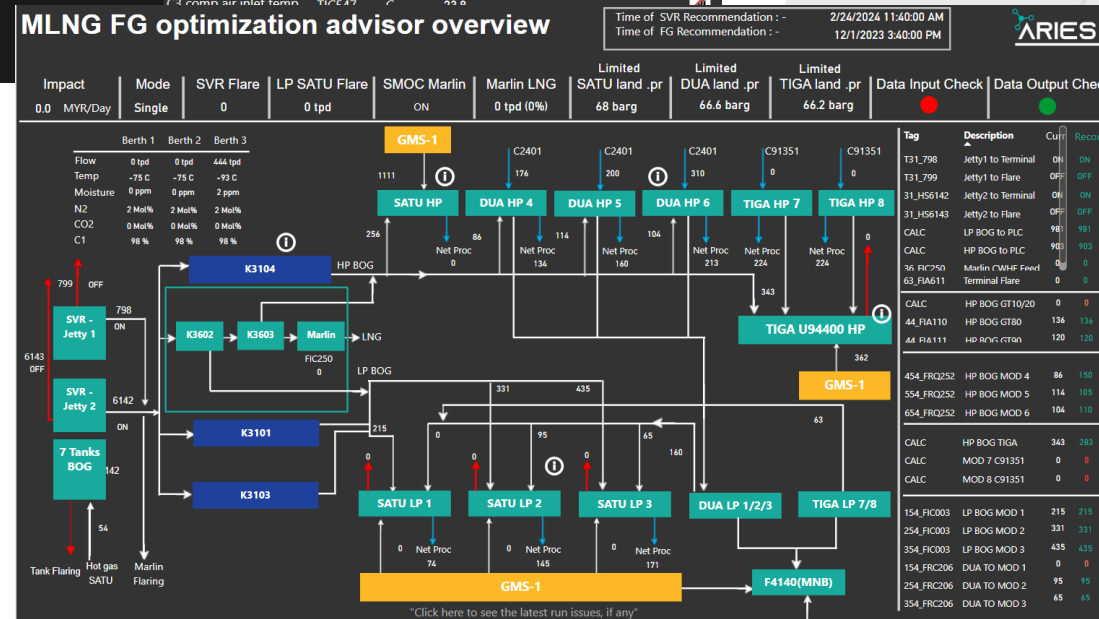
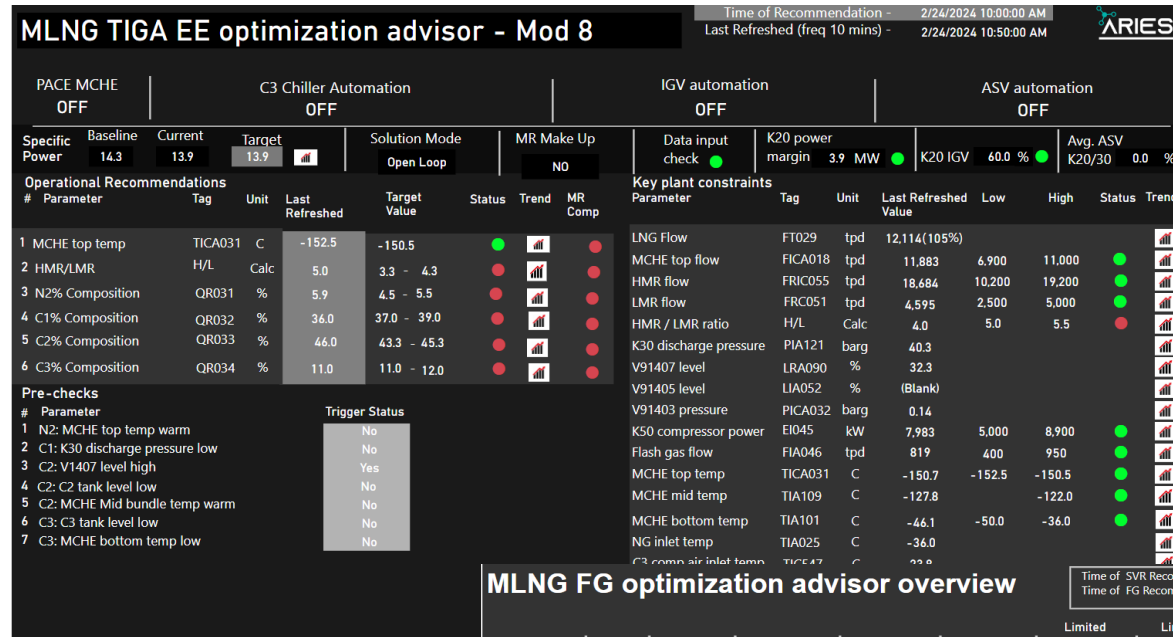
CV CONSTRAINT

NO	VARIABLE	Description	TYPE	STATUS	HIGH LIMIT	LOW LIMIT	OPT TARGET	PV
1	11QI011	Furnace Excess O2	CV CONST	GOOD	60.00	20.00	Scan Off	10.02
2	11TI524	Furnace Skin Temp	CV CONST	GOOD	80.00	50.00	Scan Off	31.91
3	11TI525	Furnace Skin Temp	CV CONST	DROP	4.00	1.50	Scan Off	64.22
4	11JCFUFIR10	Tray 10 internal reflux	CV CONST	GOOD	145.00	103.00	Scan Off	0.20
5	11JCFUIR10	Tray 10 internal reflux	CV CONST	GOOD	95.00	20.00	Scan Off	1.00

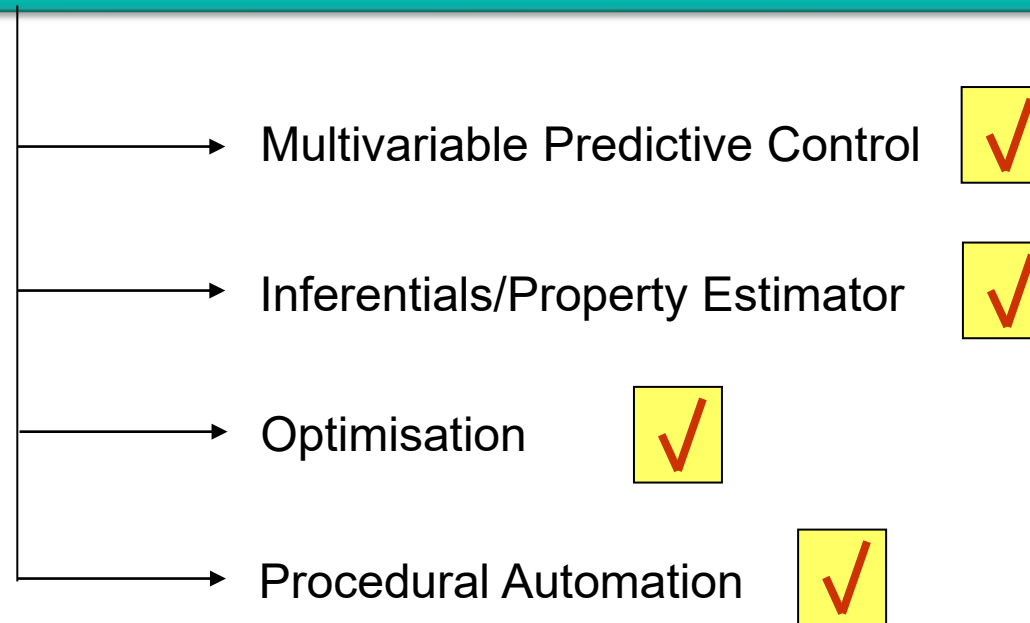
Examples of RTO display in DCS



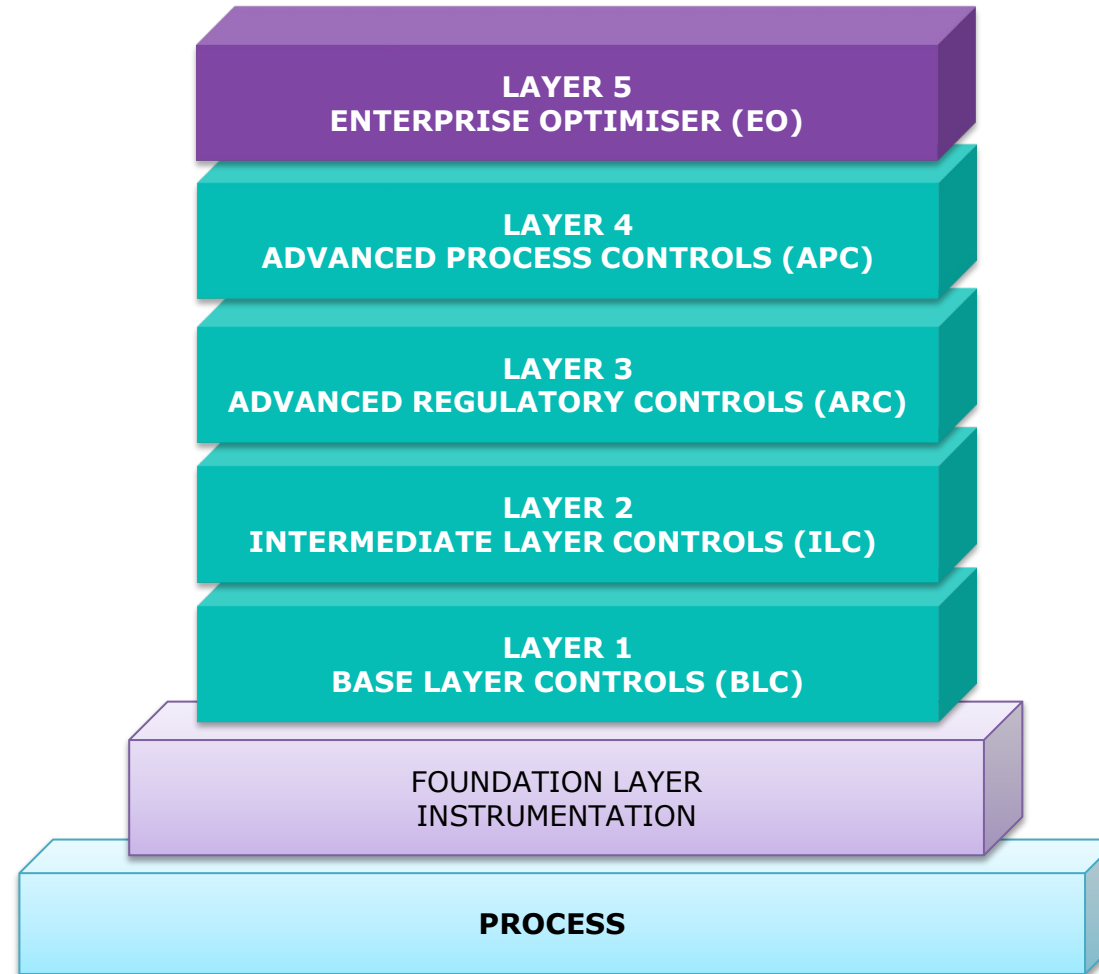
Examples of Optimiser Display via Machine Learning (using Power BI)



Advanced Process Control



Scope of SKG16.1 CTA



Are these objectives met?

1. Understand the concepts of Process Control as applied in PETRONAS (Process Control Definition, Five Layers of Control and Dynamic Stability[©])
2. Be able to explain Process Dynamics/Behaviour and their criticality in control
3. Be able to explain the “3 + 2 Steps to Understanding Base Layer Control”
4. Understand the concepts behind the Tuning and Tuning Techniques
5. Understand the different forms of Intermediate Layer controls (Cascade, Split Range, Override and Ratio Control Loops)
6. Understand the process requirement for Advanced Regulatory Controls (such as combustion, compression and distillation)
7. Aware of the concepts of APC (MPC) and Property Inference

Control Theory and Applications for Operations Management SKG16.1

For clarifications, please revert to:

Andriny Rusman, Principal Engineer, Process Control, GTS
Cheryl Poo Hwei Ching, Principal Engineer, Process Control, GTS
Tg Amansyah, Principal Engineer, Process Control, GTS
Fauziah Ali, Custodian Engineer, Process Control, GTS
Normaya Shabudin, Staff Engineer, Process Control, GTS
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Syndicate Exercise

L'or Noir Control System Design Competition



**SKG16.4 Advanced Process Control / SKG16.1 Operations Management
Control Theory and Applications Course**

Revision History

Rev	Date	Prepared By	Notes
2	8 th June 05	Prakash K.K.	Included letdown stations
3	7 th Nov 08	Prakash/Lim/ Andriny	Updated post training on 3 rd to 7 th Nov 08
4	21 Oct 11	Prakash/Lim/ Andriny	Updated post training on 17 to 21 Oct 11
5	16 Mar 12	Andriny	Updated post training on 29 Feb 12
6	23 rd Mar 12	Prakash K.K.	Major revamp post training 23 rd March 2012 to include Design Competition
7	22 th Aug 2012	Prakash K.K.	Minor amendments
8	4 th July 2013	Prakash K.K.	Minor amendments including Superheat Temp
9	12 th Feb 2014	Prakash K.K.	Minor Amendments
47	10 th April 2017	Prakash K.K.	Alignment of Revision Numbers and inclusion of RV and STG bypass
47g	7 th Sept 2018	Prakash K.K.	Inclusion of blow-offs on each header

Design Competition ...1

The Activity

- A competitive design activity has been initiated by L'or Noir Private Limited. L'or Noir is a multinational Oil and Gas company and have appointed a few design companies to design a control system for one of their plants in Southeast Asia.

Who do you represent?

- You and your selected team members represent the Control Design Team from different **competing** Engineering Design companies.

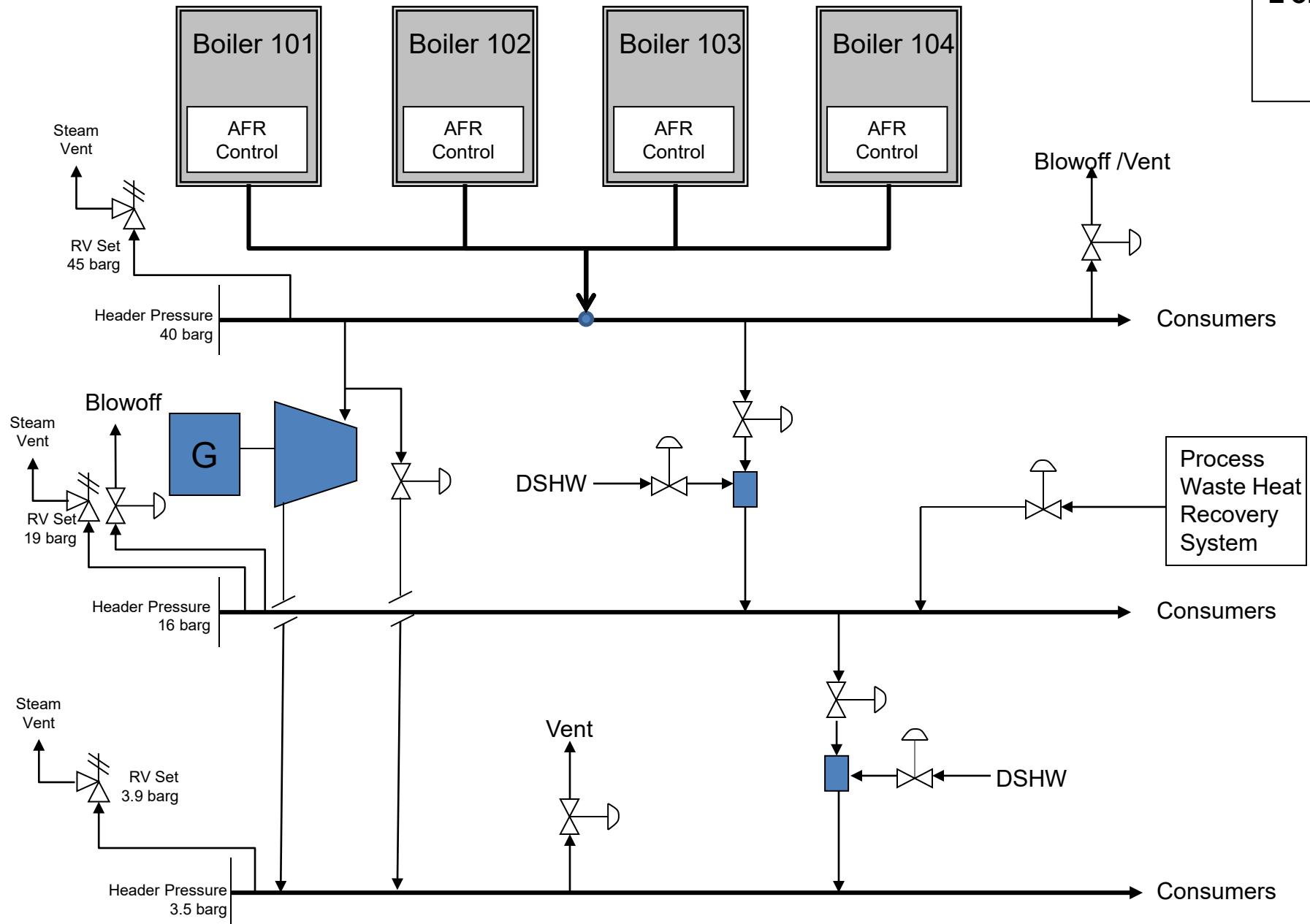
Task

- Each team has been assigned to build a control system for a three-level utility system in accordance with specific process and design criteria.
- The team will have 1 hour to develop a design that the team thinks will be effective in achieving the process objectives in a stable and sustainable manner.

Design Competition ...2

Delivery

- Upon completion of the design phase, each team will appoint an individual (the team lead) to present the design to the host team (L'or Noir Process Control Systems Head).
- L'or Noir will critique each design and determine if it meets its criteria for acceptance.
- L'or Noir may/may not decide to select one design for final implementation.



Process Overview

- L'or Noir SEA Gas Pvt. Ltd. is a 20-year-old gas processing facility located on an island in the South China Sea.
- The facility obtains both gas and condensates from offshore and onshore gas fields.
- The gas is treated and sent to a distribution system via a series of compressors. The condensate is stabilised and shipped to a refinery located in mainland Malaysia.
- The facility is primarily a steam driven facility, with steam utilised to drive the main refrigerant compressors and reboilers on the process side and also to drive turbo-generators and the Instrument Air turbine-compressor in the utility system.

Process Description ...1

- L'or Noir Gas Pvt Ltd's gas and condensate process plant is supported by an integrated Utility Plant.
- The steam is generated from 4 boilers, operated consistently in an N+1 sparing configuration.
- The plant has a three-level steam system. The pressure and temperature levels are as follows:
 - HP header pressure – 40 barg, Superheat temperature – 375 C
 - MP header pressure – 16 barg, Superheat temperature – 300 C
 - LP header pressure – 3.5 barg, Superheat temperature – 240 C

Process Description ...2

The boilers discharge into the HP steam system headers at 40 barg. RV's at the boiler drum will relief at 46 barg. There is a 2 barg pressure drop from the boiler drum to the header.

Power is generated from HP steam by utilising a back-pressure turbine which releases into the LP system at 3.5 barg. The load on the turbine is dependent on the power demand of the various electrical equipment in the plant.

Process Description ...3

- The MP system is supplied
 - from the HP system via a letdown station
 - from a new heat recovery system in the process

- The heat recovery system should be able to support demand of steam demand from the MP header at maximum process production. At the time of the preparation of this document the H&MB's have not yet been finalised and this detail is still in HOLD state.

- The LP system has two supplies:
 - Via the backpressure turbine
 - Supplementary supply of LP steam is required to meet demand in the LP system. This steam is supplied from the MP system via a letdown station.

Process Operating Requirements

➤ Operating Requirements:

Each header shall be controlled at the specified setpoints of pressure and superheat temperature. The exception to this rule is the LP header. The pressure in the LP header may be allowed to increase to 3.8 barg. RV's on this header are set at 3.9 barg.

The Waste Heat Recovery System shall be prevented from backflow from the MP Steam header

Each letdown station shall ensure that the steam entering a header has the correct superheat.

In the event of the Steam Turbo-Generator (STG) trip, the intent is for HP-LP letdown valve to open temporarily to prevent the RV's on the HP header to open

Control Design Requirements

➤ Design Requirements

- Prepare a P&ID with the required control scheme to meet the Operating Requirements
- Identify the transmitters required to control the process.
- Identify the control valve direction and the controller action (direct/reverse)
- Assume standardisation of DCS signals such that 0% controller output equals valve close
- Draw in all the necessary control blocks as necessary for e.g. multipliers, etc.
- Consider start-up scenarios and any control scheme that may be required during system start-up

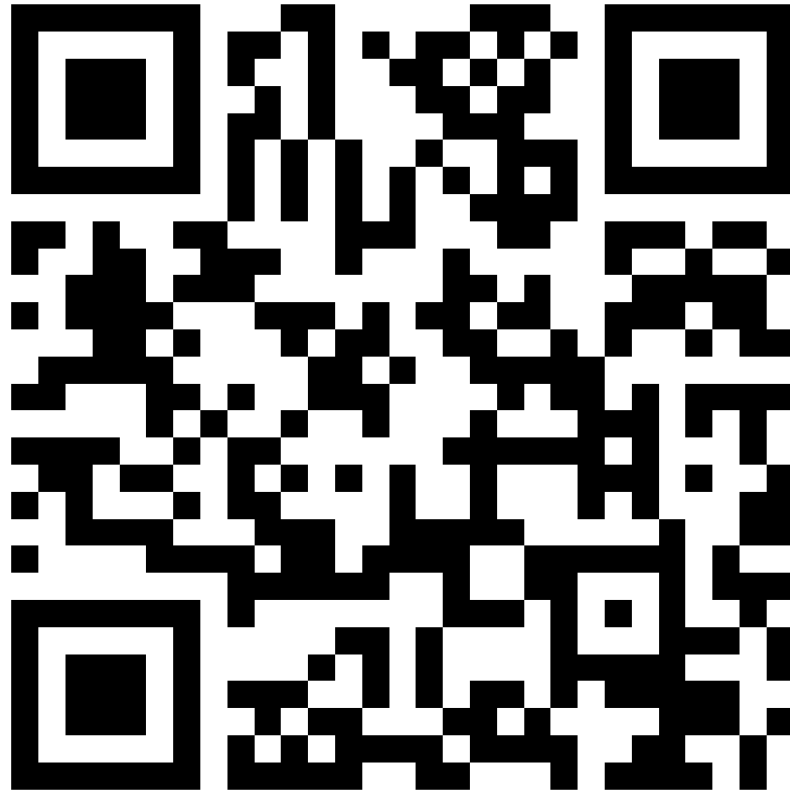
➤ Presentation

- State the name of your design company
- You shall need to convince the Chief Engineer, Process Control of Le Noir
- It is the Chief Engineers discretion to select a scheme for his facility

END SLIDE

L'or Noir
Control System Design Competition

Final Quiz



- <https://forms.office.com/r/jUXYi3evVS>